Techno-Economic Viability Study Report of 120 KLD Ethanol Distillery Project of Crescendo Industries Private Limited at Mahewa, Rewa of Madhya Pradesh

By,

NITCON Limited

317 - A, 3rd Floor D 21, Corporate Park, Sector 21 Dwarka,

New Delhi-110077

Dated – February 20, 2026

Issuance Certificate

M/s Crescendo Industries Private Limited (CIPL), incorporated on 28th June 2022, is setting up a 120 KLD grain-based ethanol manufacturing facility along with a 3.00 MW captive cogeneration power plant at Village Mahewa, Teonthar Tehsil, Rewa District, Madhya Pradesh. The project will produce fuel-grade ethanol, DDGS, and CO_2, with ethanol to be supplied to Oil Marketing Companies under the Government of India's Ethanol Blending Programme (EBP). The plant will utilize broken rice, maize, and other surplus or damaged grains as feedstock.

The first instalment of the sanctioned loan was disbursed by IREDA on 27th September 2024. As per the financing terms, the project timeline is 18 months from the date of commencement of work, which began upon receipt of the first instalment in September 2024. Accordingly, the original target Commercial Operation Date (COD) was 31st March 2026. However, a delay is anticipated, and the revised COD has been projected as 30th June 2026.

The delay in implementation result in a cost overrun of Rs. 27.39 Crores, primarily on account of higher Interest During Construction (IDC) and escalation in manpower and execution costs, without any material change in project scope or technology. Consequently, the total project cost has increased from Rs. 144.46 Crores to Rs. 171.85 Crores.

The project was initially funded through a term loan of Rs. 115.57 Crores from IREDA and promoter equity of Rs. 28.89 Crores. To bridge the funding gap, the Company has proposed an additional term loan of Rs. 21.91 Crores along with promoter equity.

In this regard, Company has appointed NITCON Limited to undertake the Techno-Economic Viability study of the said distillery project.

The finding of the assessment as undertaken by NITCON has been presented in form of this draft report for review of Lenders/IREDA and M/s Crescendo Industries Private Limited.

For and on behalf of NITCON Limited

Name – Mr. Aditya Karki
Designation – Senior Project Analyst

Table of Contents

Executive Summary

Name of Company	Crescendo Industries Private Limited ('CIPL' or 'The Company')
Date of incorporation	28th June 2022
CIN	U24304UP2022PTC166562
Constitution	Private Limited Company
Nature of activity	Manufacturing of fuel grade Ethanol
Registered Office	Plot No.7 and 8, Khata No. 49A, B Chak Malhori Road, Malhour Chinhat, Lucknow, Uttar Pradesh 226028
Project	120 KLD Grain Based Distillery along with 3 MW Co-generation power plant
Project Location	Rakba no. 38/1/3(S), 38/1/1(S), 38/2(S), 38/3(S) Mahewa Village, Teonthar tehsil, Rewa District, Madhya Pradesh
Company Background	M/s Crescendo Industries Private Limited ("CIPL" or "the Company") is a private limited incorporated on 28th June 2022. It is registered at Registrar of Companies, Kanpur with registered address at Plot No.7 and 8, Khata No. 49A, B Chak Malhori Road, Malhour Chinhat, Lucknow, Uttar Pradesh 226028. Its authorized share capital is Rs. 28,90,00,000 and its paid-up capital is Rs. 28,89,00,000, as on 14th February 2026. And the Company is setting up a grain-based ethanol manufacturing facility with an installed capacity of 120 kilolitre per day (KLD) along with a 3.00 MW captive cogeneration power plant, at Rakba No. 38/1/3(S), 38/1/1(S), 38/2(S), and 38/3(S), Village Mahewa, Teonthar Tehsil, Rewa District, Madhya Pradesh. The Project plant will produce Fuel-Grade Ethanol, Distillers Dried Grains with Soluble (DDGS), and carbon dioxide (CO_2). Ethanol produced at the facility will be sold to Oil Marketing Companies (OMCs) such as Bharat Petroleum Corporation Limited (BPCL), Hindustan Petroleum Corporation Limited (HPCL), and other eligible OMCs under the Ethanol Blending Programme (EBP) in line with the National Policy on Bioenergy, 2018 of the Government of India (GoI). DDGS will be sold to feed manufacturers as well as in the open market as a protein-rich animal feed supplement. The by-product CO_2 is proposed to be supplied to aerated beverage manufacturers and/or fertilizer units.

The plant will primarily utilize broken rice/ maize and other damaged or surplus grains and food materials available in the region as feedstock for ethanol production. The ethanol produced will mainly be used for blending with petrol as a renewable biofuel, supporting the Government of India's objectives of energy security and reduction in fossil fuel dependency.

To meet the internal power requirement of the facility a 3.00 MW cogeneration power plant being installed to fulfil the total power requirement, in line with the recommendations stipulated under Environmental Clearance (EC). Power generation will be based on high-pressure steam generated from a 25 TPH boiler operating at 67 ATA, which will supply steam both for the ethanol production process and for power generation through a steam turbine generator (STG).

The original cost of project was as below –

All Figures in Rs. Crore	
Particulars	**Total**
Land and Land Development	1.00
Building and Civil Works	20.88
Plant and Machinery Cost	102.58
Miscellaneous Fixed Assets	0.50
Preliminary and Pre-operative	3.00
Interest During Construction	11.54
Contingency	4.96
Margin Money for Working Capital	-
Total Project Cost	**144.46**

Based on the above table and information received from the Company, it is noted that the project has received a term loan – 1 sanction of Rs. 115.57 Crores from IREDA (Lender), and the balance amount of Rs. 28.89 Crores has been infused as equity by the promoters of the Company.

Accordingly, the project is funded in a Debt–Equity Ratio of 4:1, with debt of Rs. 115.57 Crores and promoter's equity contribution of Rs. 28.89 Crores.

Land details

Based on the site visit and information received it is noted that the Company has purchased land admeasuring of 76,200 Sq.M. or 7.62 hectare at Mahewa Village, Teonthar tehsil, Rewa District, Madhya Pradesh and Land use conversion certificate has been obtained from "The Madhya Pradesh Bhu-Rajsva Sanhita".

Project's Current status

As discussed, the Company is setting up a 120 KLD grain-based distillery. The project implementation commenced in September 2024 and was initially targeted to achieve COD by 31st March 2025. However, owing to unforeseen circumstances beyond the control of the Company, the project execution was delayed by approximately 03 months, resulting in cost overruns and revised COD has been assumed as on 30th June 2026.

The overall project cost has increased by Rs. 27.39 Crores, primarily due to escalation in Interest During Construction (IDC) arising from the extended implementation period, higher manpower and wage costs, and certain additional works necessitated during execution. It is noted that the cost overrun is time-related in nature and not attributable to any major change in project scope or technology. The revised project cost and means of finance have been exhibited below –

Cost of Project –

All Figures in Rs. Crore					
Particulars	Original Cost	Already incurred	31-Mar-26	31-Mar-27	Total
Land and Land Development	1.00	0.97	1.00	-	1.00
Building and Civil Works	20.88	18.58	21.85	1.90	23.75
Plant and Machinery Cost	102.58	96.55	98.61	14.22	112.83
Miscellaneous Fixed Assets	0.50	0.50	0.44	0.07	0.50
Preliminary and Pre-operative	3.00	3.00	2.61	0.39	3.00
Interest During Construction	11.54	9.88	13.05	3.47	16.52
Contingency	4.96	0.72	1.00	3.96	4.96
Margin Money for Working Capital				9.29	9.29
Total Project Cost	**144.46**	**130.19**	**138.56**	**33.30**	**171.85**

Means of finance –

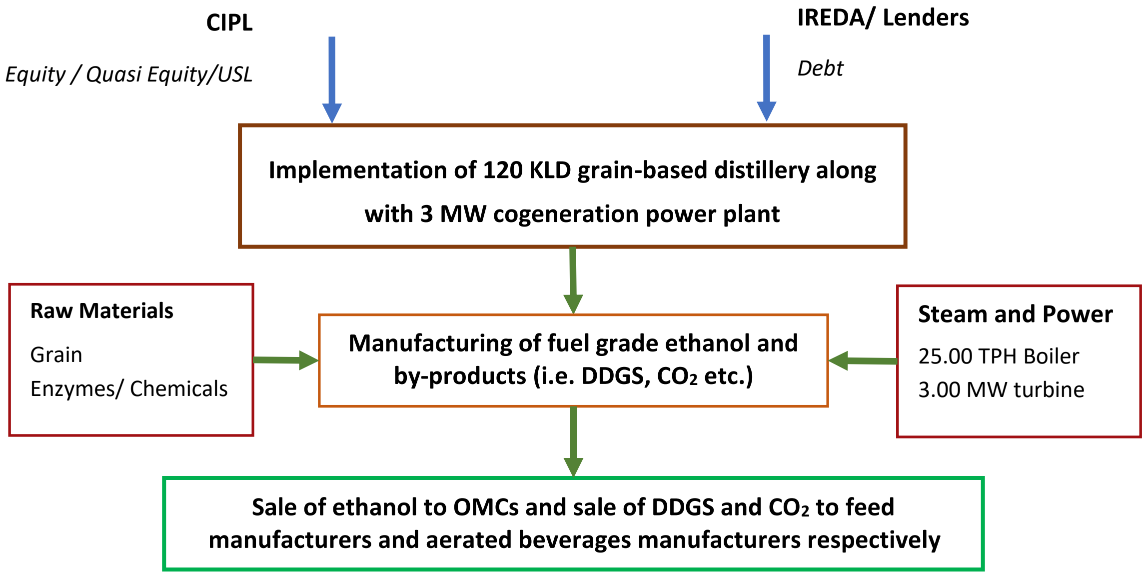
All Figures in Rs. Crore					
Particulars	Original Cost	Already incurred	31-Mar-26	31-Mar-27	Total
Promoters Contribution	28.89	28.89	28.89	5.48	34.37
Term Loan 1	115.57	101.30	109.67	5.90	115.57
Term Loan 2				21.91	21.91
Total Means of Finance	**144.46**	**130.19**	**138.56**	**33.30**	**171.85**

To bridge the funding gap arising from the aforesaid cost escalation and to ensure timely completion and stabilization of operations, the Company has approached the lenders for sanction of an additional term loan – 2 of Rs. 21.91 Crores, with the balance to be infused by promoters' funds.

In this regard, the Company has approached and appointed NITCON Limited to undertake Techno Economic Study of the project. NITCON accepted the assignment, the evaluation and finding of the Consultants have been presented in form of this report

Project Configuration

The project configuration as conceptualized by the Company and as envisaged by the Consultants have been illustrated below –



Source: CIPL

Project Cost

The revised project cost of the project has been exhibited below –

All Figures in Rs. Crore						
Particulars	**Original Cost**	**Already incurred**	**Increased Cost**	**31-Mar-26**	**31-Mar-27**	**Total**
Land and Land Development	1.00	0.97	-	1.00	-	1.00
Building and Civil Works	20.88	18.58	2.87	21.85	1.90	23.75
Plant and Machinery Cost	102.58	96.55	10.25	98.61	14.22	112.83
Miscellaneous Fixed Assets	0.50	0.50	-	0.44	0.07	0.50
Preliminary and Pre-operative	3.00	3.00	-	2.61	0.39	3.00
Interest During Construction	11.54	9.88	6.64	13.05	3.47	16.52
Contingency	4.96	0.72	-	1.00	3.96	4.96
Margin Money for Working Capital			9.29		9.29	9.29
Total Project Cost	**144.46**	**130.19**	**29.05**	**138.56**	**33.30**	**171.85**

Source: CIPL and NITCON Estimates

***Note -** Based on the above table, it is noted that the original total project cost was estimated at Rs. 144.46 Crore. However, during implementation, the project has witnessed cost overruns primarily on account of underestimation of certain capital cost components and an increase in Interest During Construction (IDC) arising from time overruns. Consequently, the revised total project cost has increased to Rs. 171.85 Crore.

As on 17th January 2026, the Company incurred capital expenditure of approximately Rs. 130.19 Crore towards project implementation. The balance project cost, including the increased cost components, is proposed to be funded through additional term loan assistance from the lenders, along with the stipulated promoter contribution, as per the revised funding plan.

Means of Finance

On account of revised project cost the means of finance for the project has been presented in the table below –

All Figures in Rs. Crore					
Particulars	Original Cost	Already incurred	31-Mar-26	31-Mar-27	Total
Promoters Contribution	28.89	28.89	28.89	5.48	34.37
Term Loan 1	115.57	101.30	109.67	5.90	115.57
Term Loan 2				21.91	21.91
Total Means of Finance	**144.46**	**130.19**	**138.56**	**33.30**	**171.85**

Source: NITCON Estimates

Term Loan – 1

The broad covenants for proposed terms loan – 1 facility have been presented in the exhibit below –

Description	Details
Term Loan Amount	Rs. 115.57 Crores
Interest Rate	12.34% (Rate of Interest from Existing TL of IREDA)
Door to Door Tenure	7.25 Years or 87 Months
Implementation Duration	03 months from Financial Closure (assumed as on 31st March 2026)
Date of Commercial Operations	30th June 2026
Post Construction Moratorium	12 Months
Repayment Duration	6 Years and 24 Quarters
Repayment Mode	24 Accelerated Quarterly Instalments
First Instalment	July, 2027
Last Instalment	April, 2033

Source: NITCON Estimates

Term Loan – 2

The broad covenants for proposed terms loan – 2 facilities have been presented in the exhibit below –

Description	Details
Term Loan Amount	Rs. 21.91 Crores
Interest Rate	12.34% (Rate of Interest from Existing TL of IREDA)
Door to Door Tenure	7.25 Years or 87 Months
Implementation Duration	03 months from Financial Closure (assumed as on 31st March 2026)
Date of Commercial Operations	30th June 2026
Post Construction Moratorium	12 Months
Repayment Duration	6 Years and 24 Quarters
Repayment Mode	24 Accelerated Quarterly Instalments
First Instalment	July, 2027
Last Instalment	April, 2033

Source: NITCON Estimates

Key Financial Parameters

The key financial parameters of the Project, have been presented in the exhibit below –

All Figures in Rs. Crore								
Description	31-Mar-27	31-Mar-28	31-Mar-29	31-Mar-30	31-Mar-31	31-Mar-32	31-Mar-33	31-Mar-34
Revenue	160.07	250.84	271.59	279.96	286.96	294.04	300.99	308.00
Total Operating Costs	153.26	210.12	225.85	231.51	237.14	243.06	249.14	255.37
Operating Profit	6.82	40.72	45.74	48.45	49.83	50.98	51.85	52.63
Operating Profit Margin	4.26%	16.23%	16.84%	17.30%	17.36%	17.34%	17.23%	17.09%
Contribution	8.81	43.85	49.12	51.94	53.40	54.64	55.61	56.48
Contribution Margin	5.50%	17.48%	18.09%	18.55%	18.61%	18.58%	18.47%	18.34%
BEP Sales	394.54	166.07	152.85	137.37	122.96	106.72	89.36	77.04
BEP Capacity Utilisation	246.48%	66.21%	56.28%	49.07%	42.85%	36.29%	29.69%	25.01%
Cash Break Even	305.47	128.71	116.74	102.17	87.87	71.58	54.02	41.43
Cash Break Even Margin	190.83%	51.31%	42.99%	36.49%	30.62%	24.34%	17.95%	13.45%
Net Profit Margin	-8.06%	5.91%	7.91%	9.37%	8.46%	9.20%	9.93%	10.39%
Equity Share Capital	34.37	34.37	34.37	34.37	34.37	34.37	34.37	34.37
Unsecured loan from Promoters	11.89	15.89	15.89	15.89	15.89	15.89	15.89	-
Reserves and Surplus	-12.90	1.91	23.39	49.62	73.91	100.97	130.86	162.87
Tangible Net Worth (TNW)	33.36	52.17	73.65	99.88	124.17	151.23	181.12	197.24
Term Loan	115.57	106.57	91.57	73.57	53.57	30.57	7.57	-
Debt Equity Ratio	3.46	2.04	1.24	0.74	0.43	0.20	0.04	-
Total Outside Liability (TOL)	201.59	191.46	179.20	159.76	137.31	110.90	83.53	59.92
TOL/ TNW	6.04	3.67	2.43	1.60	1.11	0.73	0.46	0.30
Closing Cash Balance	2.00	2.13	11.65	23.28	29.57	36.31	50.56	72.95
DSCR	-	1.49	1.44	1.49	1.37	1.36	1.52	5.06
Minimum DSCR	1.36							
Average DSCR	2.77							
NPV	137.74							
IRR	22.44%							
Cost of Capital	10.15%							

Source: NITCON Estimates

Based on the assessment undertaken by NITCON, the key output parameters are –

- The Net Present Value (NPV) of the Project is Rs. 137.74 Crores, which is above zero and hence indicates the project is financially viable.
- Also, Internal Rate of Return (IRR) at 22.44% is higher than Weighted Average Post-Tax Cost of Capital (WACC) at 10.15%, both parameters indicating that the Project is financially viable preposition.
- Meanwhile the average Debt Service Coverage Ratio (DSCR) of the Project has been estimated at 2.77, indicating fair term loan repayment capacity of the Project and the Company.

Sensitivity Analysis

The Consultants undertook a sensitivity analysis of the Project and the outcome of the same has been presented in table below –

Scenario	NPV	IRR	WACC	Min DSCR	Avg. DSCR
	Rs. Crores	%	%	Ratio	Ratio
100% Broken Rice	137.74	22.44%	10.15%	1.36	2.77
5% decrease in capacity utilisation	47.85	15.17%	10.33%	1.00	2.03
5% decrease in selling prices	46.54	15.05%	10.34%	1.00	2.02
5% increase in raw material & stores cost	72.73	17.32%	10.28%	1.13	2.24
1% increase in interest rates	124.93	22.36%	10.77%	1.33	2.68
2% increase in interest rates	113.09	22.31%	11.39%	1.31	2.60
10% increase in hardware cost	124.52	20.80%	10.18%	1.32	2.67
100% Broken Rice	137.74	22.44%	10.15%	1.36	2.77

Source: NITCON Estimates

Based on the review of the above table, Consultants understand that the project remains viable under all above scenarios.

Conclusions

Market Assessment

The Consultants have reviewed the data available from various sources, including the data available from various Government Departments, Ministry of Agriculture and Farmer Welfare etc. to understand the availability of feedstock in terms of surplus and broken rice and maize for the proposed 120 KLD grain-based distillery project of Crescendo Industries Private Limited. Based on the research, the consultants have concluded the following –

- India’s fuel demand continues its upward trajectory, further straining the national exchequer. In the 2024-25 fiscal year, India’s crude oil import bill remained a critical economic variable, hovering near USD 135-140 billion, as the country continues to rely on imports for approximately 87% of its domestic consumption
- the 2021 NITI Aayog roadmap initially projected a USD 4 billion annual saving, current estimates suggest that achieving the E20 target nationwide by 2025-26 could save India upwards of USD 5.5 billion to 6 billion in foreign exchange annually, depending on global oil price benchmarks.
- Fuel Grade Ethanol (purity > 99.5%) is used for blending with Petrol as transportation fuel. Ethanol is a better blend component for Petrol due to higher Research Octane Number of 108.5, which helps in improved engine power and performance. It also helps in significantly reducing other pollutants (Carbon Monoxide, Sulphur Oxides, Nitrogen Oxides, Hydrocarbon and Particulate Matter) in vehicle exhaust when compared with fossil fuels.
- In 2024, India is estimated to maintain an average ethanol blending rate of approximately 11.5 percent. Previously, India achieved its Ethanol Blending with Petrol Program (EBP) target of E-12 in April 2023 for the Ethanol Supply Year (ESY) (December-November) 2022/2023. As such, post revised its 2023 ethanol blending rate with petroleum estimate upward to 12 percent due to increased use of sugarcane syrup, B-heavy molasses, damaged food grains, and surplus rice available from the Food Corporation of India (FCI). Post forecasts ethanol supplies for the EBP to increase by the end of the current ESY 2023/2024 but expects the average blend rate will still be below the E-20 national target for 2025.

- To meet the total ethanol demand by 2025-26, both the sugar and grain sectors would need to supply close to 7 billion liters of ethanol each.
- As per the meeting of Committee of Secretaries on 13th November 2020, DFPD informed that to fulfil 20% ethanol requirement by 2025 will be met from sugar as well as grain-based Ethanol. Further, based on reports published in various sources, it is understood that Government of India is pondering increasing the blending limit to 30% by 2030, in a stepped-up manner.
- India has vast resource of livestock and poultry, which play a vital role in improving the socio-economic conditions of rural masses. There are about 303.76 million bovines (cattle, buffalo, *Mithun* and yak), 74.26 million sheep, 148.88 million goats, 9.06 million pigs and about 851.81 million poultry as per 20th Livestock Census in the country.
- The global animal feed market reached USD 518.46 billion in 2023 and is expected to grow to USD 650.76 billion by 2030, exhibiting a CAGR of 3.3% during this period.
- India has established strong markets for Cattle and Poultry feeds. Both are growing at 15% and 8-9% respectively have grounds for huge potential in future.
- As a 120 KLPD distillery requires approximately 265 Tons and 290 Tons of rice/broken rice and Maize per day respectively, depending on the starch content and yield efficiency (considering 456 liter/ton of ethanol form rice and 415 litre/ton of ethanol from maize/corn). Annually (assuming 330 operational days), the plant requires roughly 87,500 to 95,700 Tons of rice and maize respectively. And even using only 1% of the state's Maize or Rice production would comfortably feed the facility.

Technical Assessment

- The Company is setting up grain-based distillery of 120 KLD along with a co-generation plant at Rakba Nos. 38/1/3(S), 38/1/1(S), 38/2(S), and 38/3(S), Village Mahewa, Teonthar Tehsil, Rewa District, Madhya Pradesh.
- The Company has already acquired 76,200 Sq.M. or 7.62 Hectares of land for the project, which is under its possession.
- The company has appointed VAPCO Engineers Private Limited as EPC contractor for the entire project, further the Company has procured steam turbine for co-generation power plant from Triveni Turbine Limited.
- The Company is installing a 3 MW captive co-generation plant designed to utilize rice husk as the primary fuel source.
- The Fresh water requirement in the proposed project will be around 1300KL per day.
- The Company proposes setting up a Water Treatment Plant at the site to handle the water discharged from the process.
- The proposed ethanol manufacturing plant will require around 150 personnel for the successful operation of the plant.
- The project site is enclosed on all sides by a brick masonry boundary wall supported on RCC columns and topped with barbed wire fencing.
- On an overall basis, the Company has achieved approximately 75% of physical progress with the total project.
- Two borewells equipped with pumps of 5 HP capacity each have been installed at site and are being utilized for groundwater extraction to meet construction and sanitation water requirement.

Financial Assessment

- The average EBDITA of the Project is 15.46%, while the average Net Profit margin is 6.64% during the tenure of the loans.
- The Net Present Value (NPV) of the Project is Rs. 137.74 Crores, which is above zero and hence indicates the project is financially viable.
- Also, Internal Rate of Return (IRR) at 22.44% is higher than Weighted Average Post-Tax Cost of Capital (WACC) at 10.15%, both parameters indicating that the Project is financially viable preposition.
- Meanwhile the average Debt Service Coverage Ratio (DSCR) of the Project has been estimated at 2.77, indicating fair term loan repayment capacity of the Project and the Company.

Based on the market assessment, technical assessment and financial assessment it is understood by the consultants that the projects remain viable under adverse case scenarios and found to be Techno- Economically and Financially Viable Project.

Company Background

M/s Crescendo Industries Private Limited is incorporated as a private limited company on 28th June 2022 and registered with the Registrar of Companies, Kanpur. The Company's registered office is located at Plot No. 7 and 8, Khata No. 49A, B Chak Malhori Road, Malhour Chinhat, Lucknow, Uttar Pradesh – 226028. As on 14th February 2026, the Company has an authorized share capital of Rs. 28,90,00,000 and a paid-up share capital of Rs. 28,89,00,000.

Subsequent to incorporation, the Company proposed to set up a grain-based ethanol manufacturing facility with an installed capacity of 120 KLD, along with a 3.00 MW captive cogeneration power plant. The project is being established at Rakba No. 38/1/3(S), 38/1/1(S), 38/2(S), and 38/3(S), Village Mahewa, Teonthar Tehsil, Rewa District, Madhya Pradesh.

The proposed plant is designed to manufacture Fuel-Grade Ethanol, Distillers Dried Grains with Solubles (DDGS), and carbon dioxide (CO_2). Ethanol produced at the facility is intended to be supplied to Oil Marketing Companies (OMCs) such as BPCL, HPCL, and other eligible OMCs under the Ethanol Blending Programme, in accordance with the National Policy on Bioenergy, 2018 issued by the Government of India. DDGS is proposed to be sold to animal feed manufacturers and in the open market as a protein-rich feed supplement, while the CO_2 is proposed to be supplied to aerated beverage manufacturers and/or fertilizer units.

The plant will primarily utilize broken rice, maize, and other damaged or surplus grains and food materials available in the region as feedstock for ethanol production. The ethanol produced will be used mainly for blending with petrol as a renewable biofuel, thereby supporting the Government of India's objectives of energy security and reduction in dependence on fossil fuels.

To meet the internal power requirement of the facility, a 3.00 MW captive cogeneration power plant is being installed in line with the stipulations of Environmental Clearance (EC). Power generation will be based on high-pressure steam generated from a 25 TPH boiler operating at 67 ATA, which will supply steam both to the ethanol manufacturing process and to a steam turbine generator (STG) for power generation.

For implementation of the project, the Company has acquired land admeasuring 76,200 Sq. M. (7.62 Hectares) at Village Mahewa, Teonthar Tehsil, Rewa District, Madhya Pradesh. The required land-use conversion certificate has been obtained under the provisions of the Madhya Pradesh Bhu-Rajsva Sanhita.

The original project cost, as appraised, was estimated at **Rs. 144.46 Crores**, the project was funded in a debt–equity ratio of 4:1. and the Company received a term loan sanction of Rs. 115.57 Crores from Indian Renewable Energy Development Agency Limited (IREDA), with the balance amount of **Rs. 28.89 Crores** infused by the promoters as equity.

The implementation of the project commenced in September 2024, with the original Commercial Operation Date (COD) targeted for 31st March 2026. However, due to unforeseen circumstances beyond the control of the

Company, the project execution was delayed by approximately 3 months, and the revised COD has been considered as on 30th June 2026.

As a result of the extended implementation period, the overall project cost has increased by Rs. 27.39 Crores with revised overall projects cost of **Rs. 171.85 Crores**. The cost escalation is primarily attributable to higher Interest During Construction (IDC), increased manpower and wage costs, and certain additional works required during execution. It is noted that the cost overrun is time-related in nature and not on account of any significant change in project scope or technology.

To bridge the funding gap arising from the aforesaid cost escalation and to ensure timely completion and stabilization of operations, the Company has approached the lenders for sanction of an additional term loan – 2 of **Rs. 21.91 Crores**, with the balance **Rs. 5.48 Crores** proposed to be met through further promoter contribution.

In this context, the Company has appointed NITCON Limited to undertake a Techno-Economic Study of the project. NITCON Limited has accepted the assignment, and the observations, evaluation, and findings of the Consultants have been presented in the form of this report.

Board of Directors

Mr. Alkendra Pratap Singh

Mr. Alkendra Pratap Singh is the founding member of the Company and holds a Bachelor of Technology degree from the University of Pune. He has over 12 years of experience in the construction and infrastructure sector and has been a promoter and director of a multi-dimensional construction group. He has successfully executed several projects in the thermal power sector and has worked with reputed organizations such as NTPC, BHEL, GE, Doosan Power, and Mitsubishi. In the Company, he is responsible for overall project management, technical supervision of the project, and liaison with various departments and authorities concerned.

Mr. Ramendra Pratap Singh

Mr. Ramendra Pratap Singh holds a Bachelor of Commerce degree from the University of Pune and an MBA qualification. His core expertise lies in financial management, accounting, and taxation. He has successfully managed and overseen multiple business ventures across sectors such as mining, transportation, and educational institutions. In the Company, he is responsible for overseeing the overall financial management, including budgeting, accounting, taxation, and financial controls.

Mr. Shakti Singhi

Mr. Shakti Singh is an MBA graduate with over 17 years of experience in the real estate sector. He has been involved in the development and execution of various residential and infrastructure projects, primarily in and around Lucknow. His extensive experience in project execution, combined with his managerial skills and entrepreneurial acumen, makes him a key contributor to the Company’s operations and strategic initiatives.

Mr. Vaibhav Aggarwal

Mr. Vaibhav Aggarwal is a graduate of the S.P. Jain School of Global Management, Sydney, and holds a Bachelor of Business Administration along with a master’s degree in Entrepreneurship. He has been associated with the construction industry since 2018 and has developed productive workspaces for several reputed corporate clients, including Maruti, Coca-Cola, Schneider, Incuspaze, and America Tower Company. He is also the co-founder of a prominent co-working space in Lucknow, Summit Space, and brings modern project execution and entrepreneurial experience to the Company.

Key technical personnel

Mr. Sandeep K. Maurya

Mr. Maurya is a chemical engineer with over 12 years of experience in the design, commissioning, and execution of ethanol and sugar plants. He has worked with reputed organizations such as Praj Industries Limited, KBK Chem Engineering Pvt. Ltd., and Radico Khaitan Limited, covering process design, commissioning, business development, and plant operations. His experience spans end-to-end project execution, including process engineering, vendor coordination, and operational optimization of ethanol plants.

Mr. Sudhir Tiwari

Mr. Tiwari is a chemical engineer with over 17 years of experience in commissioning and operation of ethanol and sugar plants. He has extensive exposure to distillation, fermentation (grain and molasses), evaporation, ETP, captive utilities, and plant erection and instrumentation. He has successfully commissioned multiple ethanol plants across India and brings strong commissioning and stabilization capability to the project.

Mr. Niraj Kumar Trigun

Mr. Trigun is a boiler and turbine engineer with over 10 years of experience in power plant operations. He has hands-on experience with WHRB, AFBC, gas-fired, and travelling grate boilers, as well as DCS platforms such as Honeywell, ABB, and Emerson. He brings strong expertise in safe, efficient operation and maintenance of captive power plants

Shareholding Pattern

The shareholding pattern of the Company as on 14th February 2026 has been provided in the exhibit below –

S. No.	Name of Shareholders	No. of Shares	% of Share Holding
1.	Mr. Alkendra Pratap Singh	72,25,000	25.00%
2.	Mr. Ramendra Pratap Singh	72,25,000	25.00%
3.	Mr. Shakti Singh	72,25,000	25.00%
4.	Mr. Vaibhav Aggarwal	72,25,000	25.00%
TOTAL		**2,89,00,000**	**100.00%**

Source: CIPL

About NITCON

NITCON Ltd is a Technical Consultancy Organisation (TCO), set up in the year 1984, with an aim to extend Consultancy services for growth and development of Industrial Units. It is an ISO 9001:2015 certified company.

The concept of TCOs is based on a World Bank recommendation to Government of India way back in 1970s as well as the concurrence of Reserve Bank of India to create appropriate structures and systems to extend dependable and affordable professional consulting services to Indian SMEs for giving pace to industrialization and with the objective of facilitating overall industrial development of the country. With this background, Government of India had initiated and proposed setting up of a TCO in each State. It is a Board managed company having nominees of various Banks and All India/ State Financial Institutions on its Board of Directors. NITCON has all along been a self-sustaining, dividend paying organisation.

Keeping in view the majority shareholding by government institutions, NITCON accounts are audited by Auditors appointed by Comptroller and Auditor General (CAG) of India, New Delhi, Over the past 3½ decades, NITCON has widened the range of services being offered to Institutions/ Industries/ Private promoters.

Scope of Work

Market Assessment of feedstock, final product (Ethanol) and by-product (DDGS)

- Market Size
- Demand Drivers
- Demand-Supply Assessment
- Current Trends/Issues/Challenges
- Supply Chain Analysis

Technical Feasibility

- Technical Assessment
    - Site visit & data collection
    - Technology options, together with a list of potential technology licensors and their broad credentials
    - Project configuration along with process description
    - Overall block flow diagram as well as process schematic diagram for all process units
    - Plant and Machinery requirement (P&M)
    - Mode of transportation of raw material & products / by-products. Detail out current facilities available at site and additional / augmentation of infrastructure required for transportation of raw material & products / by-products.
    - Capacity details of feedstock, products, by-products, different units of the plant, etc.
    - Quality processes and standards
- Project implementation mechanism including implementation schedule along with bar chart, general guidelines for evaluation of technology & technology EPC partner, etc.
- Capacity, Material balance, product mix

- Description of raw materials, utilities, chemicals, Catalysts, off-sites, auxiliary facilities and effluent data
- Manpower requirement
- Plant Machinery suppliers' details.

Policy and Regulatory Landscape

- List of approvals required from State and Central Government and other Authorities
- Relevant Central/ State Government Policies, including incentives, etc. and likely impact, if there is change in these policies and risk factors.

Project Cost Details

- Estimated Project capital cost with a break-up for –
    - Land and land development cost
    - Civil works, Piping & structural works
    - Plant & machinery
    - Utilities & off sites
    - Electrical system; Instrumentation & control system
    - Basic design and engineering fees, License / know-how fees, project management consultancy fee
    - Misc. Fixed assets
    - Pre- operative expenses including start-up, commissioning expenses, etc.
    - Working Capital margin,
    - Interest During Construction Period
    - Contingency cost
- Basis of cost estimates for each component and break-up for Indian & foreign exchange component. Accuracy of cost estimation should be ± 20%

Financial Assessment

- Financial analysis and recommendations
    - Revenue projections
    - Cost Assumptions- Operating cost covering feedstock, other raw materials, utilities, catalyst & chemicals, repair & maintenance, insurance, manpower, administrative & overhead expenses, product marketing expenses, cost of transportation of raw material & products / by-products, etc.
    - Projected Financial statements
    - Feasibility assessment based on IRR, DSCR, ratio analysis
    - Sensitivity Analysis with respect to capital cost, product price, feedstock price and other relevant parameters

SWOT and Risk analysis

Approach

The team from NITCON worked closely with team from CIPL for early and successful completion of the Project. The data/ information utilized by NITCON for undertaking the project was sourced from –

- In house information, available with NITCON Limited
- Information available with various consulting organizations
- Published information available with various Government Departments
- Sector/ Industry/ Country reports published by reputed consultancy organizations
- Journals/ Publication of trade association
- Discussions with industry/ sector experts

Disclaimers

This report is to be read in totality and not in parts, in conjunction with relevant documents referred to in this document. While due care has been taken during preparation of this document which, however, is subject to the certain considerations as under –

- In preparation for this Report, we might have relied upon the information/data/ details provided by the Client, have not independently verified any of such information. However, based upon the review of such information we have, wherever necessary, sought explanations for the key trends and salient features in respect thereof. The team from our office has tried its level best to check the details/records & whenever not possible, estimates were considered, while placing in report.
- Given the importance to our work of the information and representations supplied to us by the management of the Client, we shall not be responsible for any losses, damages, costs, or other consequences, if information material to our work is withheld or concealed from or misrepresented to us.
- Due care has been taken by us to validate the authenticity and correctness of sources used to obtain the information; however, neither we nor any of our respective partners, directors, officers, employees, consultants, or agents, provide any representations or warranties, expressed or implied, for any direct or consequential loss arising from any use of the information, statements or forecasts contained in the report.
- The information and images (if any) provided or analysed in the report have been collated from various industry sources, including web resources, public-domain information sources, and our internal databases. We have ensured reasonable care to validate the data presented in the report; however, we have not conducted an audit, due diligence, or independent verification of such information. It is also to be noted that the images presented (if any) are pictorial representations of the overall concept and are in no way intended to represent any concrete imagery for the proposed development.
- The analysis & conclusions drawn are based on factors related to prevailing business & industry conditions and we presume that company has taken reasonable care while providing details to our team. The information contained in the report is based on judgmental estimates and assumptions, about circumstances and events. Accordingly, we cannot provide any assurance that the projected results would be attained in this ever-changing dynamic market environment.
- The projections placed in report are considering specific time frame. Delay in implementation shall impact the assessment made by our team & reported in this document.

- This report has been prepared for your internal use, on your specific instructions, solely for the purpose of the project and must not be used or relied upon for any other purpose. This report is strictly confidential, and no part thereof may be reproduced or used by any other party other than you, except as otherwise agreed between you and us. If you are permitted to disclose a report (or a portion thereof), you shall not alter, edit or modify it from the form we provided.
- Some assumptions invariably will not materialize, and unanticipated events and circumstances may occur. Therefore, the actual performance in any areas forecasted/ projected will vary from the forecast/ projection and the variations may be material. We cannot express any form of assurance on the likelihood of achieving the forecast/ projection or on the reasonableness of the used assumptions. Any such forecast/ projection is presented as part of the appraisal and is not intended to be used separately.
- This report has not considered issues relevant to any third parties. Use of this report by any third party for whatever purpose should not, and does not, absolve such third party from using its own due diligence in verifying the Report's contents. If any third party chooses to rely upon any of the contents of this report they do so entirely at their own risk, and we shall have no responsibility whatsoever in relation to any such use. We accept no duty of care or liability of any kind whatsoever to any such third party, and no responsibility for damages, if any, suffered by any third party because of decisions made, or not made, or actions taken, or not taken, based on this document, unless expressly agreed between you, us and such third party in writing.
- This report supersedes any previous oral presentations or summaries we may have made in connection herewith. Neither we nor any of our affiliates worldwide are responsible for revising or updating this report because of events or transactions occurring after the date of this report. Any updates or second opinions on this Report cannot be sought by the management from external agencies (including our affiliates) without our prior written consent.
- The report is prepared in good faith as we have no present or planned future interest in the company. Further we have no direct/indirect interest in project.

Industry Assessment

*/ Note -The industry assessment chapter is bifurcated into two segments comprising of –

- Ethanol Industry in India
- Animal Feed Industry Analysis for DDGS

Section 1 - Ethanol Industry in India

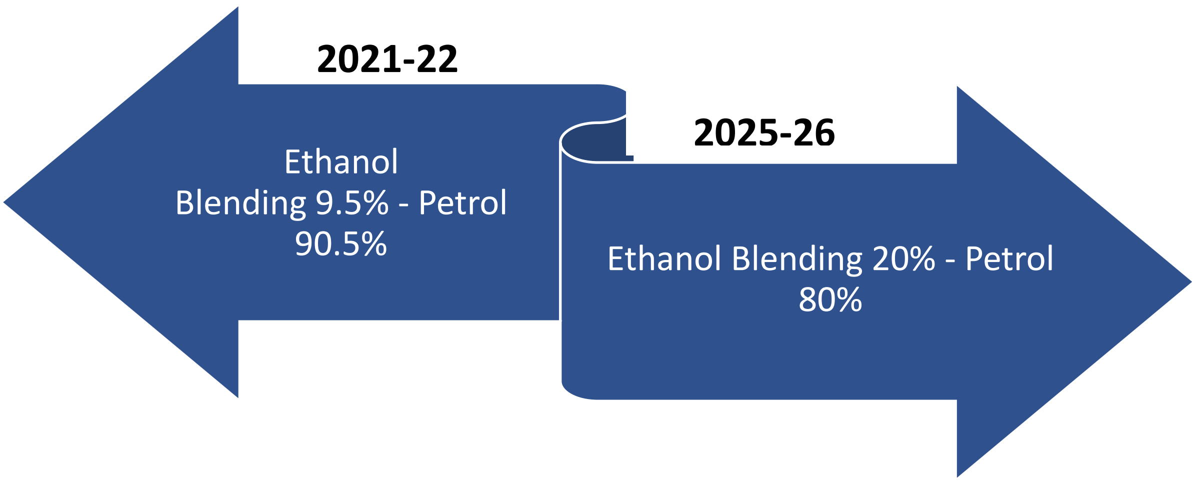
Introduction

India's fuel demand continues its upward trajectory, further straining the national exchequer. In the 2024-25 fiscal year, India's crude oil import bill remained a critical economic variable, hovering near USD 135-140 billion, as the country continues to rely on imports for approximately 87% of its domestic consumption. Amidst persistent geopolitical volatility and price fluctuations in the Brent crude market, the Government of India (GOI) has accelerated its "Aatmanirbhar Bharat" energy strategy.

A cornerstone of this strategy is the Ethanol Blended Petrol (EBP) Programme. While the 2021 NITI Aayog roadmap initially projected a USD 4 billion annual saving, current estimates suggest that achieving the E20 target nationwide by 2025-26 could save India upwards of USD 5.5 billion to 6 billion in foreign exchange annually, depending on global oil price benchmarks.

Beyond the fiscal benefits, the transition is a climate imperative. Following the record-breaking heatwaves of recent years that pushed several regions toward the "wet-bulb temperature" limit of human survivability, the shift to cleaner-burning fuels has moved from a policy goal to a public health necessity. Ethanol-blended petrol significantly reduces tailpipe emissions of carbon monoxide (CO), hydrocarbons (HC), and nitrogen oxides (NOx) compared to pure fossil fuels.

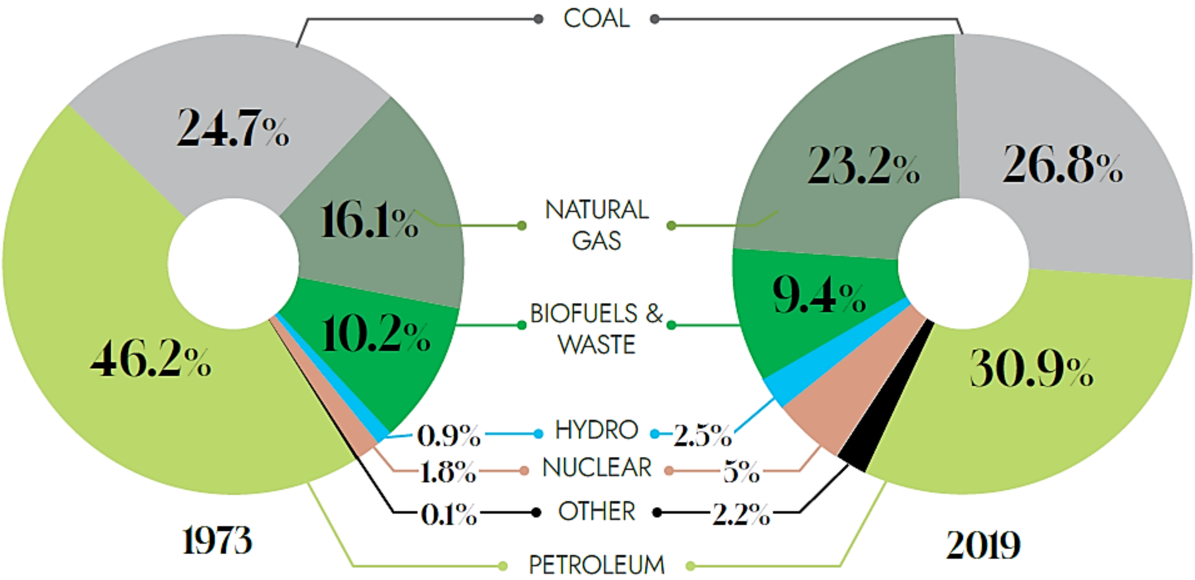
India's Petrol Consumption Outlook



Since 2018, India has moved rapidly and impressively with its ethanol production and fuel blending plan. From less than 2 billion litres in 2019-20, the country supplied about 4.1 billion litres of ethanol for fuel blending in 2021-22 (Ethanol supply year, ESY1). This pulled up the average rate of blending in the country, from 5 percent in 2019-20 to about 9.5 percent in 2021-22. GOI has now advanced its target to achieve 20 percent ethanol blending (E20) from 2029-30 to 2025-26.

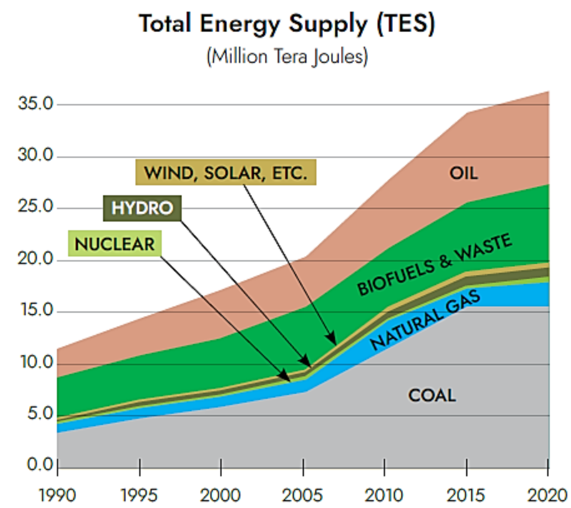
Global vs India's Energy Mix

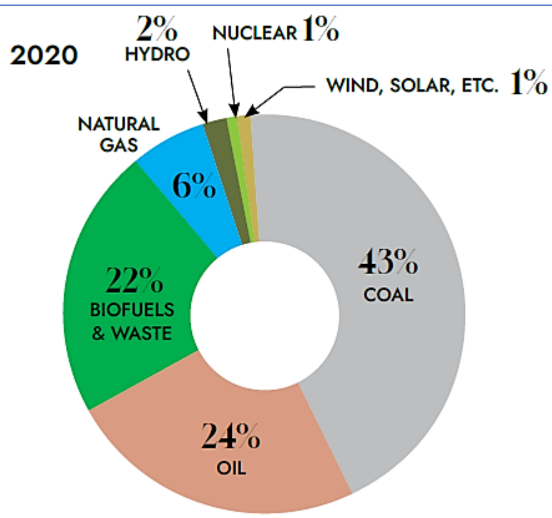
According to the International Energy Agency (IEA), the global energy mix has been changing with time. In the 46-year period between 1973 and 2019, dependence on crude oil decreased (from 46.2 percent of the Total Energy Supply (TES) in 1973 to 30.9 percent in 2019). In the same period, reliance on coal increased (from 24.7 percent to 26.8 percent). Interestingly, global energy dependence on biofuels and waste decreased from 10.2 percent to 9.4 percent in this period. Natural gas, hydroelectric power, and nuclear sources have become more important. However, despite an increase in the share of energy from renewable sources, about 86 percent of the world's energy supply continued to be met from non-renewable sources in 2019.



Source: IEA

Compared to the world average, India's energy basket is greener. The country has been diversifying from non-renewable sources of energy to renewable sources rather fast.





Source: IEA, Key World Energy Statistics

In 2020, energy from biofuels (22 percent) had become the third largest source of energy, trailing closely the contribution from oil (24 percent). However, India's dependence on coal, is still very high at 43 percent. Interestingly, natural gas for the world (23.2 percent) and biofuels for India (22 percent) are of similar importance in the respective energy supply baskets.

Biofuels are defined as plant matter used either directly or in converted form (for example charcoal, electricity, or fire). Biofuels used in the transportation sector include ethanol, biodiesel, renewable diesel, and bio-jet. Biofuel production for transportation was placed at approximately 157.4 billion litres in year 2020 (IEA Renewable 2021). The share of ethanol was about 69 percent, followed by biodiesel (27 percent) and renewable diesel (4 percent).

Indian policymakers have been increasingly focusing on bioethanol. Ethanol can be produced from grains, sugarcane, and agricultural/industrial waste using either First Generation (1G) or Second Generation (2G) technologies. 1G technology produces ethanol directly from food stocks (crops); whereas advanced biofuel technologies such as 2G can produce ethanol from non-food crops, industrial wastes, and lignocellulose feedstocks.

The NITI Aayog recommends that 2G technology be the preferred option for the production of ethanol (NITI 2021). According to a recent report from World Food Programme, India is still home to one-fourth of the total undernourished people in the world (WFP 2022). In such a scenario, manufacturing biofuels from 2G technologies (where by-products of crops can be used as feedstock) should be the preferred option compared to 1G technologies (where the crop is the feedstock).

Proposed Products Description

It is noted that Company is setting up 120 KLD, grain-based distillery project, with zero liquid discharge. The products proposed to be manufactured through the unit are –

a) Ethanol (Fuel Grade)
a) DDGS (Dry Cake as Cattle Feed)
b) Liquid CO_2/ Dry Ice
c) Distillers Corn Oil

d) Fuel Oil

So effectively the primary products of the proposed project will be Ethanol in variety of forms and purities, which are classified as Bioethanol/ Rectified Spirit/ Denatured Spirit. Since the Company proposes to manufacture fuel grade Ethanol from grains as a feedstock, hence there will be by products, which will be produced along with the main product, and these include Dry Cake (when grains are used as input material) and Liquid Carbon Dioxide.

Dry Cake is used to manufacture cattle feed, while liquid Carbon Dioxide can be bottles and sold off as it is or can be converted into dry ice and sold off. However, it is proposed to outsource the Carbon Di-oxide/ Dry Ice section of the Project on BOOT basis to a third party and hence detailed market assessment of the same has not been covered as part of this report.

Main Product Description - Ethanol or Bioethanol

Bioethanol is produced from biomass which has sugar containing materials (like sugar cane juice, molasses, sugar beet etc.) and starch containing materials (such as corn, cassava, rotten vegetables such as potatoes, damaged food grain etc.). Ethanol is generally produced from the fermentation of C-5 and C-6 sugars (mostly xylose and glucose) using classical or GMO yeast strains such as Saccharomyces cerevisiae.

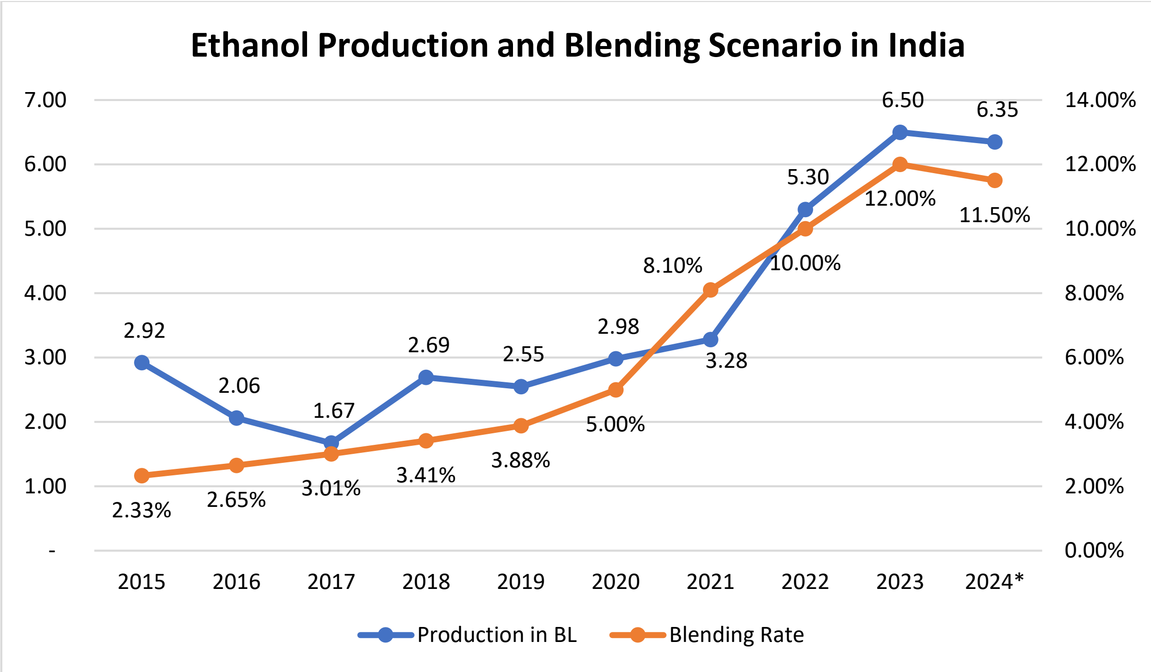
Fuel Grade Ethanol (purity > 99.5%) is used for blending with Petrol as transportation fuel. Ethanol is a better blend component for Petrol due to higher Research Octane Number of 108.5, which helps in improved engine power and performance. It also helps in significantly reducing other pollutants (Carbon Monoxide, Sulphur Oxides, Nitrogen Oxides, Hydrocarbon and Particulate Matter) in vehicle exhaust when compared with fossil fuels.

Lifecycle analysis of Ethanol produced using feedstocks derived from agriculture route, indicates significant reduction in Green House Gas emission as compared to conventional fossil fuels. Ethanol Blended Petrol (EBP) programme is initiated by Government of India to promote use of indigenously produced Bioethanol as transportation fuel along with Petrol (Motor Spirit or Gasoline) in order to reduce Carbon Dioxide CO2 emission as well as reduce import of fossil fuel.

Ethanol Industry Scenario in India

Production/Supply

In 2024, India is estimated to maintain an average ethanol blending rate of approximately 11.5 percent. Previously, India achieved its Ethanol Blending with Petrol Program (EBP) target of E-12 in April 2023 for the Ethanol Supply Year (ESY) (December-November) 2022/2023. As such, post revised its 2023 ethanol blending rate with petroleum estimate upward to 12 percent due to increased use of sugarcane syrup, B-heavy molasses, damaged food grains, and surplus rice available from the Food Corporation of India (FCI). Post forecasts ethanol supplies for the EBP to increase by the end of the current ESY 2023/2024 but expects the average blend rate will still be below the E-20 national target for 2025. This is owing to the ongoing shortage in major ethanol feedstocks due to adverse weather conditions. In 2024, imported ethanol will continue to backfill India’s unmet demand within the industrial, alcoholic beverage, and medicinal grade industries.

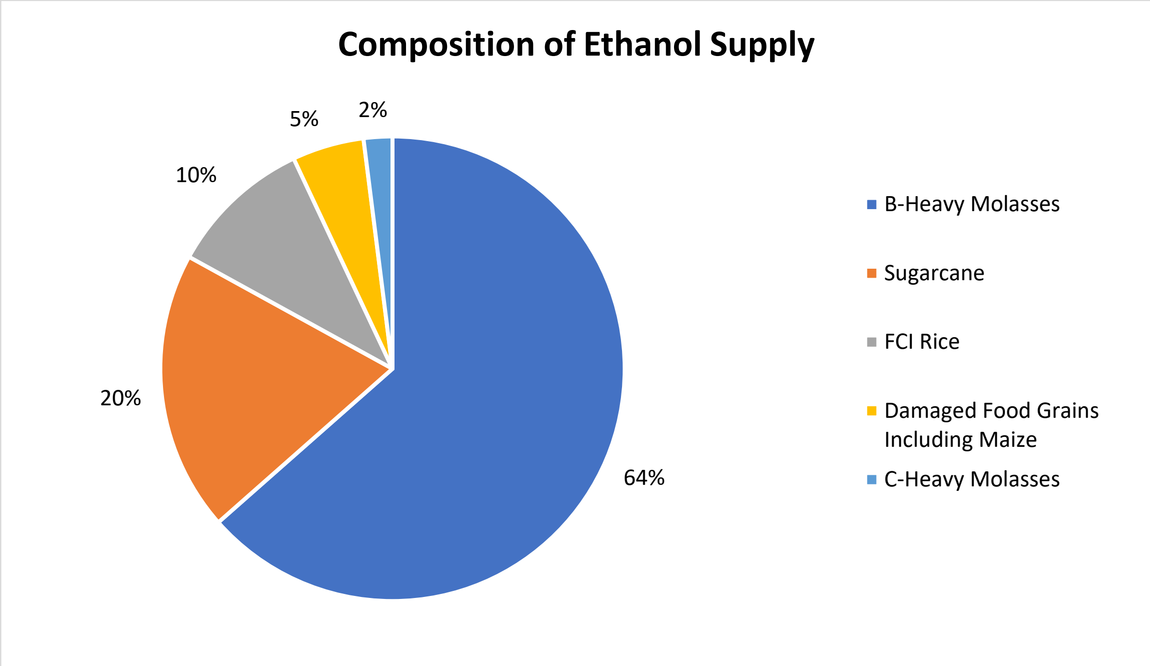


Source: GOI

FAS New Delhi expects India's ethanol production for 2024 to reach 6.35 billion litres (BL), a two percent drop from last year due to a projected decline in sugarcane production and a depleting rice grain supply. Total ethanol consumption is expected to rise to 7.1 BL out of which ethanol for fuel consumption is estimated at 6.2 BL. This estimate accounts for India's restrictions on sugar feedstocks and broken rice for fuel ethanol to avoid inflationary food prices, amidst a low sugar production year. India is expected to have a feedstock shortage and the ethanol blending rate for 2024 is expected to drop to 11.5 percent. However, the favourable EL-Nino monsoon conditions will boost the production of feedstock and could push the blending rate to 12.50%.

Out of total ethanol supplied for fuel blending in 2022-23, about 84 percent was produced from sugarcane juice and molasses (B-Heavy and C-Heavy)— about 11 percent using *surplus rice* from the FCI, and the remainder of about 6 percent from damaged grains, including rice from the open market as well as maize.

Ethanol from sugarcane can be produced via three routes: (i) through sugar syrup or cane juice, (ii) from B-heavy molasses, and (iii) from C-heavy molasses. Before 2018-19, ethanol was mainly produced from C-heavy molasses (NITI Aayog 2021). With the change in regulations in 2018-19 that allowed the use of B-heavy molasses, sugar syrup, and grains (such as rice and maize) for ethanol production, increasing amounts of ethanol are being produced from B-heavy molasses and the sugar juice route, with C-heavy molasses only accounting for 2 percent of ethanol supply in ESY 2021-22.



Source: GOI

Ethanol Pricing Mechanism

Based on the type of feedstock used, the purchase price of ethanol by OMCs is fixed by the GOI. For ethanol produced from sugarcane and its sources, the prices are fixed by the Cabinet Committee on Economic Affairs (CCEA), and, for grain-based ethanol, the prices are fixed by OMCs (NITI Aayog 2021), while there is no differential pricing for ethanol produced from different food grains (broken rice, wheat, maize etc.). Pricing for ethanol produced from molasses-based feedstocks is done using the Administrative Pricing Mechanism (APM), introduced in ESY 2014-15 (NITI Aayog 2021). Under this regime, GOI approves the price for molasses-based ethanol on the basis of market conditions in the ethanol and oil markets, availability of feedstocks in the domestic markets and the import substitution requirement (NPB 2018). Since 2018-19, the GOI has been deciding differential pricing for ethanol produced from C-heavy molasses, B-heavy molasses, and sugarcane juice/syrup. The differential pricing is based on a variety of factors (NITI Aayog). For sugarcane juice/syrup, prices are based on the Fair and Remunerative prices (FRP) for sugarcane. To determine the administered price of ethanol from cane juice, the cost of conversion to ethanol and capital costs are added to arrive at the ex-mill price.

For B-heavy molasses, ex-mill price of ethanol is calculated by considering the normative cost of producing sugar and capital costs of ethanol production. For C-heavy molasses, rates are calculated on the basis of the price of molasses and ex-mill price of sugar. For ESY 2022-23, ex-mill ethanol price from C-heavy molasses, B-Heavy molasses, and sugarcane juice has been fixed at Rs. 49.41, Rs. 60.73 and 65.61 per litre respectively.

Ethanol Price Trend by Feedstock						
Year	C-Heavy Molasses	B-Heavy Molasses	Sugarcane/ Syrup/Sugar	Damaged Food Grains	Maize	Surplus Rice (FCI)
2015-16	42.00	NA	NA	NA	NA	NA
2016-17	39.00	NA	NA	NA	NA	NA
2017-18	40.85	NA	NA	NA	NA	NA
2018-19	43.46	52.43	59.19	47.13	NA	NA

Ethanol Price Trend by Feedstock						
Year	C-Heavy Molasses	B-Heavy Molasses	Sugarcane/ Syrup/Sugar	Damaged Food Grains	Maize	Surplus Rice (FCI)
2019-20	43.75	54.27	59.48	50.36	NA	NA
2020-21	45.69	57.61	60.65	51.55	51.55	56.87
2021-22	46.66	59.08	63.45	52.92	53.45	56.87
2022-23	49.41	60.73	65.61	55.54	56.35	58.50
2023-24*	56.28	60.73	65.61	64.00	71.86	58.50

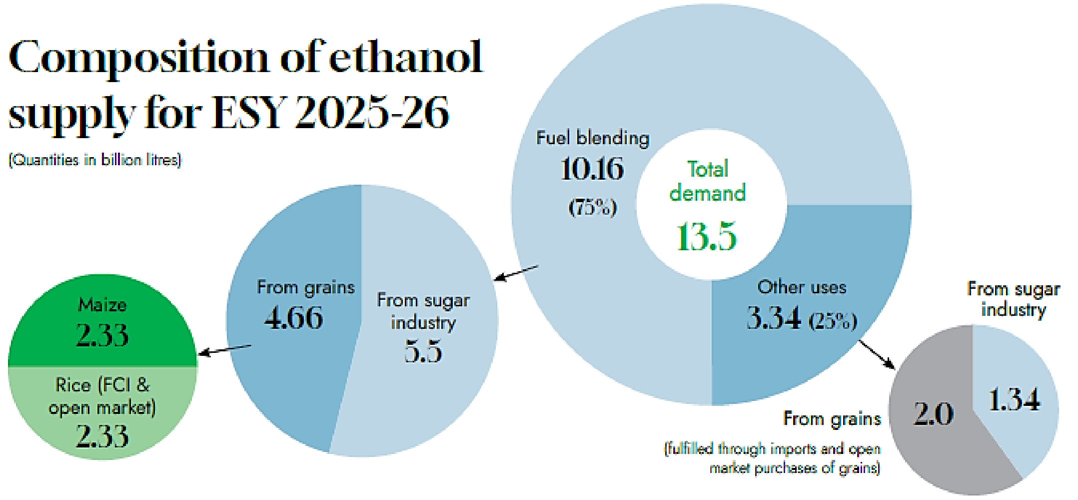
Source: BPCL, PIB and NITI

NITI AAYOG's Roadmap to E20 by 2025-26

In June 2021, NITI Aayog published the roadmap to achieving the E20 blending mandate by 2025-26, making it the guiding document for further policy action. The roadmap projects the demand of ethanol and identifies sources of its feedstock. The ethanol demand for fuel-blending is based on the country's projected petrol demand, which in turn is pegged to the expected vehicle population in the country by 2025-26. The estimates also account for likely impact of the uptake of electric vehicles (EVs) on fuel demand. As per NITI's roadmap, Indian petrol demand in 2025-26 is projected to be 50.8 billion litres. An E20 mandate translates to an ethanol demand of about 10.16 billion litres by 2025-26. In addition, the country's potable alcohol and pharmaceutical industry would continue to need about 3.34 billion litres of ethanol annually. In total, this implies that by 2025-26 the country would be needing about 13.5 billion litres of ethanol.

India's priority on ethanol and achieving E-20 by 2025 has limited sugar exports. In response to two consecutively low sugarcane production years, India initially imposed a restriction on sugar exports from June 1 to October 31, 2022. Subsequently, India amended this restriction to be effective for an indefinite period (See: India Restricts Sugar Exports Beyond October 31 for Indefinite Period). For the current sugar Marketing Year (MY) (October - September) 2023/2024, to mitigate inflationary pressure on domestic consumption, the government has limited the use of sugarcane and its derivatives for ethanol production to 2.37 MMT. The one exception is for C-heavy molasses, which has a sugar recovery rate that is comparatively smaller than the other two derivatives, sugarcane juice and B-heavy molasses.

Similarly, rice production is also expected to be 2 percent lower than the previous year due to water stress. As such, India banned the use of broken rice for ethanol production due to the depleted food grain stock. The availability of these two major fuel ethanol feedstocks is expected to decrease by approximately 20 percent for the current year, compared to the previous year. Incidentally, the government is taking initiatives to increase the maize/corn production in India for fuel ethanol by offering a Minimum Support Price (MSP) for procurement. The approach is expected to increase maize/corn procurement for fuel ethanol and increase production in 2024/2025. India's policies have attempted to augment domestic production, by continuing the prohibition of imported ethanol for fuel blending along with the aforementioned actions.



To meet the fuel-blending demand of 10.16 billion litres, close to 55 percent or 5.5 billion litres is expected to come from the sugar industry, and the remaining 45 percent or about 4.66 billion litres from grains and other damaged food grains sourced from either the FCI or the open market, according to projections made in the NITI Aayog document.

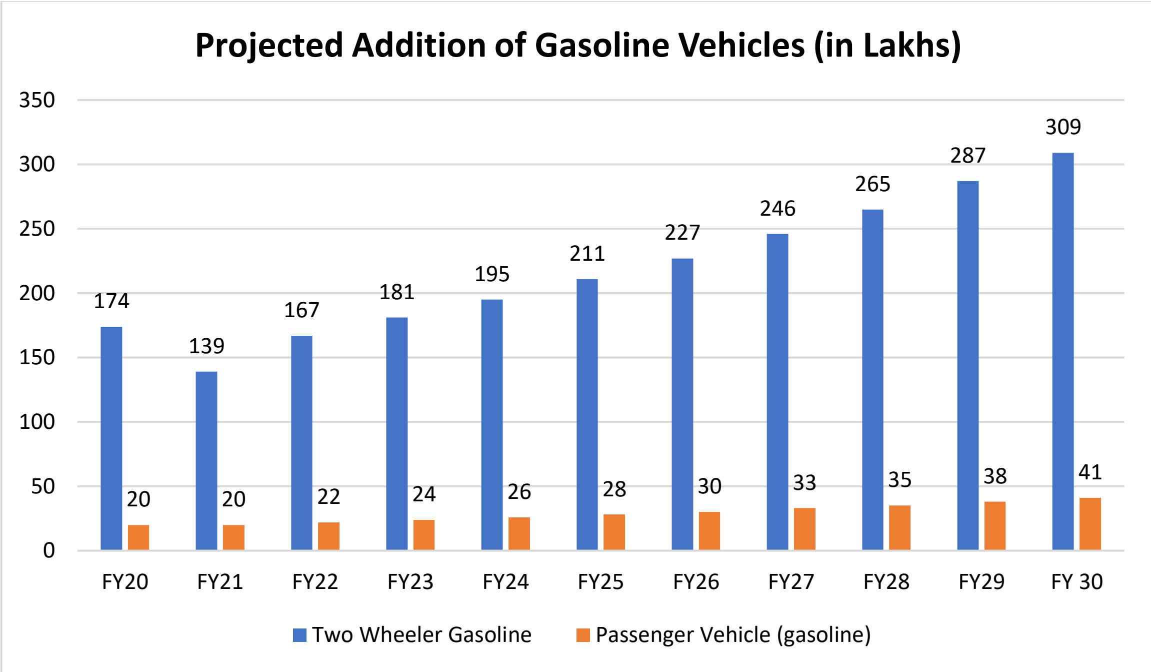
Unlike in the case of fuel blending, most of the ethanol required by the FMCG, pharmaceuticals and alcoholic beverage industry can be sourced from food grains, the NITI Aayog Roadmap states. Of the total demand of 3.34 billion litres, about 60 percent or 2 billion litres is expected to be sourced from grains, mainly rice—apart from a small component of maize, while the remaining 40 percent or 1.35 billion litres will be supplied by the sugar sector, according to the NITI Aayog estimates.

To meet the total ethanol demand by 2025-26, both the sugar and grain sectors would need to supply close to 7 billion litres of ethanol each.

Demand Scenario

Ethanol (also called ethyl alcohol, or alcohol) is an organic chemical compound with chemical formula C_2H_5OH. Besides the EBP Programme, ethanol finds competitive usage in the portable sector and the chemical & pharmaceutical industry. Demand for ethanol as a fuel is primarily driven by blending mandates, widespread availability of fuel, and compatible vehicles and fulfilment of other infrastructural requirements.

The vehicle population in the country is around 22 crore two and three wheelers and around 3.6 crore four-wheelers (SIAM). The 2 wheelers account for 74% and passenger cars around 12% of the total vehicle population on the road. The two-three wheelers consume 2/3rd of the gasoline by volume, while 4 wheelers consume balance 1/3rd by volume. The growth rate of vehicles in this segment is pegged at around 8-10% per annum. An estimate of year-wise addition of gasoline-based vehicles in the country is given in the exhibit below –



Source: Government of India

The above projections are based the following assumptions –

- V-Shape recovery in vehicle sales in FY22, followed by growth at CAGR of 8% in all segments.
- Share of petrol vehicles will be 83% of the total passenger vehicle sale.

Considering the above grow rate in the petrol driven vehicle population in India, which has been provided till 2030, the Consultants have further considered the same growth rate for total duration of period of 25 years from FY 2021-22 and assessed the expected demand for ethanol for EBP program, assuming that the blending target of 30% remains for the balance of the period. Based on these assumptions, the projected demand for Ethanol for EBP program is presented in the table below –

Ethanol Supply Year	Projected Petrol Sale (MMT)	Projected Petrol Sale (Cr. litres)	Blending (in %)	Requirement of ethanol for blending in Petrol (Cr. litres)
2023-24	33.00	4,656.00	12.50%	582
2024-25	35.00	4,939.00	15.00%	740.85
2025-26	36.00	5,080.00	20.00%	1,016.00
2026-27	37.00	5,221.00	23.00%	1,200.83
2027-28	39.00	5,503.00	26.00%	1,430.78
2028-29	40.00	5,644.00	28.00%	1,580.32
2029-30	41.00	5,785.00	30.00%	1,735.50
2030-31	42.46	5,991.84	30.00%	1,797.55
2031-32	43.97	6,204.95	30.00%	1,861.48
2031-33	45.53	6,425.63	30.00%	1,927.69
2033-34	47.15	6,654.17	30.00%	1,996.25
2034-35	48.83	6,890.83	30.00%	2,067.25
3035-36	50.57	7,135.91	30.00%	2,140.77

Ethanol Supply Year	Projected Petrol Sale (MMT)	Projected Petrol Sale (Cr. litres)	Blending (in %)	Requirement of ethanol for blending in Petrol (Cr. litres)
2036-37	52.36	7,389.70	30.00%	2,216.91
2037-38	54.23	7,652.53	30.00%	2,295.76
2038-39	56.15	7,924.70	30.00%	2,377.41
2039-40	58.15	8,206.55	30.00%	2,461.96
2040-41	60.22	8,498.42	30.00%	2,549.53
2041-42	62.36	8,800.67	30.00%	2,640.20
2042-43	64.58	9,113.68	30.00%	2,734.10
2043-44	66.88	9,437.82	30.00%	2,831.34
2044-45	69.25	9,773.48	30.00%	2,932.04
2045-46	71.72	10,121.08	30.00%	3,036.33

Source: Government of India and NITCON Estimates

In above projections it has to be noted that the petrol projections may undergo revision due to various factors like penetration of EVs, etc. and the figures are optimistic as the E20 fuel will be consumed by new vehicles from April 2023 only.

On the conservative side, the ethanol demand will be in the range of 722-921 crore litres in 2025 to meet E20 targets.

Estimation of Supply and Capacity Augmentation – Future Scenario

During the meeting of Committee of Secretaries on 13th November 2020, DFPD informed that to fulfil 20% ethanol requirement by 2025 will be met from sugar as well as grain-based Ethanol. Further, based on reports published in various sources, it is understood that Government of India is pondering increasing the blending limit to 30% by 2030, in a stepped-up manner.

Further it is noted that Ethanol has various other application apart from fuel blending requirements and these include requirement of Ethanol for industrial application, potable purpose and for medical applications as well. Taking all these applications into consideration, the expected production forecast of Ethanol for next 25 years duration, assuming that capacities will continue to be added during the study period, has been presented in the exhibit below –

Ethanol Production Projections (Cr. Litres)										
FY	For Blending			Blending in %	For Other Uses			Total		
	Grain	Sugar	Total		Grain	Sugar	Total	Grain	Sugar	Total
2023-24	208	490	698	12.50%	180	110	290	388	600	988
2024-25	438	550	988	15.00%	190	110	300	628	660	1,288
2025-26	466	550	1,016	20.00%	200	134	334	666	684	1,350
2026-27	568	633	1,201	23.00%	210	141	351	778	774	1,552
2027-28	722	708	1,430	26.00%	220	148	368	942	856	1,798
2028-29	787	793	1,580	28.00%	231	155	386	1,018	948	1,966
2029-30	847	889	1,736	30.00%	242	163	405	1,089	1,052	2,141
2030-31	889	933	1,823	30.00%	254	171	425	1,143	1,105	2,248
2031-32	934	980	1,914	30.00%	267	180	447	1,201	1,160	2,360

Ethanol Production Projections (Cr. Litres)										
FY	For Blending			Blending in %	For Other Uses			Total		
	Grain	Sugar	Total		Grain	Sugar	Total	Grain	Sugar	Total
2031-33	981	1,029	2,010	30.00%	280	189	469	1,261	1,218	2,478
2033-34	1,030	1,081	2,110	30.00%	294	198	492	1,324	1,279	2,602
2034-35	1,081	1,135	2,216	30.00%	309	208	517	1,390	1,343	2,733
3035-36	1,119	1,169	2,287	30.00%	318	214	532	1,437	1,383	2,820
2036-37	1,158	1,204	2,362	30.00%	328	221	548	1,486	1,424	2,910
2037-38	1,199	1,240	2,438	30.00%	337	227	565	1,536	1,467	3,003
2038-39	1,240	1,277	2,518	30.00%	348	234	582	1,588	1,511	3,099
2039-40	1,284	1,315	2,599	30.00%	358	241	599	1,642	1,556	3,198
2040-41	1,329	1,355	2,684	30.00%	369	248	617	1,698	1,603	3,301
2041-42	1,375	1,395	2,771	30.00%	380	256	636	1,755	1,651	3,406
2042-43	1,423	1,437	2,861	30.00%	391	264	655	1,815	1,701	3,516
2043-44	1,473	1,480	2,954	30.00%	403	271	674	1,876	1,752	3,628
2044-45	1,525	1,525	3,050	30.00%	415	280	695	1,940	1,804	3,744
2045-46	1,578	1,571	3,149	30.00%	428	288	716	2,006	1,859	3,864

Source: Government of India and NITCON Estimates

The Consultants have considered the following while estimating the forecasted production level of Ethanol –

1. The production of grain and sugar-based fuel grade ethanol will grow at CAGR of 33% during the per annum till FY 2029-30, in line with the projections as provided by Government Departments.
2. Subsequently the production of fuel grade ethanol from grain and sugar-based feedstock will rationalise and grow at CAGR of 5.00% till FY 2034-35 and subsequently will grow at CAGR of 3.50% till 204-46.
3. Meanwhile the Ethanol production for other application from grain and sugar-based feedstock will continue to grow at a CAGR of 5.30% and 5.00% till 2029-2030, in line with the projections as available from various Government sources.
4. Subsequently the production of non-fuel grade Ethanol for other application will grow at 5.00% till FY 2034-35 and then rationalise to 3.50% CAGR till FY 2045-46.

During the entire period, the production or supply of Ethanol will lead the demand for ethanol for both fuel grade and non-fuel grade application, barring the initial phase between 2023 to 2026, when most of the projects as announced will slowly become online and commence production in full scale.

The following table details the Grains and Molasses based Ethanol Production Capacity necessary to meet the Production Projections above –

Capacity Augmentation (Cr Litres Per Annum)			
Year	Capacity Requirement		
	Grain	Molasses	Total
2023-24	450	725	1,175
2024-25	700	730	1,430
2025-26	740	760	1,500
2026-27	865	859	1,724
2027-28	1,047	951	1,998
2028-29	1,131	1,054	2,185

Capacity Augmentation (Cr Litres Per Annum)			
Year	Capacity Requirement		
	Grain	Molasses	Total
2029-30	1,210	1,168	2,378
2030-31	1,258	1,215	2,473
2031-32	1,309	1,263	2,572
2031-33	1,361	1,314	2,675
2033-34	1,416	1,366	2,782
2034-35	1,472	1,421	2,893
3035-36	1,516	1,464	2,980
2036-37	1,562	1,508	3,069
2037-38	1,609	1,553	3,161
2038-39	1,657	1,599	3,256
2039-40	1,707	1,647	3,354
2040-41	1,762	1,701	3,463
2041-42	1,819	1,756	3,576
2042-43	1,878	1,813	3,692
2043-44	1,940	1,872	3,812
2044-45	2,003	1,933	3,936
2045-46	2,068	1,996	4,064

Source: Government of India and NITCON Estimates

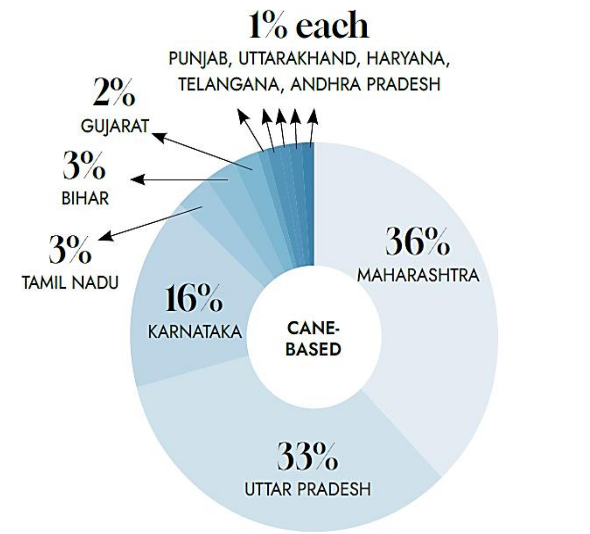
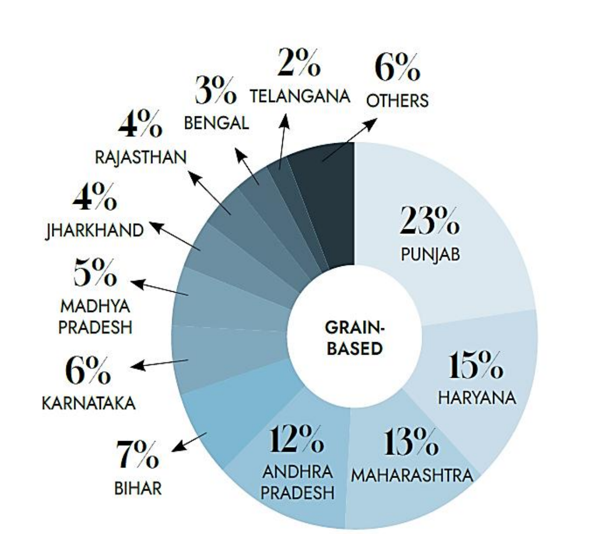
Minor shortfall in capacity in any year can be compensated as sugar mills are also using sugar rich feed stocks like B heavy molasses / sugar syrup which produces 20% more ethanol of rated capacity. Now many mills have started using these feed stocks in place of C heavy molasses. The capacity utilization in these 5 years vary from 84 percent to 90 percent, which is on account of the fact that the older distilleries based on sugar/ molasses as feedstock will experience lower capacity utilisation level as compared to newer capacities, which are expected to achieve capacity utilisations in range of 90% to 95% level, with 95% level being peak capacity utilisation level.

The above projections are based on below observations –

- Molasses based distilleries can produce 20% additional ethanol if sugar rich feed stocks like B- heavy molasses are used as the same capacity can cater the higher demand of ethanol.
- Total planned capacity is 4064 crore litres per annum, distribution between grain and molasses may change depending on various factors.

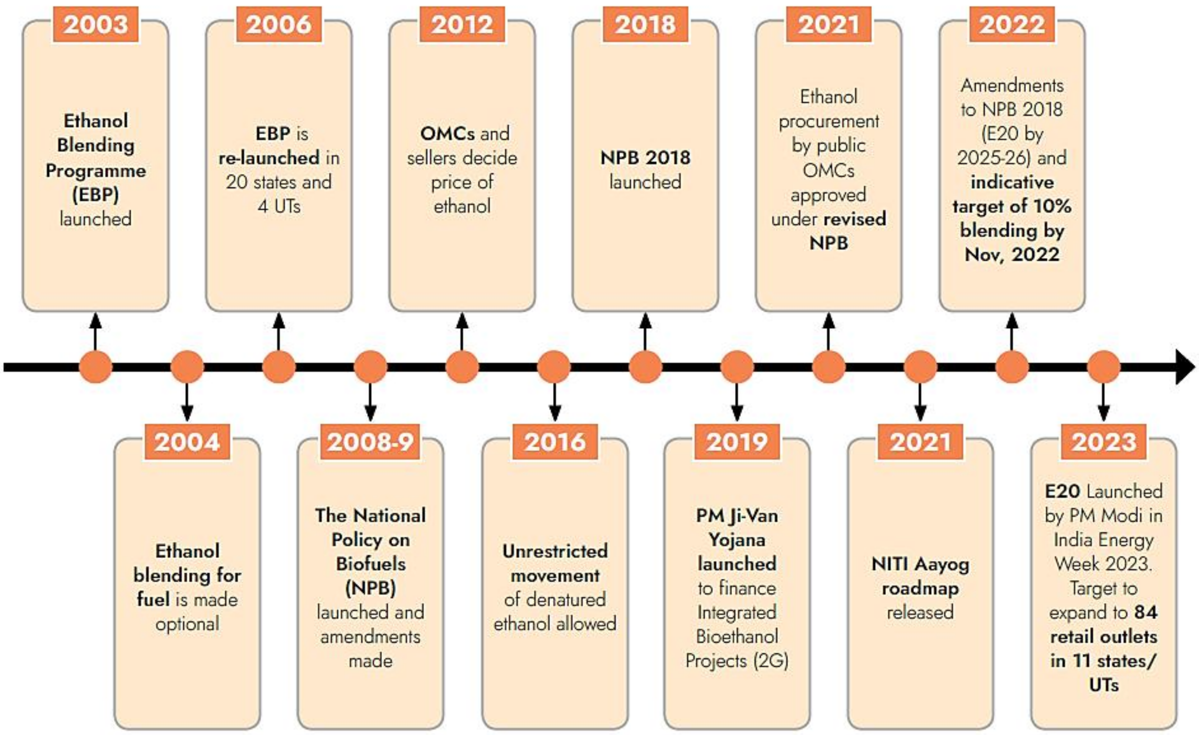
State-wise Cane and Grain Based Distillation Capacity

For molasses-based ethanol production, 12 states account for the total distillation capacity. The highest share is of Maharashtra (36 percent), followed by Uttar Pradesh (33 percent), Karnataka (16 percent), Tamil Nadu (3 percent), Bihar (3 percent), and Gujarat (2 percent). Andhra Pradesh, Haryana, Telangana, Uttarakhand, Punjab, and Madhya Pradesh account for about 1 percent each.

Molasses Based Ethanol Production	Grain Based Ethanol Production
	

Of the total grain-based distillation capacity, 94 percent of the capacity is in 11 states. Punjab has the highest share of 23 percent followed by Haryana (15 percent), Maharashtra (13 percent), Andhra Pradesh (12 percent), Bihar (7 percent), Karnataka (6 percent), Madhya Pradesh (5 percent), Jharkhand (4 percent), Rajasthan (4 percent), West Bengal (3 percent), and Telangana (2 percent).

Evolution of Indian Ethanol Policy



Source: NITCON Research

Broadly, there are two types of ethanol i.e. denatured ethanol and un-denatured. Denatured ethanol has denaturant added to it, rendering it unfit for human consumption. It is used as fuel, or as inputs for medical and industrial purposes to produce sanitizers, etc. while undenatured ethanol is mainly used as a concentrate for alcoholic beverages.

Between the financial years 2015-16 and 2021-22, India was a net importer of denatured ethanol. In FY 2021-22, the country imported 0.49 billion litres of denatured ethanol valued at USD 344 million. Imports increased from 2016-17 and peaked at 0.59 billion litres in 2020-21, official statistics show. India's denatured ethanol imports are mainly sourced from the USA and Brazil. The import of ethanol for fuel blending has been restricted since 2019. However, in her Union Budget speech for the year 2023-24, Finance Minister Nirmala Sitharaman proposed to exempt basic customs duties on imported denatured ethanol (PIB 2023). This measure is likely to improve the availability of denatured ethanol for industrial use by chemical and other industries, releasing greater amounts to meet the ethanol-blending mandates.

Government Policies and Initiatives

India's 2018 National Policy on Biofuels was a significant step towards promoting the production and use of biofuels in the country.4 The policy aimed to reduce India's dependency on fossil fuels, promote sustainable development, and address environmental concerns. It established the framework that is still relevant today, leading to an upward trajectory for the fuel ethanol industry and market after a long history of policies that failed to advance the sector. Under the current framework, the Ethanol Blending with Petrol Program (EBP) in India has been instrumental in boosting ethanol production from various feedstocks like sugarcane juice, B-heavy molasses, C-heavy molasses, broken rice, damaged grains, and maize. In contrast, India's biodiesel policies have failed to advance its goals.

In the 2018 National Biofuels Policy, the Indian government lays out its targets for ethanol blend rates in gasoline. The aim is to significantly increase the use of ethanol as a renewable fuel source. In 2023, India reached its highest level with a national 12 percent blending rate. Previously it reached E-10 in June 2022 for the 2022/2023 year. Looking ahead, India aims to reach 15 percent in 2023/2024, and is targeting 20 percent (E-20) by 2024/2025. To achieve these targets, the government is focusing on the following four areas –

- **Increasing Domestic Production -** The emphasis is on boosting domestic production of ethanol from diverse feedstocks. This includes first-generation (1G), second-generation (2G), and third-generation (3G) biofuels, which utilize different raw materials and conversion processes to produce ethanol. The Cabinet Committee of Economic Affairs (CCEA) approved the Indian government's initiative Pradhan Mantri JI-VAkN Yojana which aims at accelerating the development of 2G ethanol capacity in India, and provide viability gap funding to support the establishment of 2G ethanol projects. Excluding union territories, OMCs are responsible for nationwide blending ethanol into gasoline.

  Relatedly, India's amendment to the Sugarcane Control Order in May 2021, marked a significant shift in policy that helped to boost domestic production.5 It allows for the development of independent ethanol manufacturing plants. The amendment restricts procurement from informal sources to ensure quality control and standardization in the ethanol production processes.6 It also regulates the sugar and ethanol industries to ensure fair practices, consumer protection, and compliance with government policies. Additionally, India's financial assistance program to sugar mills for ethanol production increased by 12 percent in the Indian Fiscal Year (IFY) 2024/2025, compared to IFY 2023/2024.

Allocation	FY 2022-23 Realized Outlays	FY 2023-24 Initial Budget	FY 2024-25 Revised Budget	% Change Budget
Scheme for extending financial assistance to sugar mills to enhance and augment ethanol production capacity	Rs. 1.75 billion	Rs. 4 billion	Rs. 4.5 billion	12.5

- **Diverse Application of Feedstocks -** Under the 2018 National Biofuel Policy, EBP encourages using different feedstocks like molasses, broken rice, damaged food grains, and corn/maize for ethanol production. To meet E-20 by ESY 2024/2025, the Indian government actively persuaded sugar-mills and distilleries to divert excess sugar to ethanol. This worked when India had an ample sugarcane production during the past six years.

  In February 2024, the Cabinet Committee on Economic Affairs updated its Fair and Remunerative Price (FRP) for sugarcane for MY 2024/2025 from USD 3.79/quintal (INR 315/quintal) to USD 4.09/quintal (INR 340/quintal), based on a recovery rate of 10.25 percent. The revised FRP is to be implemented on October 1, 2024, Marketing Year (MY) 2024/2025, which is 8 percent higher than the current year 2023/2024.

  However, due to a low sugar production this year, the Indian government has put 2.37 MMT limit on the use of sugarcane and its derivatives for ethanol production and continued the sugar export ban to correct the domestic sugar market. Initially the restriction was set at 1.7MMT, but in April, due to an excess production of B-heavy molasses, the Indian government allowed an additional diversion of 0.67 MMT of B-heavy molasses for ethanol production. Furthermore, FCI stopped supplying broken rice for ethanol in response to higher market prices due to the lower rice production for the current MY. As a result, OMCs increased procurement prices of damaged food grains and maize to attract more use for ethanol production.
- **Regulated Supply Chains with Effective Tender Pricing -** One of the key objectives of this initiative is to prevent the diversion of ethanol produced by mills for localized purchasing. Through long-term agreements with OMCs, these entities can secure fixed rates for their ethanol, providing them with financial stability and predictability. Additionally, the program aims to address logistical challenges by providing secure transportation options for ethanol and ensuring timely payments to ethanol suppliers. This helps to streamline the supply chain and facilitate the smooth flow of ethanol from producers to OMCs. As per the policy and key to its success, procurement prices have increased over time, reflecting the government's commitment to more effectively incentivize ethanol production.
- **Expansion into Various Sectors –** The utilization of ethanol blends extends beyond just automotive use. The government intends to promote its use in machinery, stationary power applications, and portable power applications, thereby diversifying the market for ethanol and creating new opportunities for its production and consumption.

Ethanol Blended Petrol (EBP) Programme

1. Government has been implementing Ethanol Blended Petrol (EBP) Programme throughout the country except Union Territories of Andaman Nicobar and Lakshadweep islands, wherein OMCs sell petrol blended with 10% ethanol.
2. To increase indigenous production of ethanol the Government since 2014 took multiple interventions like -
    - Re-introduction of administered price mechanism.
    - Opening of alternate route for ethanol production.

- Amendment to Industries (Development & Regulation) Act, 1951 which legislates exclusive control of denatured ethanol by the Central Government for smooth movement of ethanol across the country.
- Reduction in Goods & Service Tax (GST) on ethanol meant for EBP Programme from 18% to 5%
- Differential ethanol price based on raw material utilized for ethanol production.
- Extension of EBP Programme to whole of India except islands of Andaman Nicobar and Lakshadweep w.e.f. 1st April 2019.
- Interest Subvention Scheme for enhancement and augmentation of the ethanol production capacity by Department of Food and Public Distribution (DFPD)
- Publication of Long-Term Policy on ethanol procurement.

3. For the first time during ethanol supply year 2018-19, following raw materials apart from C heavy molasses were allowed for ethanol production viz. B heavy molasses, sugarcane juice, sugar, sugar syrup, damaged food grains like wheat and rice unfit for human consumption. Also, different ex-mill price of ethanol, based on raw material used for ethanol production, was fixed by the Government in case of sugarcane juice/sugar/sugar syrup, B heavy molasses and C heavy molasses.
4. The aforesaid actions helped in increasing ethanol procurement by PSU OMCS from 38 crore litres during Ethanol Supply Year (ESY) 2013-14 (December 2013 to November 2014) to 415.53 crore litres during FY 2021-22 thereby achieving average blend percentage of 9.50% in ESY 2021-22.
5. Under the EBP Programme, the target for ESY 2019-20 (December 2019 to November 2020) was 7% which has been progressively increased to 10% by ESY 2021-22.
6. The vision to achieve 10% ethanol blending during the year 2022-23 and to reach blending in petrol upto 30% by 2030, the biggest constraint is availability of ethanol distillation capacities and is considered as the critical actionable points. In order to address the ethanol distillation capacity constraint, Department of Food and Public Distribution (DFPD) notified a Scheme on 19th July 2018 for extending financial Assistance to sugar mills for enhancement and augmentation of the ethanol production capacity.

Section 2 - Animal Feed Industry Analysis for DDGS

Animal Feed Industry in India

Animal Husbandry and Dairying activities, along with agriculture, continue to be an integral part of human life since the process of civilization started. These activities have contributed not only to the food basket and draught animal power but also by maintaining ecological balance.

Livestock production and agriculture are intrinsically linked, each being dependent on the other, and both crucial for overall food security. Livestock sector is an important subsector of the agriculture in Indian economy. It forms an important livelihood activity for most of the farmers, supporting agriculture in the form of critical inputs, contributing to the health and nutrition of the household, supplementing incomes, offering employment opportunities, and finally being a dependable “bank on hooves” in times of need. It acts as a supplementary and complementary enterprise.

India has vast resource of livestock and poultry, which play a vital role in improving the socio-economic conditions of rural masses. There are about 303.76 million bovines (cattle, buffalo, *Mithun* and yak), 74.26 million sheep, 148.88 million goats, 9.06 million pigs and about 851.81 million poultry as per 20th Livestock Census in the country.

Species	19th Livestock Census 2012 (in millions)	20th Livestock Census 2019 (in millions)	Growth Rate
Cattle	190.90	193.46	1.34%
Buffalo	108.70	109.85	1.06%
Yaks	0.08	0.06	-25.00%
Mithun	0.30	0.39	30.00%
Sheep	65.07	74.26	14.12%
Goat	135.17	148.88	10.14%
Pigs	10.29	9.06	-11.95%
Other Animals	1.54	0.79	-48.70%
Total Livestock	**512.05**	**536.75**	4.82%
Poultry	729.21	851.81	16.81%

Source: DAHD

Animal Feed

Animal feed refers to the foods that are given to various animals in the process of animal husbandry. Traditional sources of animal feed include household food scraps and the byproducts of food processing industries such as milling and brewing. On a large scale, the food waste is first collected from its various sources like restaurants, offices, houses, markets, etc. The collected food waste is then is automatically sorted and crushed before being recycled in converters as animal feed. The recycled food waste is then given as animal feed presenting a viable alternative to the traditional animal feeds. In a small scale, leftovers such as kitchen scraps and food waste such as vegetables, leaves, fruits, etc. can be collected, treated and processed and can be fed to livestock such as cows, pigs, chickens, etc. Even zoo animals can benefit from this type of feed. Chickens can be given mixtures of waste grain and milled with other by-products to produce what is known as chicken scratch.

Classification of Animal Feed

Cocentrates
- Plant and Animal Sources
- Grains/ Milling By-Products/ Oilseed Cakes

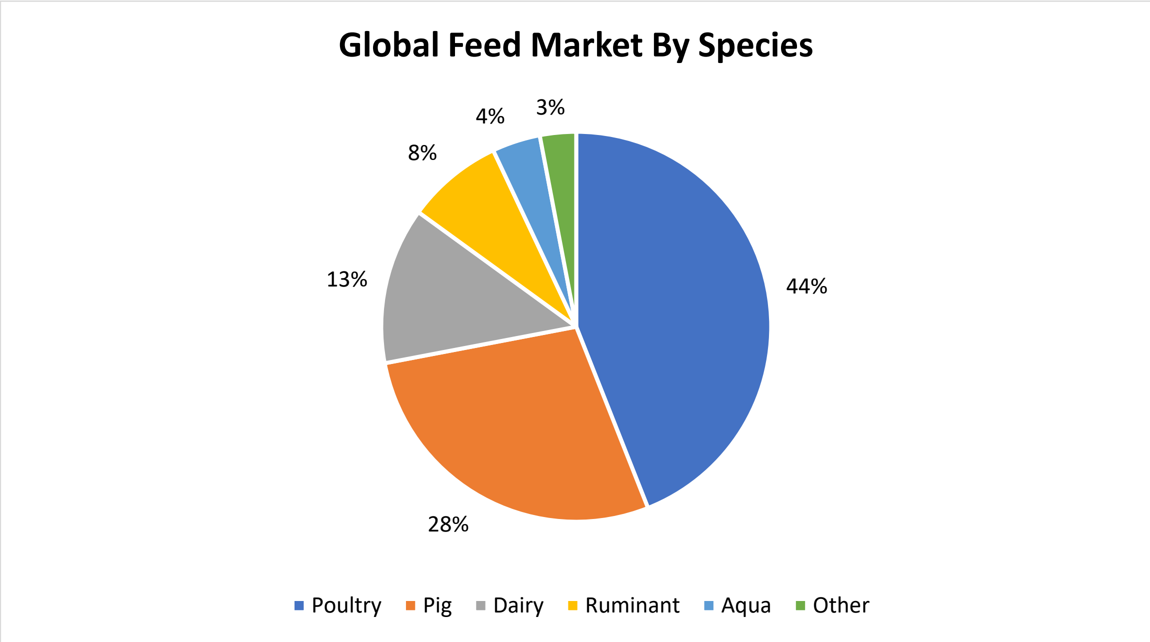
Roughages
- Dry or Green Roughage like hay/straw/husk
- Silage and Grasses like legumes and non-legumes

Unconventional Feeds
- Other Sources
- Protein sources, energy sources, miscellaneous

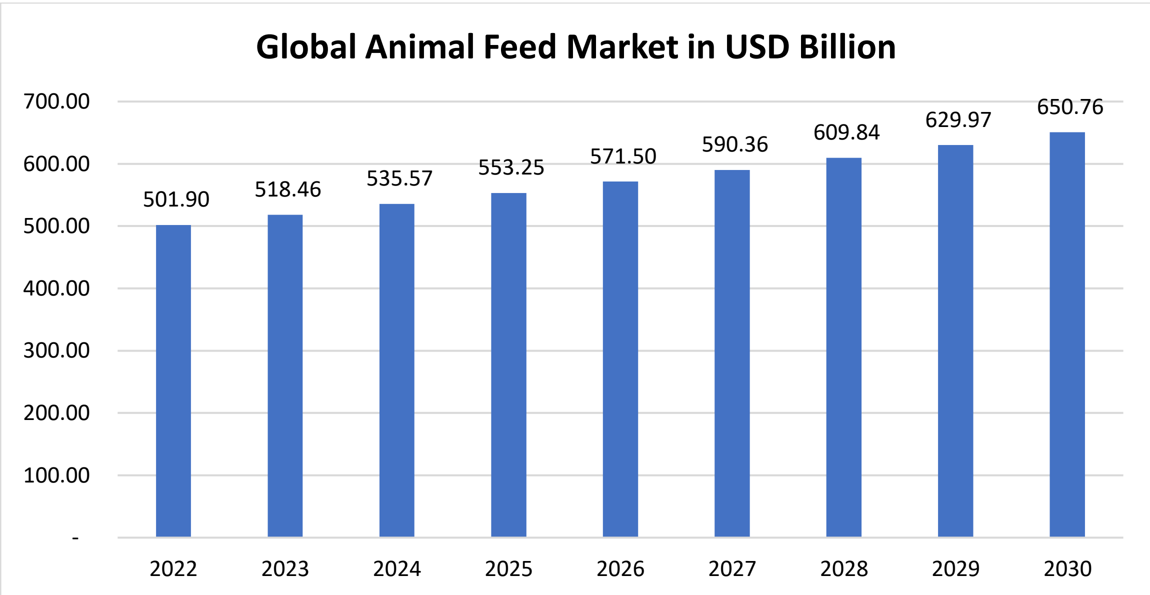
Global Animal Feed Market

Animal feed is the food given to domestic animals in the course of animal husbandry. Globally, the feed industry serves a variety of animals including poultry, dairy *(cattle, calf, and beef*), aquatic/marine, pig, pets etc. The feed demand for different animals varies with respect to socio-cultural dynamics, eating habits and economic importance of animals in the region.

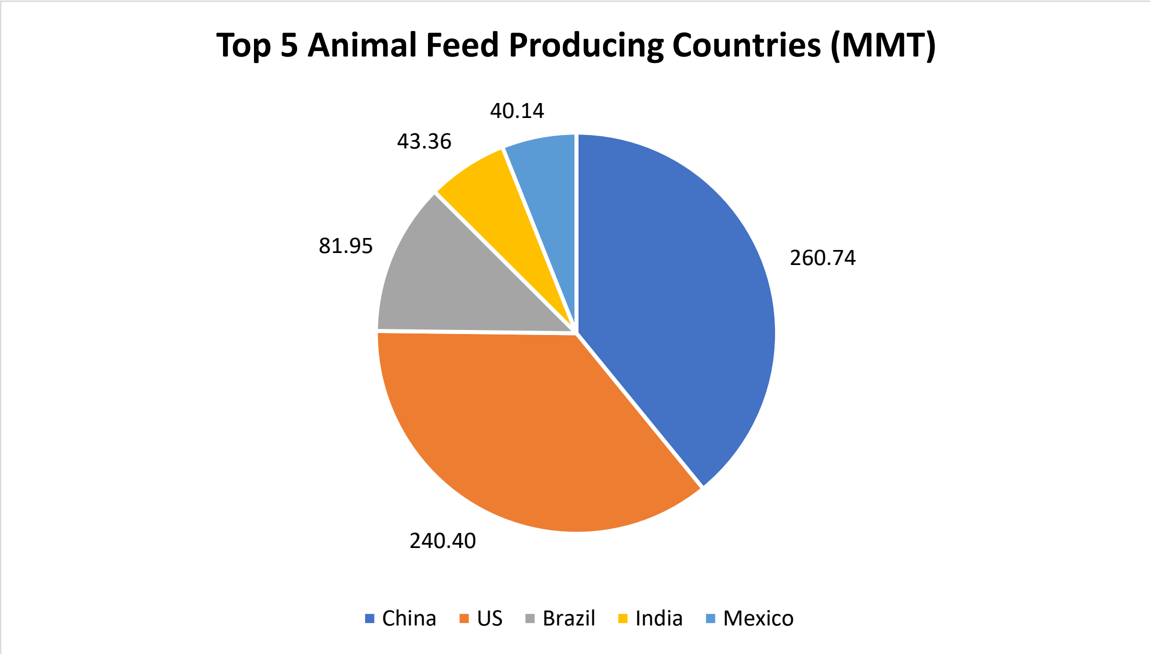
India is glaring at a critical fodder problem over the past few years, according to market reports. According to a recent coverage, India is facing a major challenge in terms of producing adequate feed and fodder for its livestock, given its shrinking land resource. Erratic fodder supply during summer/ drought creates a further gap in the supply chain. The deficit in green fodder is estimated at 11.24%. Currently, fodder is being cultivated on 8.4 million hectares (*nearly 4% of gross cropped area*), whereas experts suggest a share of 14-17%.



In most of the countries across world, the poultry segment occupies a major share of overall feed production volume, according to IFIF (International Feed Industry Federation) estimates, 2021. The global animal feed market reached USD 518.46 billion in 2023 and is expected to grow to USD 650.76 billion by 2030, exhibiting a CAGR of 3.3% during this period.



The 2023 Altech Agri-Food Outlook revealed that despite significant macroeconomic challenges that affected the entire supply chain, global feed production remained steady 2022 at 1.27 billion metric tons *(BMT)* in 2022, a decrease of less than one-half of one per cent *(0.42%)* from 2021's estimates. Data compiled from more than 140 countries and around 2,800 feed mills revealed that the top 5 feed-producing countries over the past year were –



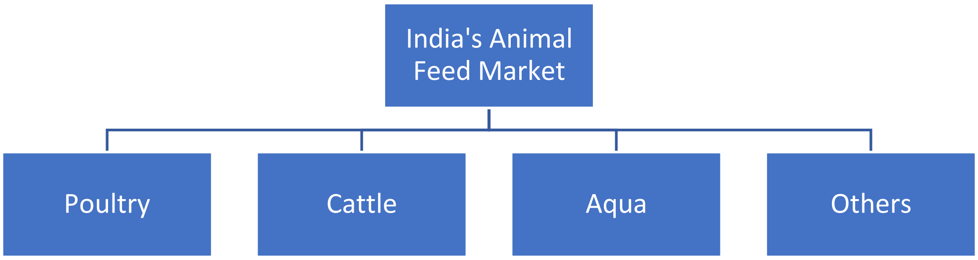
The top 10 countries produced 64% of the world's feed production. China, the US, Brazil and India account for half of the world's feed consumption. According to International Feed Industry Federation *(IFIF),* commercial feed manufacturing generates an estimated annual turnover of over USD 400 billion. The industry continues to expand in volume and value in response to increases in world population, urbanization and growing consumer purchasing power.

The above predictions can be substantiated by the estimates compiled by the Food and Agriculture Organization (FAO), i.e. by 2050, the world has to produce 60 per cent more food to feed a total population of 9.3 billion. Therefore, animal-based protein production is expected to increase along with this increase. OECD-FAO Agricultural Outlook 2021-30, projects the global meat supply to expand over the period, reaching 374 Mt by 2030. Growth in global consumption of meat proteins over the next decade is projected to increase by 14% by 2030 compared to the base period average of 2018-2020, driven largely by income and population growth.

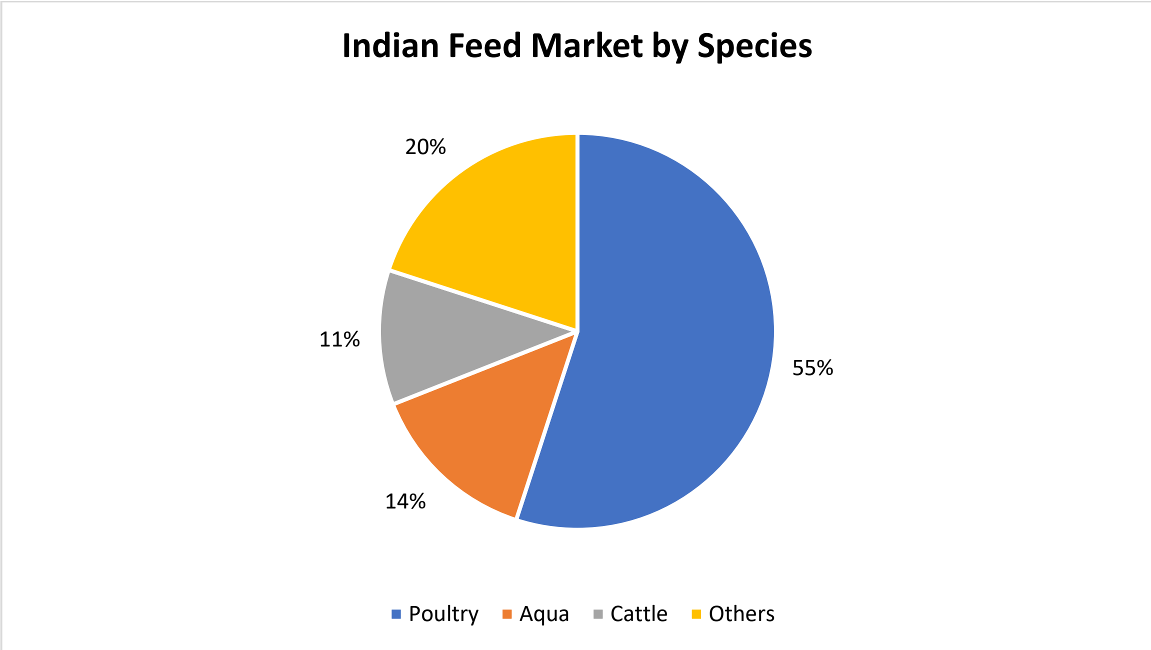
India Animal Feed Market

India's animal feed market has completed its 58 golden years in 2023. The country is known to have one of the largest and fastest-growing compound feed markets in the world. It has built a strong base for compound feeds, which are nutritionally balanced in the form of pellets. It becomes composite with 20% protein and 2-3% fat content.

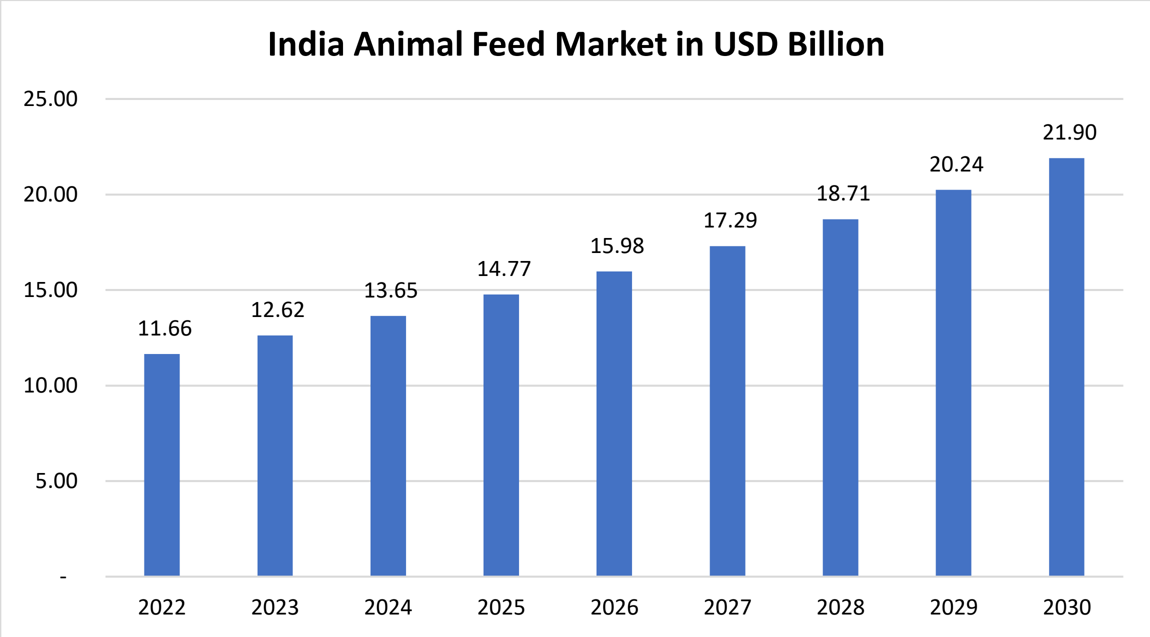
India's Animal Feed manufacturing on a commercial and scientific basis started around 1965, with the setting up of medium-sized feed plants in northern and western India. Today, the industry is undergoing a very exciting phase of growth for the next decade. It is classified into three broad categories as shown below, with the poultry feed market showing dominance.



In terms of its share, the aqua feed segment has seen the maximum growth out of the three major segments, despite being a late starter.



The India's Animal feed market grew at 3.5% CAGR over the last 5 years and the size approx. USD 12.62 billion, as mentioned in the latest report by IMARC Group. There is an expectation for the market to reach USD 21.90 billion by 2028 exhibiting a CAGR of 8.2%.



Segment Wise Market Growth

Cattle and Poultry

India has established strong markets for Cattle and Poultry feeds. Both are growing at 15% and 8-9% respectively have grounds for huge potential in future. India's Animal feed market mainly comprises grains (*like jowar, millets etc.*) and their cakes, mustard, cotton seeds and their oil cakes, de-oiled rice and soyabean. On the other hand, poultry is mainly fed maize, jowar, wheat, soy and their concentrates.

With being the fourth largest broiler producer and third largest egg producer, India has witnessed 120.6 billion egg production in the FY 2021-22, registering a 6.19% YoY growth. The steady growth of poultry consumption over the last decade has been associated with an increase in animal-based protein intake at affordable prices. This has kept feed consumption growing with the broiler industry being the strongest driver behind this development. The presence of integrators *(70% of the total industry*) and a shorter production cycle has brought feed-based efficiencies and rationalization to the industry. While poultry integrators are much stronger in regional pockets of Andhra Pradesh, Karnataka and Tamil Nadu, the much larger landscape for the poultry industry and its expansion beyond these belts provide ample opportunity for standalone feed players. The use of compound feed in the layer industry varies from 5 to 25% and is highly underpenetrated, which offers plenty of opportunities of growth.

On the same lines, India is the largest milk producer in the world, contributing 24% of global milk production in the year 2021-22. India's milk production has registered a 51% increase during the last 8 years. The level of consumption has been constantly boosting demand for Cattle feed. Western India has been registering the maximum share in cattle feed production followed by North, South and East.

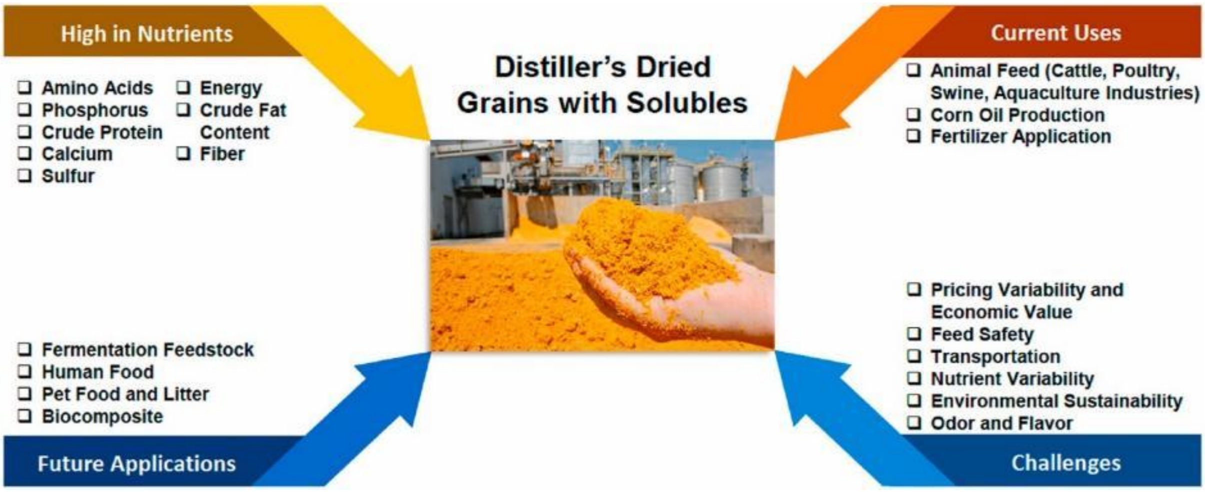
Aquaculture

India's aquaculture feed market has also expanded with a CAGR of 3.5% (2017-22) on the basis of the revenue generated. The market stood at a value of around US$ 1.4 billion in 2020 and is expected to grow at a CAGR of 8% from 2023-28. One of the major determinants of the growth of the aqua feed market in India is attributed to the surging demand for shrimp globally. As of FY 2022, there were approximately 30 high-tech feed mills operating to cater to the rising export and domestic demand. India is the second largest aquaculture country in the country in world accounting for 8% of global production. With an annual growth of 10.34% in fish production in 2021-22, the sector grew at a CAGR of 8% over the last 8 years.

Established industry players like *Anmol Feeds* find the Aqua feed industry a pretty under-researched area. For other major Aqua feed manufacturers like *UNO feeds, and Avanti feeds,* the main problem lies with the procurement of raw materials to produce the feed, especially when the genetically modified soybean and maize are not allowed to be used by the Indian government whereas it is allowed in the other countries like Vietnam and Thailand. Narsimha Rao, Managing Partner, UNO Feeds mentions that it makes it difficult to capture the market since the production gets costly by up to 30-40% and makes Indian cattle feed exports expensive. According to aqua feed manufacturers, there is a misconception that it creates health issues.

Role of DDGS in Animal Feed Market

Identifying alternative fuels produced from plant biomass is essential due to the potential depletion of non-renewable energy sources plus the negative impact that burning them causes to the environment. **Distillers' dried grain with soluble (DDGS)** is a residual product obtained from the process of bioethanol fermentation. This process involves the use of dry milling technology on grains that are high in starch, such as corn, wheat, and barley. The present surge in bioethanol is driven by the demand for sustainable liquid fuels, particularly in the transportation industry. Due to its high content of crude protein, fat, fibre, vitamins, and minerals, DDGS is presently employed as feed for aquaculture, livestock, and poultry. DDGS has been utilized as a feedstock in the manufacturing of value-added goods through microbial fermentation in recent years. DDGS is mainly used as animal feed in cattle, poultry, swine, and aquaculture industries. It is known to be a good source of protein, calcium, phosphorus, and sulphur, which are beneficial for the environment and animal health



Dried distillers' decoction can take several forms depending on the technology employed in its production: wet distillers' grain (WDG), wet distillers' grain with soluble (WDGS), and high-protein wet distillers' grains (HPWDG), as well as dried distillers' grain (DDG), dried grain with soluble (DDGS), and high-protein dried distillers' grains (HPDDG).

Uses of DDGS for Different Animals

- **Cattle -** Calves need a lot of protein as they grow, and DDGS is a great way to meet that demand while also providing calories for their creep diets. Producers can improve cattle performance and lengthen the grazing season by feeding distillers dried grains with soluble (DDGS) to cattle on high forage diets. Depending on the situation and the animal's requirements, DDGS may be fed at a rate of 0.25 to 1.0% of body weight. DDGS, as a rich source of protein, can replace part of the total mix ration (TMR) to feed dairy cattle, while it can also be a source of energy to replace some of the grains in dairy cow feed. Milk yield, protein, and lactose content were all improved when DDGS was used to replace 20% of the barley grain or silage in the diet. There is no need to worry about acidosis in the rumen because rumen pH was not affected.
- **Poultry -** The simplest approach to combat the rising prices of soybean meal and yellow corn is to incorporate DDGS into chicken diets. DDGS is a rich source of xanthophylls, particularly Lutein and Zeaxanthin. Maximum 25% incorporation of DDGS in broiler feed mix ration is advised. By boosting the yellow and red pigmentation of skin and egg, DDGS in chicken feed increases consumer demand for food products like eggs and broilers. Chickens fed a diet with a higher percentage of DDGS gained weight more efficiently.

Demand for DDGS in Animal Feed Market

The increasing competition with the human diet resulting in less availability of raw materials has forced the manufacturers to shift to other materials along with conventional sources. Due to this, price volatility affects feedstock production.

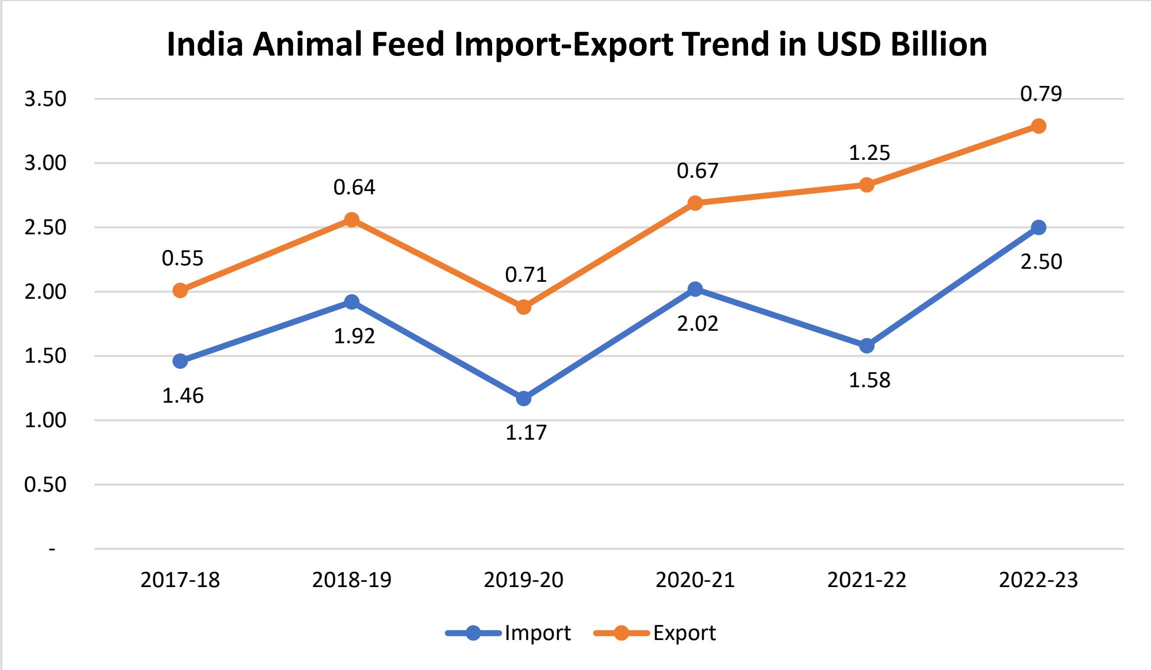
Along with nutrient-rich additives, by-products of the ethanol industry i.e DDGS getting utilized in animal feed production (7-10%) and is regarded as a star ingredient for future requirements. But the farmers are not well aware of its usage in the feed. Apart from some major players, even the industry has information lacunas.

DDGS is the byproduct produced in ethanol production which contains 90% of water content, i.e. on drying up 1 kg of by-product, only 80-90 grams of DDGS is produced which cost Rs. 28 per kg coming out to be equally costly as the conventional grains.

The wet DGS is easily available to the farmers set up near distilleries and is easily fed due to the cost advantage. But the healthier option, the Dried DGS usually isn't that cost-effective, as drying the matter is a costly affair. Therefore, DDGS does hold the potential to support the conventional methods of feed production with booming grain-based ethanol production in India but the above issues should be taken into consideration to boost the usage.

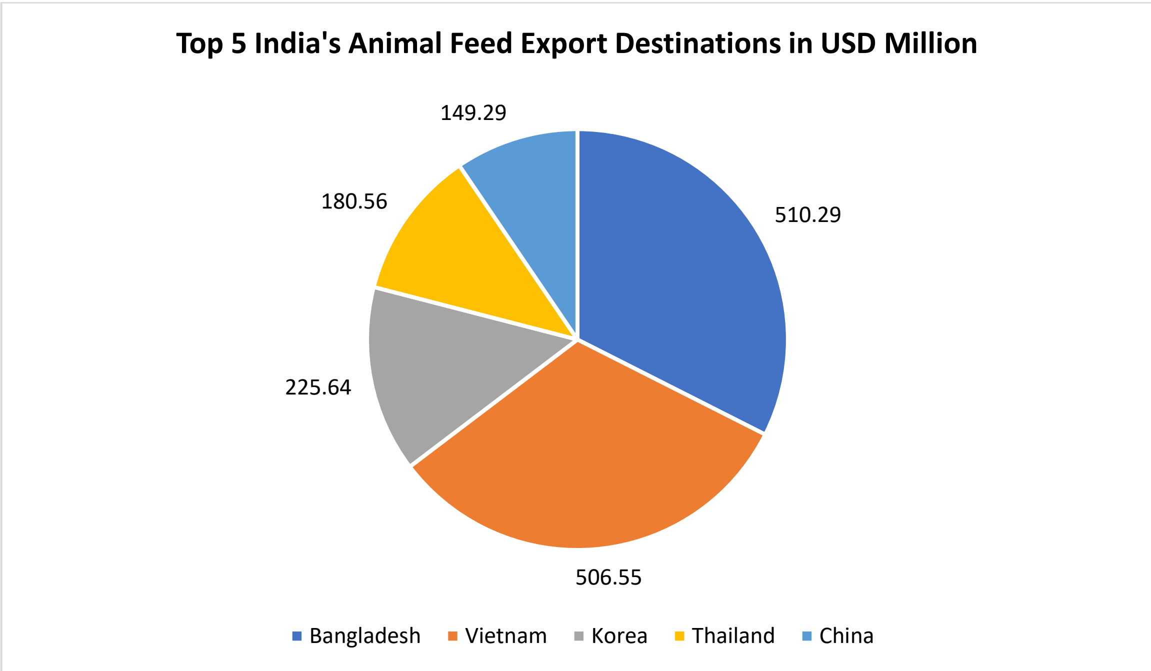
Import Export Trend in Animal Feed Market of India

India established a total trade of US$ 3.29 billion in Animal Feed *(HS 23)* in 2022-23. While India's cattle feed exports rose to USD 2.5 billion (0.1% YoY growth) during the same year, imports dipped to USD 790.21 million (decline by 36.6% YoY).



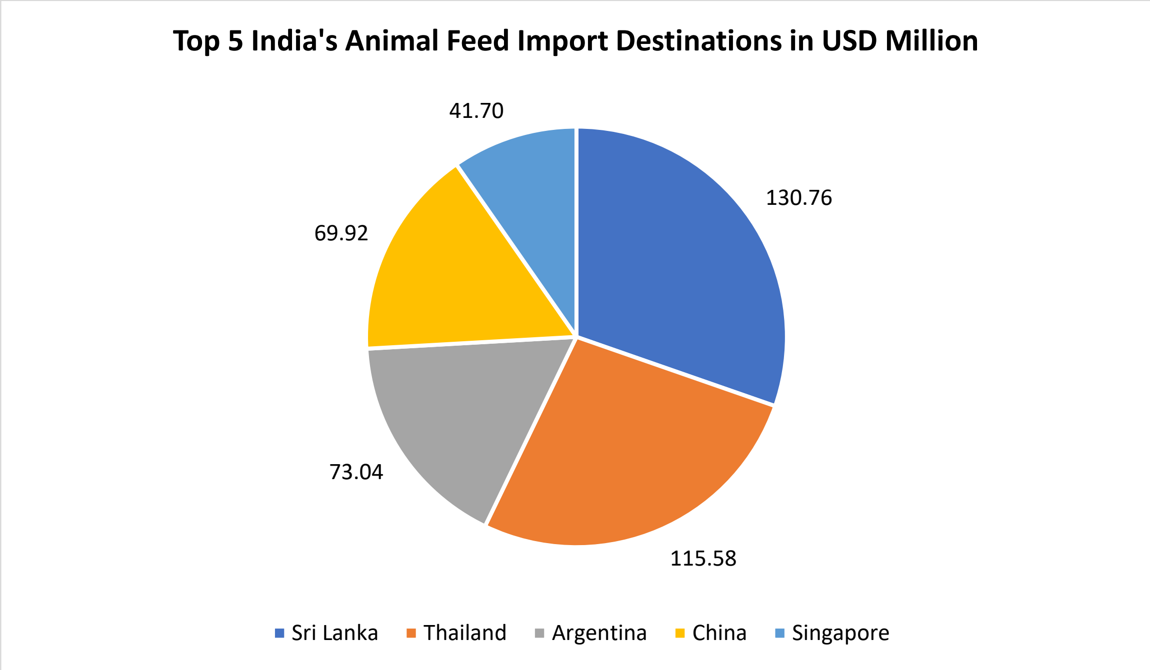
Source: Ministry of Commerce

Among the top 10 Indian cattle feed exports markets, China *(97.7%),* Thailand *(28.3%),* Indonesia *(25%)* and Vietnam *(19.6%)* have registered highest CAGR in the last 5 years. India witnessed an 11.4% CAGR in exports over a span of 5 years. The country's cattle feed exports touched an all-time high in these 5 years.



Among the top 10 Indian cattle feed exports markets, China *(97.7%),* Thailand *(28.3%),* Indonesia *(25%)* and Vietnam *(19.6%)* have registered highest CAGR in the last 5 years. India witnessed an 11.4% CAGR in exports over a span of 5 years.

Among the top 10 import sources, Argentina *(181.9%)* established the highest CAGR over a span of 5 years, followed by South Africa *(78%),* Singapore (26%) and Belgium *(25.1%).*



Future Outlook

The world's animal feed market is projected to reach new heights within the next five years with the right policies in place. Apart from established partners, Indian industry finds huge potential in exports to other nations lacking in native productions like UAE, China, Russia and Brazil. India's share of 1.3% in the world's total cattle feed exports can be further boosted due to increased domestic and international demand by bringing in some government interventions and increased investments in research and developments for feed manufacturing.

Raw material availability in Madhya Pradesh

Agriculture Scenario

In the fiscal years 2023-24 the production of major crops increased by 0.20%. Analysing the crop production by category, cereals production was reduced by 1.91%, pulses experienced a significant increase, and oilseeds showed a growth of 7.32%. In the realm of horticultural crops, there was a growth in spice production from 52.62 lakh metric tonnes in 2022-23 to 53.59 lakh metric tonnes in 2023-24. Vegetable production also rose from 236.41 metric tonnes to 242.62 metric tonnes during the same period. Also, fruit production increased from 95.10 lakh metric tonnes in 2022-23 to 95.54 lakh metric tonnes in 2023-24. Similarly, during 2022-23, milk production increased by 5.88%, reaching 201.22 lakh tonnes. Egg production increased by 9.48%, reaching approximately 31,850.38 lakh eggs in 2022-23. Meat production also rose by 9.37%, reaching about 138.95 thousand tonnes in the same period.

During 2023-24, the contribution of the crop sector to GSVA1 in Madhya Pradesh was 9.10 and 1.84% at current and constant prices (2011-12), respectively. Similarly, the contribution of the livestock sector to GSVA in MP was 13.86 and 8.60% at current and constant prices (2011-12), respectively. The contribution of the fisheries to GSVA in MP was 16.10 and 14.11% at current and constant prices (2011-12), respectively.

Rainfall

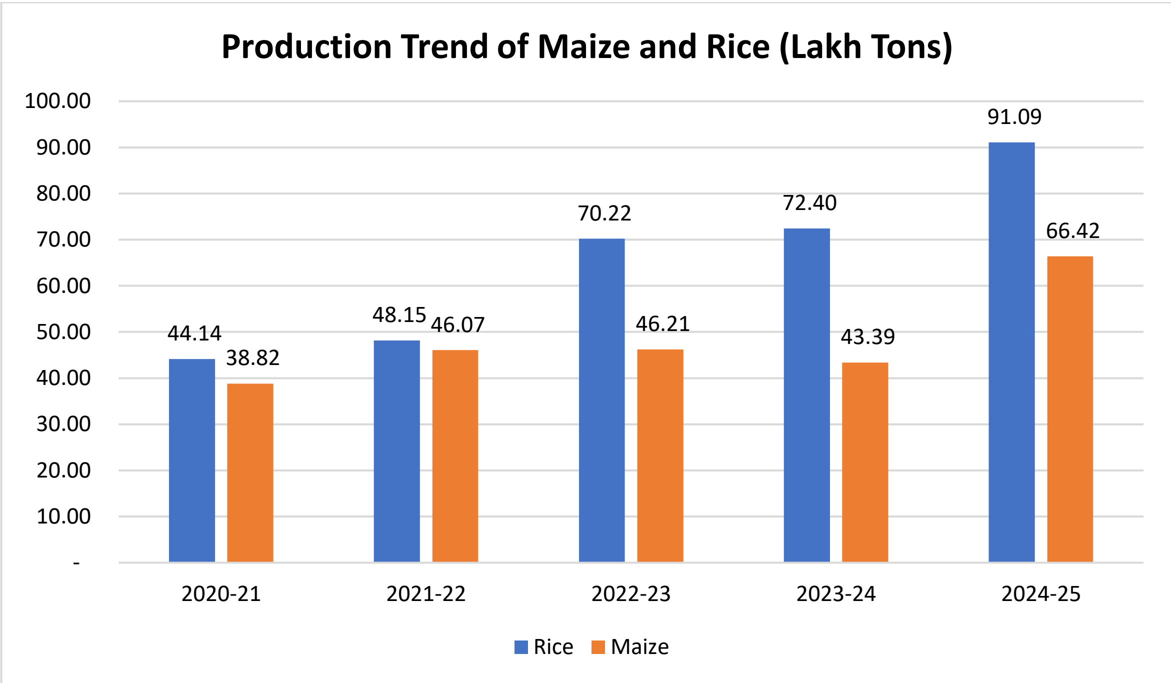
The typical monsoon season in Madhya Pradesh spans from June to September, featuring an average precipitation of 930.9 mm. In the year 2022 (June to September), the documented rainfall amounted to 1131.8 mm, marking an approximately 22% surplus compared to the regular average rainfall.

In the corresponding period of 2023, the total recorded rainfall was 935.8 mm, which is nearly 1% higher than the customary average of 930.9 mm. Notably, June 2023 witnessed a substantial 47.80% increase in rainfall compared to June 2022. Moving into July 2023, there was an 8.68% reduction in rainfall compared to July 2022. August 2023 experienced a more pronounced decrease, with a 29.05% decline in rainfall compared to August 2022. Similarly, September 2023 saw a 17.32% decrease in rainfall compared to September 2022.

Crops Production

Revered as one of the nation's principal breadbaskets, Madhya Pradesh stands proudly as an agrarian state, where the fertile soil and diverse climatic zones foster a flourishing agricultural landscape, leading to the cultivation of a variety of crops. Major crops include wheat, rice, soybeans, pulses, oilseeds, cotton and sugarcane. The state's climate and soil conditions are conducive to soybean cultivation; hence, it has emerged as one of the major soybean-producing regions. Horticulture, featuring fruits like oranges, guavas, and bananas, has gained significance.

Production of Rice and Maize in the state of Madhya Pradesh has been exhibited below –



Based on the above production data of Madhya Pradesh, it can be noted that there is ample amount of raw material availability for a 120 KLPD (Kilo Liters Per Day) grain-based distillery.

As a 120 KLPD distillery requires approximately 265 Tons and 290 Tons of rice/broken rice and Maize per day respectively, depending on the starch content and yield efficiency (considering 456 liter/ton of ethanol form rice and 415 litre/ton of ethanol from maize/corn). Annually (assuming 330 operational days), the plant requires roughly 87,500 to 95,700 Tons of rice and maize respectively. And even using only 1% of the state's Maize or Rice production would comfortably feed the facility.

As the plant is designed on dual (Rice/maize) feedstock technology and MP is producing massive quantities of both crops, this allows the plant to switch between Rice and Maize based on seasonal pricing.

Conclusions

The Consultants have reviewed the data available from various sources, including the data available from various Government Departments, Ministry of Agriculture and Farmer Welfare etc. to understand the availability of feedstock in terms of surplus and broken rice and maize for the proposed 120 KLD grain-based distillery project of Crescendo Industries Private Limited. Based on the research, the consultants have concluded the following –

- India's fuel demand continues its upward trajectory, further straining the national exchequer. In the 2024-25 fiscal year, India's crude oil import bill remained a critical economic variable, hovering near USD 135-140 billion, as the country continues to rely on imports for approximately 87% of its domestic consumption
- the 2021 NITI Aayog roadmap initially projected a USD 4 billion annual saving, current estimates suggest that achieving the E20 target nationwide by 2025-26 could save India upwards of USD 5.5 billion to 6 billion in foreign exchange annually, depending on global oil price benchmarks.
- Fuel Grade Ethanol (purity > 99.5%) is used for blending with Petrol as transportation fuel. Ethanol is a better blend component for Petrol due to higher Research Octane Number of 108.5, which helps in improved engine power and performance. It also helps in significantly reducing other pollutants (Carbon Monoxide, Sulphur Oxides, Nitrogen Oxides, Hydrocarbon and Particulate Matter) in vehicle exhaust when compared with fossil fuels.
- In 2024, India is estimated to maintain an average ethanol blending rate of approximately 11.5 percent. Previously, India achieved its Ethanol Blending with Petrol Program (EBP) target of E-12 in April 2023 for the Ethanol Supply Year (ESY) (December-November) 2022/2023. As such, post revised its 2023 ethanol blending rate with petroleum estimate upward to 12 percent due to increased use of sugarcane syrup, B-heavy molasses, damaged food grains, and surplus rice available from the Food Corporation of India (FCI). Post forecasts ethanol supplies for the EBP to increase by the end of the current ESY 2023/2024 but expects the average blend rate will still be below the E-20 national target for 2025.
- To meet the total ethanol demand by 2025-26, both the sugar and grain sectors would need to supply close to 7 billion liters of ethanol each.
- As per the meeting of Committee of Secretaries on 13th November 2020, DFPD informed that to fulfil 20% ethanol requirement by 2025 will be met from sugar as well as grain-based Ethanol. Further, based on reports published in various sources, it is understood that Government of India is pondering increasing the blending limit to 30% by 2030, in a stepped-up manner.
- India has vast resource of livestock and poultry, which play a vital role in improving the socio-economic conditions of rural masses. There are about 303.76 million bovines (cattle, buffalo, *Mithun* and yak), 74.26 million sheep, 148.88 million goats, 9.06 million pigs and about 851.81 million poultry as per 20th Livestock Census in the country.
- The global animal feed market reached USD 518.46 billion in 2023 and is expected to grow to USD 650.76 billion by 2030, exhibiting a CAGR of 3.3% during this period.
- India has established strong markets for Cattle and Poultry feeds. Both are growing at 15% and 8-9% respectively have grounds for huge potential in future.

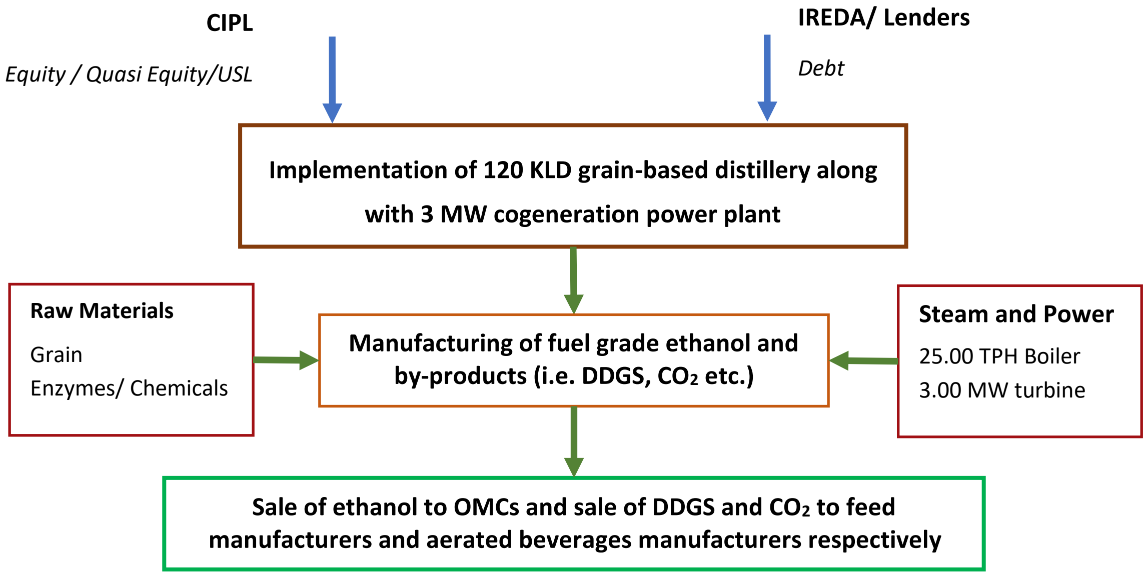
- As a 120 KLPD distillery requires approximately 265 Tons and 290 Tons of rice/broken rice and Maize per day respectively, depending on the starch content and yield efficiency (considering 456 liter/ton of ethanol form rice and 415 litre/ton of ethanol from maize/corn). Annually (assuming 330 operational days), the plant requires roughly 87,500 to 95,700 Tons of rice and maize respectively. And even using only 1% of the state's Maize or Rice production would comfortably feed the facility.

Technical Assessment

NITCON's team conducted a physical site visit of M/s Crescendo Industries Private Limited's manufacturing facility in Rewa District, Madhya Pradesh. In this chapter, NITCON evaluates the proposed manufacturing facility of the company with reference to plant infrastructure/project configuration, production capacity.

Project Configuration

The project configuration as envisaged by the Company and as understood by the Consultants has been presented in the exhibit below –



Source: CIPL

Land Details

The Company has already acquired 76,200 Sq.M. or 7.62 Hectares of land for the project, which is under its possession. The land use conversion has been duly obtained under the Madhya Pradesh Bhu-Rajsva Sanhita, and the current land use is classified as industrial and details of the same has been provided below –

Rakba	Seller	Purchaser	Area (Hec.)
38/1/3(S)	Mr. Abhiman Patel	M/s Crescendo Industries Private Limited	0.76
38/1/1(S)			2.29
38/2(S)	Mr. Alkendra Pratap Singh	M/s Crescendo Industries Private Limited	0.81
38/3(S)			3.76
Total			**7.62**

Source: CIPL

Land Location

The said project of M/s Crescendo Industries Private Limited includes 120 kiloliters per day (KLPD) grain-based ethanol distillery along with 3 MW power co-generation facility and Zero Liquid Discharge (ZLD) system is located at Rakba Nos. 38/1/3(S), 38/1/1(S), 38/2(S), and 38/3(S), Village Mahewa, Teonthar Tehsil, Rewa District, Madhya Pradesh. The latitude and longitude of the project site are 24°53'43.68"N and 81°46'4.01"E respectively. The location of the site has been provided in the exhibit below –

Source: Google Earth

The location of site with respect to other important nearby areas has been provided in the exhibit below –

Source: Google Earth

Location Analysis

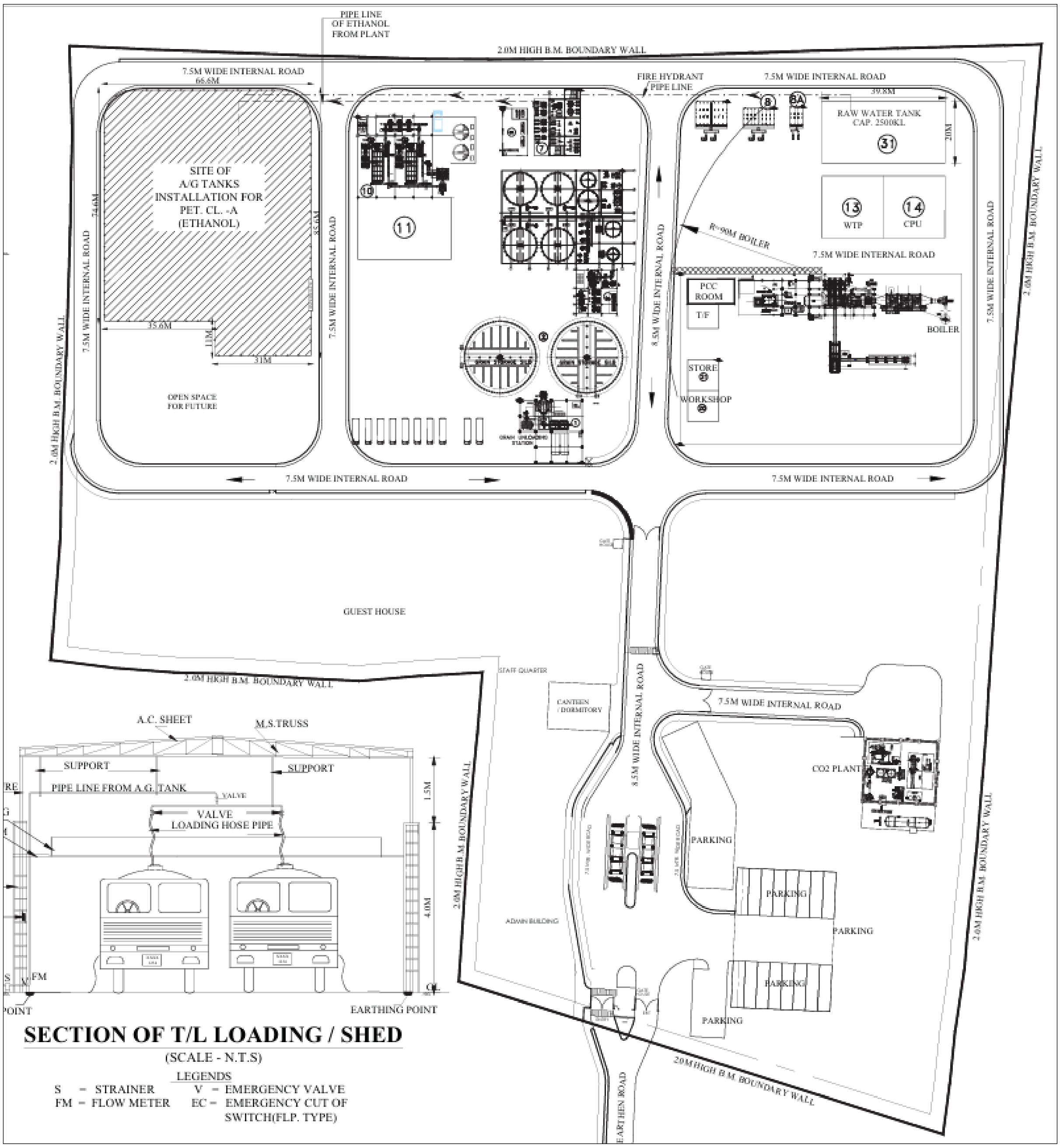
The distance of key demand centers from the plot have been provided below –

Description	Distance (in Km)
Nearest Highway	National Highway - (9.50 Km)
Nearest Town	Mahewa (0.50 Km)
Nearest District	Rewa, Madhya Pradesh (80.00 Km)
Nearest Railway Station	Shankargarh Railway Station (52.00 Km)
Nearest Airport	Prayagraj Airport (90.00 Km)
Nearest Seaport	Paradip Port Odisha (950.00 Km)

Source: Google Earth

Plant Layout

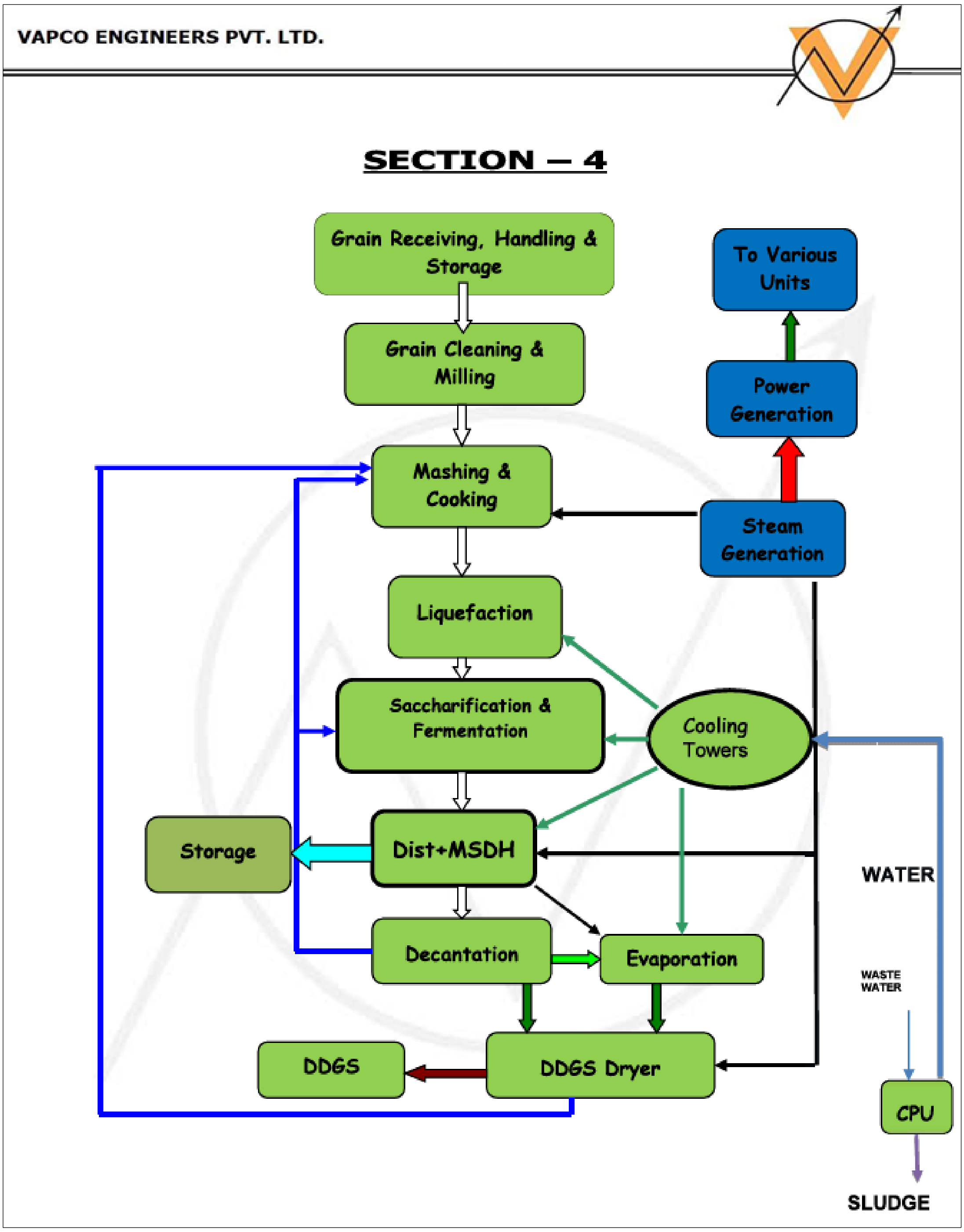
The plant layout as shared by the Company has been attached below –



Source: CIPL

Process Flow Block Diagram

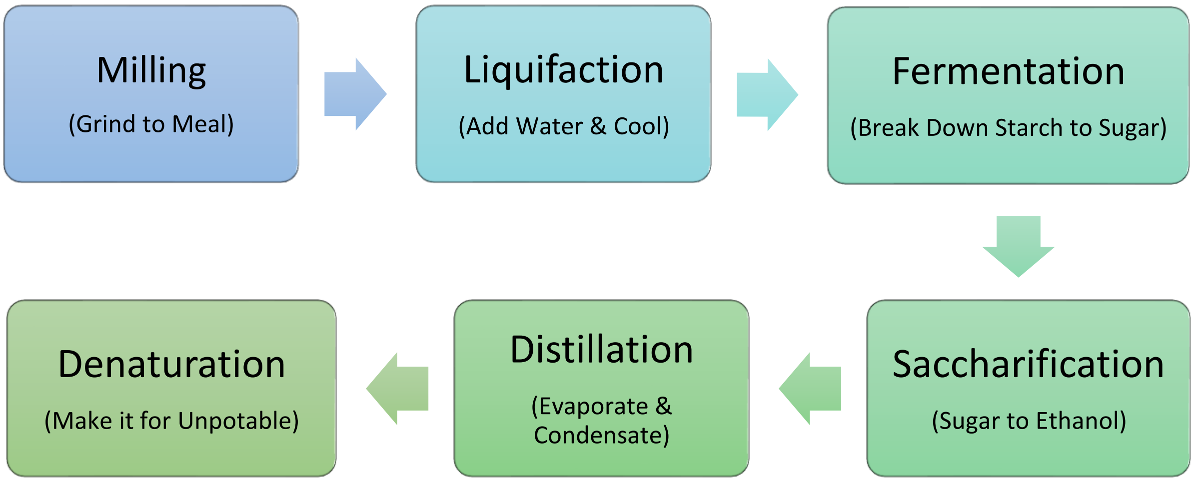
The proposed block diagram of the upcoming ethanol plant as per the Company's plant supplier i.e. Vapco Engineers Private Limited has been presented below –



Source: CIPL

Ethanol Manufacturing Process

Most ethanol is produced from starch-based crops by dry- or wet-mill processing. Nearly 90% of ethanol plants are dry mills due to lower capital costs. Dry milling is a process that grinds Grain into flour and then slurried with water to form a “mash”. Enzymes are added to the mash to convert starch to sugar. The mash is cooked, then cooled and transferred to fermenters. Yeast is added and the conversion of sugar to alcohol begins also Carbon dioxide (CO_2) and Cattle feed is produced as by product. A basic manufacturing process of Ethanol has been presented below –

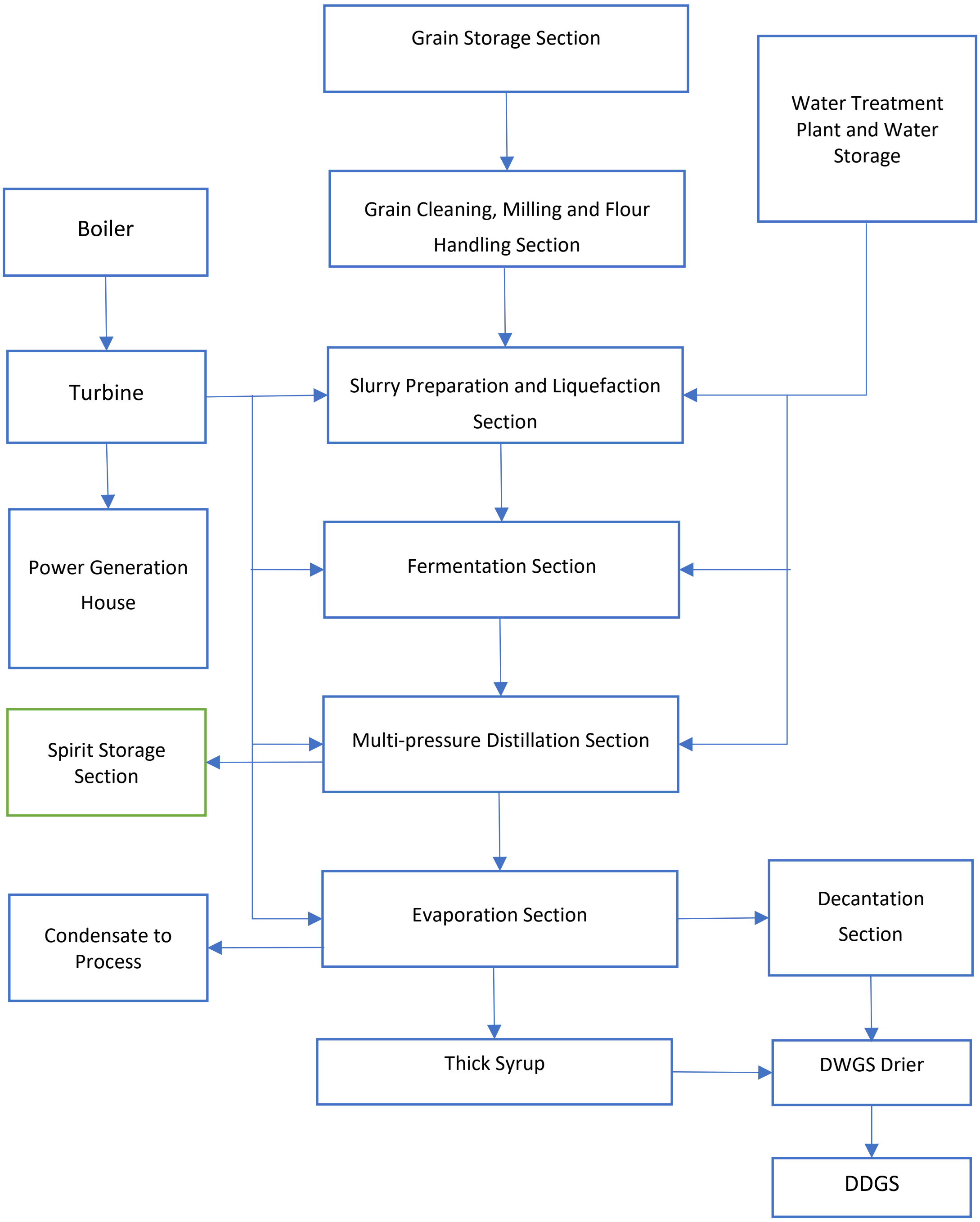


Source: NITCON Limited

In India Ethanol is mainly manufactured using either Molasses or Grain, the Grain based Distillery Process and Molasses Based Distillery process has been explained below –

Grain Based Distillery Process

Major raw material required for grain-based distillery operation would be mainly Rice, broken rice, and Corn. The process flow chart and manufacturing process for the grain-based distillery process are explained below –



1. Grain Cleaning, Milling and Flour Handling

Grains are unloaded at grain hoppers and stored in the Grain Silos through screw conveyers and bucket elevators. From grain silos they are conveyed for pre-cleaning before milling. Grains pass through Vibrating Screen, De-stoner and Magnetic Separator to remove dust and stones. Pre-cleaned Rice is fed at a controlled rate to the Mill. In this unit operation, Rice is broken down into small particles (flour). Oversized particles are segregated with the help of Vibratory Screen.

Silos for Grain Storage

2. Slurry Preparation /Liquefaction

Grain Flour is fed to Seal Conveyer and then to Pre-masher and mixed with recycle Streams and Thin stillage after Analyser column Flash Tank in Premasher. Slurry from Pre-masher is taken to Slurry Mixing Tank where temperature is maintained by sparging direct steam. The retention time in the tank is for one hour. Liquefying Enzyme is partly added to Slurry Mixing Tank.

The Slurry from the Slurry Mixing Tank is transferred through Slurry Transfer Pump to Jet Cooker, where steam is added to raise the temperature of slurry. From Jet Cooker, the slurry is transferred to Retention Vessel.

Part of Liquefying enzyme is added in the Final Liquefaction Tank. The Retention time in Final Liquefaction tank is for two hours. In Slurry Mixing and Final Liquefaction Tank the temperature is maintained between 85-88 Deg. C. The liquefaction is carried out in Slurry Mixing and Final Liquefaction Tank. The Liquefied Slurry is then cooled up to 34 Deg. C. in two stages and transferred to Fermentation Section. In the first stage heat is removed by using the heat of the Liquefied Slurry to pre-heat the Fermented Mash before feeding to Distillation Section. In the second stage heat is removed in a heat exchanger by using cooling water on other side. Wide gap plate heat exchangers are used for heat transfer.

3. Saccharification and Instantaneous Fermentation (SIF)

A part of the Liquefied Slurry is fed to Pre-Fermenter, where a part of Saccharifying Enzyme is added to convert the Liquefied Starch to Reducing Sugars. The required quantity of Active Yeast as specified by supplier is added to prepare the Yeast Cell Mass. Filtered Process Air from Air blower is sparged with measured quantity to Pre-Fermenter for yeast growth.

The heat generated due to exothermic reaction of ethanol formation is removed by forced circulation cooling in external heat exchangers (wide gap plate heat exchanger) and the temperature of mash is maintained temperature in the Fermenter. The propagated cell mass is then transferred to Fermenter to provide Yeast Cell Mass. One Pre-fermenter is provided in the Fermentation section.

Balance Liquefied Slurry is taken to the Fermenter. Yeast Cell Mass from Pre-fermenter and rest of the Saccharifying Enzyme is also added to the Fermenter. The heat generated due to exothermic reaction of ethanol formation is removed by forced circulation cooling in external heat exchangers (wide gap plate heat exchanger) and the temperature of fermented mash is maintained temperature in the Fermenter. The fermentation reaction time depends upon the specification of feed stock.

CO2 produced during the process is fed directly to CO2 scrubber. Clean in place (CIP) system is provided for cleaning of Pre-Fermenter and Fermenters after every batch. An overall fermentation cycle time of about ~50 to 60 hours is provided considering the time for CIP. Four Fermenters and One Beer Well with flat slopping bottom and conical top with top entry agitator is provided in the system.

4. Distillation

In the Distillation Plant Alcohol from Fermented Mash is concentrated up to 93-94 %v/v. This is further concentrated in MSDH Plant to produce Anhydrous Alcohol with 99.8%v/v concentration.

The whole stillage from Distillation Plant is fed to Decantation section before feeding it to Evaporation plant. Here thin stillage is concentrated to produce concentrated stillage.

Distillation plant consists of following distillation columns as listed below –

a. DG column

Preheated Fermented Mash from Fermented Mash preheater is fed in a measured quantity to DG Column in Distillation section. The purpose of this column is to strip off non-condensable and more volatile impurities from the Fermented mash.

Analyser Column vapours are used as energy source for DG column. The Fermented Mash from DG column is transferred to Analyser column.

b. Analyser Column

This Column is operated under vacuum. Vapours from Column top are fed Rectifier Column. The alcohol concentration in the vapours varies with the alcohol concentration in the Fermented Mash feed. The purpose of Mash Column is to strip off alcohol from Fermented Mash. The alcohol vapours from column top are with Alcohol and Water.

The rest of the fermented mash depleted alcohol flows down the Analyser Column and is taken out as Whole Stillage from Analyser Column bottoms. Vapours from Rectifier column is Mechanically recompressed in MVR & used as energy source to drive Analyser Column by using Thermosyphon Reboiler.

c. Rectifier Column

The vapours from mash column are directly fed to bottom of Rectifier column transferring the mass as well as energy from mash column to Rectifier column. Alcohol is enriched at the top and is drawn out as Hydrous Alcohol at about 93 - 94%v/v. The Rectifier Column vapours are Mechanically recompressed in MVR & used as energy source to drive Analyser Column and condensate taken to Rectifier Reflux Tank. Liquid from the Rectifier Reflux Tank is pumped back to Column as reflux. Fusel oil build up is avoided in the Column by withdrawing side streams (Fusel Oil). This side stream is sent to Fusel Oil Decanter where these streams are diluted with water and Higher Alcohol rich layer is separated and sent to Fusel Oil Product Tank. Fusel Oil Washings from decanter are recycled back to Rectifier Column. Part of Rectifier Vapours is sent to MSDH section for further dehydration.

Regeneration Liquid from Dehydration Section is fed to Rectifier Column to recover ethanol from the liquid.

d. Exhaust Column

Bottom stream containing maximum amount of water will flow from the rectifier column to the exhaust column. MSDH top vapours used as an energy source for exhaust column. The spent lees will be taken out from the exhaust column.

5. Molecular Sieve Dehydration (MSDH) Plant

Vapours from the Rectifier Column are passed through Superheater where these vapours are superheated. Energy for superheating is supplied by steam condensation on the shell side of the Superheater.

Superheated Alcohol Feed Vapours are then sent to Sieve Beds. The Sieve Beds operate in a cyclic manner. These beds are provided to allow for sieve regeneration in continuous operation. While one bed is in Dehydration Mode, the other is in Regeneration Mode. Depending on feed and product specifications, dehydration-regeneration exchange takes place approximately every few minutes. The Hydrous Feed Vapours are passed through the bed under Dehydration Mode. The Sieve Bed adsorbs moisture present in Hydrous Alcohol Vapours and Dehydrated Product Vapours (with moisture content less than 0.2% v/v) are obtained from bottom of the bed.

The Dehydrated Product Vapours are then passed through Regeneration Preheater for heat recovery. Partial vapours are fed to Exhaust reboiler as source of energy and balance vapours are then passed through Product Condenser where these are condensed with the help of Cooling Water. Condensed Anhydrous Alcohol Product is collected in Product Receiver. The Anhydrous Alcohol from Product Receiver is pumped to Product Cooler where it is cooled with the help of Cooling Water and then sent for Dehydrated Ethanol Storage.

Vacuum is pulled in the system with the help of Vacuum Eductor. Regeneration Stream is used as motive fluid for Vacuum Eductor. The Regeneration Stream coming from the Regeneration Condenser is pumped, preheated in Regeneration Preheater and fed to the Rectifier Column for recovery of ethanol. Regeneration liquid is recycled back to Rectifier Column for Alcohol Recovery.

After each cycle, the beds are interchanged, that is, the bed in Dehydration Mode will be switched over to Regeneration Mode and the bed on Regeneration Mode will be switched over to Dehydration Mode, with the help of automation system.

6. Decantation Section

Decanters (Centrifugal) are used to separate the suspended fibrous material present in the Whole Stillage. The solid product is collected as Wet Cake, while the thin slop is partially recycled to Liquefaction Section and rest of it is taken to the Evaporation Section for concentration.

7. Evaporation

Three effect falling film system (3Working+1Standby) with one forced circulation effect (1Working+1Standby) – Thin Slop from Distillation along with CIP waste solution of Evaporation section fed to Falling Film Effect-I. Partially Concentrated Syrup from Falling Film Effect - I will be fed to Falling Film Effect – II. Then more concentrated slurry fed to Falling film effect III. Fourth Falling film effect will be used as Stand by effect. Finisher for concentration up to 35.00 % w/w. The concentrated Slop from Finisher will be sent to Product Tank.

Vapours generated in Dryer section is Mechanically vapor recompressed (MVR) and used as heating source of Force circulation Effect. Vapours generated in all three Falling films is collected and mechanically vapour recompressed (MVR) to use as heating source back to all Falling Film effect.

Process Condensate generated will be collected in Process Condensate Tank. Some of the process condensate will be sent to Liquefaction section and remaining Process Condensate will be sent to Process Condensate Treatment Plant. Necessary vacuum required will be generated by a Water Ring Vacuum Pump.

Concentrated Syrup from Evaporation will be sent to Dryer Section for Drying.

8. Dryer

The Feed to dryer is handled with various conveyors and mixers with partial recycling of dried product. The Dryer is a Tube bundle dryer and operates on Steam which is fed to tube side. Product from Dryer is conveyed through conveying system and is collected in Silo. The Drying operation also involves bag filters, bucket elevators, Blowers, cyclone separators for operation.

Vapours from Dryers are passed through cyclone separator to separate the vapours from dust through induced draft blower. This Blower then sends the vapours to DCC for Dryer Vapor Recovery.

9. DCC

Dryer Vapors are sent to DCC system where dryer vapors come in direct contact with recirculation liquid (Water). The DCC liquid from DCC is sent to flash tank where the vapors are flashed at lower pressure. DCC liquid is recirculated to DCC system to condense the incoming dryer vapors. Purge from the DCC is taken out from DCC recirculation system.

10. PCTP, WTP & ATFD

Condensate From Evaporation and Dryer Section is sent to process condensate treatment plant after That treated water used in further Process & Reject will send to Evaporation section.

Wastewater from various Sections is collected and treated in wastewater treatment plant and sent to further use.

Plant and Machinery List

The list of equipment proposed to be installed in the project are presented in the exhibit below –

Section	S.No.	Equipment Item	Quantity
I. Grain Handling & Storage	1	Receiving Hopper	1
	2	Grain Silo	2
	3	Bucket Elevator #1	1
	4	Grain Pre-cleaner	1
	5	Bucket Elevator #2	1
	6	Chain Conveyor	2
	7	Central Electric Slide Gate with Actuator	Lot
	8	Manual slide gate valve	Lot
II. Milling Section	1	Bucket Elevator #1	1
	2	Grain Cleaner	1
	3	Magnetic Separator	1
	4	De-stoner	1
	5	Bucket Elevator #2	1
	6	Surge Bin	1
	7	Weigh Feeder	1
	8	Hammer Mill	2+1
	9	Bucket Elevator #3	1
	10	Buffer Flour Hopper	1
	11	Aspiration system (Air blower, bag house)	1

Section	S.No.	Equipment Item	Quantity
III. Liquefaction	1	Pre-masher	1
	2	Mash Tank	1
	3	Mash Tank Agitator	1
	4	Cooker Feed Pump	1+1
	5	Jet Cooker	1
	6	Retention Vessel	1
	7	Flash Vessel	1
	8	Main & Vent Condenser	2
	9	Hot Water Tank	1
	10	Hot Water Pump	1+1
	11	Liquefaction Tank	1
	12	Liquefaction Tank Agitator	1
	13	Mash Cooler Pump	1+1
	14	Mash Coolers	2+2
	15	Enzyme Tank	1
	16	Enzyme Pumps	2+1
	17	Nutrient Tank	1
	18	Nutrient pumps	1+1
	19	Lime/Caustic Tank with pump & agitator	1
IV. Pre-fermentation	1	Pre-fermenters with Agitators	1
	2	Pre-fermenter Circulation Pump	1
	3	Pre-fermenter Cooler	1
	4	Pre-fermenter Cleaner	1
	5	Air Blower with HEPA filters	1+1
V. Fermentation	1	Fermenters with Agitators	3
	2	Fermenter Coolers	3
	3	Fermenter Re-Circulation Pumps	3
	4	Fermenter Cleaners	3
	5	CO2 Scrubber	1
	6	Antifoam Tank	1
	7	Antifoam Pump	1+1
	8	Beer well with Agitator	1
	9	Beer Feed Pump	1+1
	10	Caustic tank	1
	11	Caustic pump	1
	12	CIP tank	1
	13	CIP pump	1+1
VI. Distillation	1	Degassing Column	1
	2	Beer Stripper Column	1
	3	Rectification Column	1
	4	Aldehyde Column	1
	5	Vent Scrubber	1
	6	Beer Stripper Column Reboiler	1
	7	Rectification Column Reboiler	1
	8	Aldehyde Main Condenser	1
	9	Aldehyde Vent Condenser	1
	10	Rectification Vent Condenser	1

Section	S.No.	Equipment Item	Quantity
	11	Rectification Column Feed Preheater	1
	12	Rectifier Column Feed Tank	1
	13	Rectification Column Reflux Tank	1
	14	Steam Condensate Tank	1
	15	Distillation CIP Tank	1
	16	Distillation CIP Pumps	1+1
	17	Spent Wash Circulation Pump	1+1
	18	Spent Wash Pump	1+1
	19	Rectification Column Lees Pump	1+1
	20	Rectification Column Reflux Pump	1+1
	21	Steam Condensate Pump	1+1
	22	Vacuum Pump	1+1
VII. MSDH Plant	1	Super heater	1
	2	Molecular Sieve Vessels	2
	3	Lean Feed Preheater	1
	4	Product Condenser	1
	5	Product Tank	1
	6	Product Pump	1+1
	7	Regeneration Condenser	1
	8	Regeneration Cooler	1
	9	Regeneration Tank	1
	10	Ejector	1
	11	Regeneration Pump	1+1
VIII. Decantation	1	Spent Wash Tank	1
	2	Spent Wash Tank Agitator	1
	3	Decanter Feed Pump	1+1
	4	Decanters	2+1
	5	Thin Stillage Tank	1
	6	Thin Stillage Transfer Pump	1+1
IX. Integrated Evaporator	1	Falling Film Evaporators	3
	2	Forced Circulation Evaporator	1
	3	Vapor Separator	4
	4	Circulation Pumps	4+2
	5	Process Condensate Vessel	1
	6	Process Condensate Pump	1+1
	7	Surface Condenser	1
	8	Syrup tank	1
	9	Syrup pump	1
	10	Vacuum Pump	1
X. DDGS Dryers	1	Paddle Mixer	2
	2	Screw Feeder	2
	3	Dryer shell (fixed)	2
	4	Steam Tube Bundle (Rotating)	2
	5	Product Chute with rotary valve	2
	6	Back mixing / sieving / crusher system	2
	7	Drive Mechanism (Motor/Gear box)	2
	8	Induced draft fan	2

Section	S.No.	Equipment Item	Quantity
	9	Product Cooler	1
	10	Cyclone separator & rotary valve	Lot
	11	Heat Recover system	1
XI. DDGS Storage	1	Diversion Chute for silo	1
	2	Silo	1
	3	Rotary Valve	1
	4	Semi-Automatic Bagging system	1
XII. Water Treatment	1	Softening plant (35 m3/hr)	1
	2	DM water plant (15.0 m3/hr)	1
	3	Multi Grade Filter (85 m3/hr)	1
XIII. Utilities & Storage	1	Cooling Tower (1500 m3/hr)	1
	2	Cooling Tower Pumps	1+1
	3	Cooling Tower (Fermentation)	1
	4	Cooling Tower Pumps (Fermentation)	1+1
	5	Instrument Air Compressor	1+1
	6	AA Receivers (150 m3)	3
	7	AA Bulk Storage tank (2000 m3)	2
	8	Issue cum Denaturant tank	1
	9	FO tank (15.0 m3)	1
	10	Transfer pumps (Product Storage)	6
	11	Weigh Bridge (60 MT)	1
	12	DG Set (750 KVA)	1
	13	Boiler (25 TPH)	1
	14	Back Pressure Turbine (3 MW)	1
	15	EOT crane	1

Source: CIPL

Technology/Service Providers

Based on the information and discussion with the executives during site visit of the Company it is understood that the company has appointed VAPCO Engineers Private Limited as EPC contractor for the entire project, further the Company has procured steam turbine for co-generation power plant from Triveni Turbine Limited and the details of same has been given below –

Engineers Private Limited (EPC contractor)

VAPCO Engineers Private Limited is a specialized engineering firm headquartered in Navi Mumbai, India, that operates as a comprehensive EPC (Engineering, Procurement, and Construction) contractor for the process industry. The company has carved out a significant niche in the biofuel and distillery sectors, where it provides end-to-end turnkey solutions ranging from conceptual design and manufacturing to final commissioning. Their expertise spans multiple sectors, including starch and derivatives, dairy, chemicals, and pharmaceuticals, making them a versatile partner for complex industrial infrastructure projects.

Triveni Turbine Limited

Triveni Turbine Limited (TTL) is a global leader in the industrial steam turbine market, specifically dominating the sub-100 MW segment. Based in Noida with specialized manufacturing facilities in Bengaluru, TTL is part of the diversified Triveni Group and holds a commanding presence in the Indian market while exporting to over 80 countries. The company specializes in the design and manufacture of backpressure and condensing steam turbines that are essential for power generation and combined heat and power (CHP) applications. TTL is a key player in the renewable energy transition, as their turbines are frequently utilized in biomass-based power plants, waste-to-energy projects, and district heating. Beyond manufacturing, the company provides extensive aftermarket support through its refurbishment and maintenance services, ensuring the long-term operational efficiency of rotating equipment across various global industries.

NITCON Observation

From the above profiles of Contractor and vendor the Consultants note that the Company has appointed reputed players from the industry and will not face any potential challenges regarding implementation and efficiency. And the appointment of VAPCO Engineers Private Limited as the EPC contractor provides the project with specialized, turnkey expertise essential for distillery operations, Furthermore, the selection of Triveni Turbine Limited for the co-generation unit ensures the deployment of proven, high-efficiency technology.

Utilities

Power

The Company is installing a 3 MW captive co-generation plant designed to utilize rice husk as the primary fuel source. The facility employs a back-pressure turbine to generate electricity while simultaneously extracting steam for industrial heating within the plant. This dual-purpose configuration ensures high thermal efficiency by utilizing a single fuel source for both power and process heat. The overall power consumption for the plant is estimated to be as below –

Section	Operating in KWh	Consumed Power
Grain Handling and Milling Section	400	300
Liquification Section	100	75
Fermentation Section	350	263
Distillation Section	120	90
Ethanol Section	60	45
Decantation Section	100	75
DDGS Dryer Section	350	263
Dryer Vapour Integration Section	450	338
Cooling Tower Section	650	488
Product Storage Section	15	11
Water Treatment plant	100	75
CPU Section	180	135
Plant and Yard Lighting	20	15
Misc	10	8
Boiler Section	400	300
Turbine Section	60	45
C02 Plant	250	188
Grain	105	79
Total Connected Load	**3,720**	**2,790**
Turbine Capacity		**3,000**

Source: CIPL

As per the information, the Estimated Captive Consumption of the Power would be approximate 2790 KWh which would be fulfilled by the Captive Co-gen Plant with the capacity of 3 MW. Furthermore, DG set of 250 KVA would also be part of the alternate power backup arrangement.

Fuel

Ethanol manufacturing requires intensive use of steam, hence for the generation of steam a husk-based boiler of capacity 25.00 TPH is being installed, hence the husk will be used as fuel for boilers which will be procured from respective suppliers/sellers.

Here it should be noted that the primary fuel for the purpose of firing the boiler has been considered as rice husk/ agro waste, which can be procured from multiple sources within the catchment area.

Water

Based on the information received it is understood that water requirement in the proposed project will be around 1300KL per day. However, the water will be recycled and the actual makeup water requirement on account of evaporation will be around 530 KLD. The Consultants have considered water requirement for the Project at 4.00 KL/ KL of Ethanol produced for purpose of financial evaluation of the Project.

Steam Generation Plant

Ethanol manufacturing requires too much steam for the process of liquefaction, fermentation, distillation dehydration, and DDGS Dryer. For the same the proposed plant is equipped with a rice husk/ agro waste-based boiler of capacity 25 TPH. Steam supplied at pressure of min. 3.50 kg/cm2 (g) and temperature of 148°C at inlet of respective section (MP Steam - Dry, saturated, desuperheater and regulated steam).

Here it should be noted that the primary fuel for the purpose of firing the boiler has been considered as rice husk/ agro waste, which can be procured from multiple sources within the catchment area.

Water Treatment Plant

As per the guideline laid down by the Ministry of Environment, Forest and Climate Control (Government of India), any new distillery setup needs to be zero-discharge setup. The Company proposes setting up a Water Treatment Plant at the site to handle the water discharged from the process.

Water Treatment facility as planned with be designed with the concept that “no effluent/treated water will be discharged outside factory premises” or in other words, the Project will be a Zero Liquid Discharge Project.

Raw Material

The major raw material for this project is Rice and broken rice, however if there is shortage of rice the project could be using Maize to produce ethanol. The raw material will be procured from nearby rice producing area. Also, it can be procured from the Food Corporation of India (FCI).

Parameter of Feedstock (Broken Rice)	Unit	Values
Water (Moisture)	%W/W	10 - 12
Total Solids	%W/W	88 - 90
Starch	%W/W	70-72
Other Organic Matter	%W/W	20 - 22

Source: CIPL

Yield

The Consultants note that rice with starch content 68-70% has been considered for the Project, with yield of 456 liters of ethanol production per ton of broken rice input.

Manpower Details

The proposed ethanol manufacturing plant will require around 150 personnel for the successful operation of the plant. Detailed list of manpower required for the project and their respective salary has been exhibited below –

Particulars	Unit	Value	Salary/Months (Rs.)
General Manager	Nos.	1	1,50,000
Finance Head	Nos.	1	90,000
Finance Manager	Nos.	2	35,000
Finance Executive	Nos.	2	25,000
Accounts Manager	Nos.	1	45,000
Accounts Executive	Nos.	2	20,000
Marketing Manager	Nos.	2	65,000
Marketing Executive	Nos.	4	35,000
HR & Admin	Nos.	2	25,000
Production and Maintenance			
Plant Manager	Nos.	1	1,00,000
Store Manager	Nos.	4	35,000
Quality Control Staff	Nos.	6	35,000
Manager (Production)	Nos.	6	50,000
Supervisors/ Foreman	Nos.	14	30,000
Security Guards	Nos.	14	12,000
Skilled Operator/ Workers	Nos.	12	25,000
Semi-Skilled Operator/ Workers	Nos.	32	18,000
Helpers / Unskilled Workers	Nos.	44	12,000
Total Manpower Requirement	Nos.	**150**	
Annual Increase in Salaries	%		2.50%
Benefits and Perks	%		15%

Source: NITCON Estimate

Here it should be noted that benefits to the tune of 15% up and above the base salaries as provided in table have been considered by the Consultants, while undertaking financial evaluation. Further an annual increase of 2.50% in the salaries has also been considered from second year of operation.

Status of Approvals and Clearances

The Consultants note that the Project is in advances stages of implementation and has acquired necessary approvals and clearances and the same has been presented in the exhibit below –

S.N.	Approval / Consent	Authority	Reference No.	Remarks
1	Certificate of Incorporation	ROC	U24304UP2022PTC166562	Received
2	Registration under GST	Commercial Tax department	23AAKCC4232D1Z6	Received

S.N.	Approval / Consent	Authority	Reference No.	Remarks
3	Non-Agricultural Conversion Certificate	Tehsildar		Received
4	Environmental Clearance (EC)	MOEF, GOI	EC22A022MP189436	Received
5	DFPD Approval	Dept. of Food and Public Distribution, GOI		Received
6	Power Connection	Local Authority		Received
7	State water allocation	Central Ground Water Department		Applied
8	PESO (Initial Approval)	Jt. commissioner of Explosives	A/P/HQ/MP/15/3218 (P610688)	Received
9	Consent to Establishment (CTE)	Madhya Pradesh PCB	CTE-60583	Received
10	Industrial Entrepreneurs Memorandum	Ministry of Commerce & Industry	636/N/SIA/IMO/2024	Received
11	NOC for Ground Water Extraction	Department of Water Resources, River Development & Ganga Rejuvenation Central Ground Water Authority	CGWA/NOC/IND/ORIG/2024/19975	Received
12	Consent to Operation (CTO)	Member Secretary, PCB		Awaited; To be obtained before COD
13	Factory License	Chief Inspector of Factories		Awaited; To be obtained before COD
14	Boiler License	Chief Inspector of Boilers		Awaited; To be obtained before COD
15	Licenses / Registration under various labour laws including 15 Contract, Labour Regulation & Abolition Act	Appropriate Authority/ Labour Department		Awaited; To be obtained before COD

Source: CIPL and NITCON

Site Visit Observations

The team member from NITCON, visited the site located at Rakba Nos. 38/1/3(S), 38/1/1(S), 38/2(S), and 38/3(S), Village Mahewa, Teonthar Tehsil, Rewa District, Madhya Pradesh, India, on 19th January 2026. The observations made during the site visit are listed below –

- The project site is enclosed on all sides by a brick masonry boundary wall supported on RCC columns and topped with barbed wire fencing.

- In addition to the existing facilities, adequate vacant land is available within the enclosed premises for future expansion.
- The Company is in the process of installing two weighbridges at the entry gate; during the site visit, one weighbridge was observed to be fully installed, while installation of the second is under progress.
- A chimney with a height of approximately 63 meters has been erected at the site.
- A 250 kVA transformer has been installed near the steam turbine area.
- On an overall basis, the Company has achieved approximately 75% of physical progress with the total project.
- The water reservoir catering to the steam turbine requirements has been completed.
- Construction activities for the staff and administrative building are currently underway.
- A temporary power connection of 50 kVA connected load has been availed through a 33 kV transmission line for construction purposes.
- Additionally, two borewells equipped with pumps of 5 HP capacity each have been installed and are being utilized for groundwater extraction to meet construction and sanitation water requirement.

Site Visit Photographs

Based on the site visit, the photos captured has been exhibited below –






Source: Site Visit

Implementation Schedule

Based on the site visit observation and discussion with the executive of the Company it is understood the execution of the project is running smooth and will achieve commercial operation by 30th June 2026 consider financial closure as on 31st March 2026 for additional TL.

Business Analysis

SWOT Analysis

Strength	Weakness
• The project site is well connected and provides easy access to raw materials as well as the skilled and unskilled manpower required for plant operations.	• The main weakness of the proposed Project is that the basic possible feedstocks are agricultural produce and hence the dependence on climatic condition.

• The financial position of the Promoters is sound, and parties involved will be in position to infuse the necessary equity capital required for setting up the Project. • There are multiple incentives and policies which have been announced by the Central and State Governments, which will be available for the proposed Project, increasing the overall viability of the Project in long run. • The technology proposed to be implemented is flexible in nature and will be able to utilise maize as alternative feedstock for manufacturing bioethanol. • Ample availability of basic feedstock i.e. broken/ damaged rice and maize/ corn in the area. • There is ample availability of land in the region for possible future expansion for the Project, in case any expansion in planned.	*During drought or flood scenario, the project can focus on receiving feedstock from other areas that its catchment area.* • The project's pricing remains subject to regulation by the Central Government and OMCs, limiting the Company's control over selling prices.
Opportunity	**Threat**
• The focus of the Central Government to move towards renewable sources of energy, with favourable policies to promoter manufacturing of bioethanol across the country. • The growing demand for the finished products proposed to be manufactured by the proposed Project i.e. Bioethanol and DDGS.	• The generic threat of slowdown/ recessionary trend in the international and domestic markets on account of pandemic like scenario recently experienced by the country and internationally. • Sighting opportunity, there are a series of new entrants entering the segment of manufacturing bioethanol from grain as well as from molasses/ sugar syrup, which increases the competition for the proposed Project.

Risk Analysis

The risk analysis, allocation and mitigation specific to the project are shown the following table -

Key Risk	Risk Carrier	Weakness
Experience and Capability	CIPL	The Consultants note that while the promoters of CIPL possess extensive experience in the infrastructure sector, they have limited direct experience in the ethanol industry. To address this gap, the Company has engaged an industry

Key Risk	Risk Carrier	Weakness
		professional with approximately 20 years of relevant experience. Further, CIPL has appointed M/s VAPCO, one of India's emerging ethanol manufacturing technology suppliers, as the EPC contractor for the proposed project, thereby mitigating the technical and execution-related risks. Additionally, the management of CIPL proposes to induct further experienced professionals from the ethanol industry to manage the day-to-day operations of the project. Hence experience related risk is not envisaged for the proposed Project.
Funding Risk	CIPL	As discussed earlier in this report, the project is at an advanced stage of implementation, with approximately 75% of the total works completed. Against this progress, the Company has an existing term loan – 1 sanction of Rs. 115.57 Crores from IREDA, and the promoters have infused equity of approximately Rs. 28.89 Crores into the project as on 17th January 2026. However, as discussed, the project has experienced delays primarily due to underestimation of certain cost components, resulting in a cost overrun of approximately Rs. 27.39 Crores. To bridge this gap, the Company proposes to raise an additional term loan – 1 of Rs. 21.91 Crores along with a further equity infusion of Rs. 5.48 Crores by the promoters. Accordingly, a funding-related risk persists for the project until financial closure for the proposed additional term loan is achieved.
Time Over-run	CIPL	Based on site visit observations and discussions with the key executives of the Company, it is understood that the project is expected to be completed within a period of 3–4 months. However, after allowing a cushion period of two months, the Consultants have assumed a construction period of 3 months from the date of financial closure, considered as 31st March 2026. Further, in the event of any time overrun arising due to reasons beyond the control of CIPL, the resultant financial

Key Risk	Risk Carrier	Weakness
		impact shall be borne by the promoters through their own sources of funds.
Cost Over-run	CIPL	The Project Cost as estimated is based on the inputs received from VAPCO, which has ample amount of experience in undertaking similar Projects successfully in recent past. Further, the Consultants have considered a contingency of 4.00% on the hardware cost, so as to take care of any escalation in commodity prices during the course of implementation of the Project. Hence cost overrun related risk is not associated with the Project. However, in case of any cost overrun on account of reasons out of control of CIPL, the increased cost will have to be borne by the Promoters from own sources of funds.
Statutory Approvals	CIPL	The Consultants note that the Project is in its advance stage of implementation and has received necessary approvals and clearances for the Project. Hence statutory approvals and clearances related risk is not associated with the Project at the moment.
Demand Risk	CIPL	Based on the market assessment undertaken by the Consultants and considering the fact that manufacturing of Ethanol falls under the Thrust Sector in India and in state of Madhya Pradesh and hence there is no perceived demand related risk associated with the Project. Further, it is noted that the market for products like DDGS is in nascent stages and catching up at a rapid rate and finding more application/ usages areas like marine feedstock, chicken/ poultry feed, piggeries apart from regular cattle feed. Hence, demand risk is not envisaged for the Project.
Pricing Risk	CIPL	As discussed in previous sections of the report, it is understood that the prices of fuel grade Ethanol are regulated by the Central Government and OMC and hence there is a pricing related risk associated with the Project.

Key Risk	Risk Carrier	Weakness
		This becomes evident while undertaking the sensitivity analysis on the Project, considering 5% increase in raw material prices and 5% decrease in selling prices, under both circumstances, project remains viable, but debt repayment capacity of the Project reduces slightly. However, here it should be noted that the Government and OMCs have taken due care in formulating the pricing mechanism for Ethanol, which has been proactive and selling prices have increased over the year, taking inflation into consideration.
Technology Risk	CIPL	The distillation technology to produce bioethanol from grains like rice and maize is an established technology and there are a series of such projects already operational across the globe and in India. In fact, VAPCO, which is appointed as EPC Contractor for the Project has rich experience in providing the technology for multiple plants already operational in India. *Hence Technology related risk not associated with the project.*
Force Majeure	CIPL/ Insurer	The lenders/ investors may insist upon the Company to take adequate insurance cover for insurable Force Majeure risks.

Business Plan

Product

As discussed in previous sections of the report, CIPL is setting up 120 KLD grain-based distillery project, which will be capable of manufacturing the following Products –

1. Fuel grade Ethanol/ Bio Ethanol
2. Distillers Dried Grain Soluble (DDGS)
3. Fusel Oil/ Impure Spirit
4. Carbon Dioxide/ Dry Ice

Price

1. Ethanol – As discussed in previous sections of the report, the price of fuel grade Ethanol is regulated by the Government. This has been the trend in the country since Independence in 1947, whereby the fuel like petrol, diesel, natural gas prices have always been regulated by the Governments. The trend is expected to continue in foreseeable future as well.
2. DDGS – The prices of DDGS are determined by the markets and hence the Company will have to follow the practice of marking the product price to market.
3. Fusel Oil/ Impure Spirit – The prices of fusel oil/ impure spirit are determined by the markets and hence the Company will have to follow the mark to market pricing policy for sales of fusel oil/ impure spirit we well.
4. Carbon Dioxide – The Company will have to market carbon dioxide to the nearby beverage manufacturing units. Surplus of carbon di-oxide produced will have to be purged.

Promotion/ Sales Strategy

The Project is proposed to follow the following promotion/ sales strategy for each of the products –

1. Ethanol – The Project/ CIPL will be interacting with various Oil Marketing Companies and apply to these, which will purchase the finished products as per the requirement from time to time.
2. DDGS – As discussed in previous sections of the report, DDGS is primarily used by Animal/ Poultry / Fish feed manufacturing units for preparation of feed for animal/ poultry/ fish and can be sold/supplied to the same.
3. Fusel Oil/ Impure Spirit – Fusel Oil/ Impure Spirit have various industrial application and there are multiple sub industrial units located in nearby area, and these will be targeted by the sales team from the Company.
4. Carbon Dioxide – The Company will have to deploy own sales team for marketing of CO2 to various carbonated soft-drink, industrial application users.

Sales Team

The Company will be appointing sales team for marketing/ sales of the first 3 products in the list discussed above. The provision for salaries of the sales team and their travelling cost has been accounted for by the Consultants while undertaking the financial assessment for the Project.

Project Cost

The revised project cost of the project has been exhibited below –

All Figures in Rs. Crore						
Particulars	Original Cost	Already incurred	Increased Cost	31-Mar-26	31-Mar-27	Total
Land and Land Development	1.00	0.97	-	1.00	-	1.00
Building and Civil Works	20.88	18.58	2.87	21.85	1.90	23.75
Plant and Machinery Cost	102.58	96.55	10.25	98.61	14.22	112.83
Miscellaneous Fixed Assets	0.50	0.50	-	0.44	0.07	0.50
Preliminary and Pre-operative	3.00	3.00	-	2.61	0.39	3.00
Interest During Construction	11.54	9.88	6.64	13.05	3.47	16.52
Contingency	4.96	0.72	-	1.00	3.96	4.96
Margin Money for Working Capital			9.29		9.29	9.29
Total Project Cost	**144.46**	**130.19**	**29.05**	**138.56**	**33.30**	**171.85**

Source: CIPL and NITCON Estimates

***Note -** Based on the above table, it is noted that the original total project cost was estimated at Rs. 144.46 Crore. However, during implementation, the project has witnessed cost overruns primarily on account of underestimation of certain capital cost components and an increase in Interest During Construction (IDC) arising from time overruns. Consequently, the revised total project cost has increased to Rs. 171.85 Crore.

As on 17th January 2026, the Company incurred capital expenditure of approximately Rs. 130.19 Crore towards project implementation. The balance project cost, including the increased cost components, is proposed to be funded through additional term loan assistance from the lenders, along with the stipulated promoter contribution, as per the revised funding plan.

Means of Finance

On account of revised project cost the means of finance for the project has been exhibited in the table below –

All Figures in Rs. Crore					
Particulars	Original Cost	Already incurred	31-Mar-26	31-Mar-27	Total
Promoters Contribution	28.89	28.89	28.89	5.48	34.37
Term Loan 1	115.57	101.30	109.67	5.90	115.57
Term Loan 2				21.91	21.91
Total Means of Finance	**144.46**	**130.19**	**138.56**	**33.30**	**171.85**

Source: CIPL and NITCON Estimates

Conclusion

Based on the technical assessment the Consultants has made following conclusion as below –

- The Company is setting up grain-based distillery of 120 KLD along with a co-generation plant at Rakba Nos. 38/1/3(S), 38/1/1(S), 38/2(S), and 38/3(S), Village Mahewa, Teonthar Tehsil, Rewa District, Madhya Pradesh.

- The Company has already acquired 76,200 Sq.M. or 7.62 Hectares of land for the project, which is under its possession.
- The company has appointed VAPCO Engineers Private Limited as EPC contractor for the entire project, further the Company has procured steam turbine for co-generation power plant from Triveni Turbine Limited.
- The Company is installing a 3 MW captive co-generation plant designed to utilize rice husk as the primary fuel source.
- The Fresh water requirement in the proposed project will be around 1300KL per day.
- The Company proposes setting up a Water Treatment Plant at the site to handle the water discharged from the process.
- The proposed ethanol manufacturing plant will require around 150 personnel for the successful operation of the plant.
- The project site is enclosed on all sides by a brick masonry boundary wall supported on RCC columns and topped with barbed wire fencing.
- On an overall basis, the Company has achieved approximately 75% of physical progress with the total project.
- Two borewells equipped with pumps of 5 HP capacity each have been installed at site and are being utilized for groundwater extraction to meet construction and sanitation water requirement.

Financial Assessment

Introduction

As discussed in previous section of the report, the Company is setting up 120 KLD grain-based distillery project. Based on the assessment of the feedstock availability and technical parameters, Consultants have considered that proposed grain-based Ethanol plant will operate on 100% broken rice as raw material and the same is considered as base case scenario for the financial evaluation of the proposed project.

The Consultants have evaluated the viability of the Project based on parameters like NPV, IRR, WACC and DSCR. The Cost Stream and the Revenue Stream as expected from the Project have been discussed in this chapter, before moving on to working capital assessment and discussion the means of finance and financial ratios, as expected.

Capacity and Capacity Utilisation

Based on the design data as provided by Company's EPC contractor i.e. VAPCO Engineers Private Limited, it is understood that the proposed Project will have an input feed requirement of 263.16Tons of broken/ damages rice, with starch content of 68-70% and based on the same, the gross ethanol output of the proposed Project has been ascertained. Based on these assumptions, the capacity and utilisation level expected from the Project has been provided in the table below –

Particulars	Unit	31-Mar-27	31-Mar-28	31-Mar-29	31-Mar-30	31-Mar-31	31-Mar-32	31-Mar-33	31-Mar-34
Days in Year	**Days**	**274**	**365**	**365**	**365**	**365**	**365**	**365**	**365**
Distillery Unit									
Alcohol Yield per Ton of RM	Litres/ Ton	456	456	456	456	456	456	456	456
Daily Max. RM Feed	Tons	263.16	263.16	263.16	263.16	263.16	263.16	263.16	263.16
Installed Capacity per Day	KL	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00
Annual Operational Working Days	Days	248	330	330	330	330	330	330	330
Annual Operational Capacity	KLPA	29,727	39,600	39,600	39,600	39,600	39,600	39,600	39,600
Capacity Utilisation Level	%	80%	80%	85%	85%	85%	85%	85%	85%

Source: CIPL and NITCON Estimates

While applying the capacity utilization levels, the Consultants had considered the following –

1. The first year of operations will be FY 2026-27, when the plant will be operational from June 2026 and hence operational days available will be 248 days for distilling unit.
2. The peak utilisation level for distillery section has been considered at 85%, based on 330 days of operations.
3. The peak utilisation level of 85% has been assumed to take into consideration the schedule and unscheduled maintenance shutdown, so as to keep the projection conservative. If preventive

maintenance philosophy is followed at the units, then slightly higher capacity utilisation might also be achieved.

Product Mix

Here it should be noted that when the grain is processed to extract alcohol, then additional by products are also produced parallelly. The by-product generation as estimated has been presented in the table below –

Particulars	Unit	Value
Ethanol	% of RM	45.60%
DDGS	% of RM	19.00%
Fusel Oil/ Impure Spirit	% of RM	0.48%
CO2	% of RM	22.30%
Waste/ Gases/ Losses (incl. Moisture)	% of RM	12.62%

Source: CIPL and NITCON Estimates

Here it should be noted that the losses in the output including the following have been covered as part of 12.62% losses as indicated in the table above –

1. Moisture loss while handling broken rice
2. Any pilferages
3. Loss in form of gases
4. Waste or foreign particles in the raw material etc.

Production Level

Based on the capacity, utilisation level and product mix as discussed above, the production level of the finished goods has been provided in the table below –

Particulars	Unit	31-Mar-27	31-Mar-28	31-Mar-29	31-Mar-30	31-Mar-31	31-Mar-32	31-Mar-33	31-Mar-34
Ethanol	KL	23,782	31,680	33,660	33,660	33,660	33,660	33,660	33,660
DDGS	Tons	9,909	13,200	14,025	14,025	14,025	14,025	14,025	14,025
Fusel Oil/ Impure Spirit	KL	250	333	354	354	354	354	354	354
CO2	KL	11,630	15,493	16,461	16,461	16,461	16,461	16,461	16,461

Source: NITCON Estimates

Stock Movement

Considering the stock movement for work in progress and finished goods of 2 days and 30 days respectively, the quantity available for sales has been presented in the table below –

Particulars	Unit	31-Mar-27	31-Mar-28	31-Mar-29	31-Mar-30	31-Mar-31	31-Mar-32	31-Mar-33	31-Mar-34
Ethanol									
Opening Balance	KL	-	3,072	3,072	3,264	3,264	3,264	3,264	3,264
Production	KL	23,782	31,680	33,660	33,660	33,660	33,660	33,660	33,660
Closing Stock	KL	3,072	3,072	3,264	3,264	3,264	3,264	3,264	3,264

Particulars	Unit	31-Mar-27	31-Mar-28	31-Mar-29	31-Mar-30	31-Mar-31	31-Mar-32	31-Mar-33	31-Mar-34
Stock Available for Sale	KL	20,710	31,680	33,468	33,660	33,660	33,660	33,660	33,660
DDGS									
Opening Balance	Tons	-	1,280	1,280	1,360	1,360	1,360	1,360	1,360
Production	Tons	9,909	13,200	14,025	14,025	14,025	14,025	14,025	14,025
Closing Stock	Tons	1,280	1,280	1,360	1,360	1,360	1,360	1,360	1,360
Stock Available for Sale	Tons	8,629	13,200	13,945	14,025	14,025	14,025	14,025	14,025
Fusel Oil/ Impure Spirit									
Opening Balance	KL		32	32	34	34	34	34	34
Production	KL	250	333	354	354	354	354	354	354
Closing Stock	KL	32	32	34	34	34	34	34	34
Stock Available for Sale	KL	218	333	352	354	354	354	354	354
CO2									
Opening Balance	KL		1,502	1,502	1,596	1,596	1,596	1,596	1,596
Production	KL	11,630	15,493	16,461	16,461	16,461	16,461	16,461	16,461
Closing Stock	KL	1,502	1,502	1,596	1,596	1,596	1,596	1,596	1,596
Stock Available for Sale	KL	10,128	15,493	16,367	16,461	16,461	16,461	16,461	16,461

Source: NITCON Estimates

Sales Quantity, Selling Prices and Sales Realisation

Sales Quantity

Based on the stock movement, the sales quantity available for sales has been presented in the table below –

Particulars	Unit	31-Mar-27	31-Mar-28	31-Mar-29	31-Mar-30	31-Mar-31	31-Mar-32	31-Mar-33	31-Mar-34
Ethanol	KL	20,710	31,680	33,468	33,660	33,660	33,660	33,660	33,660
DDGS	Tons	8,629	13,200	13,945	14,025	14,025	14,025	14,025	14,025
Fusel Oil/ Impure Spirit	KL	218	333	352	354	354	354	354	354
CO2	KL	10,128	15,493	16,367	16,461	16,461	16,461	16,461	16,461

Source: NITCON Estimates

Selling Prices

The Consultants have considered the prevailing market prices for all the products and the selling prices considered for financial evaluation have been provided in the exhibit below –

Particulars	Unit	31-Mar-27	31-Mar-28	31-Mar-29	31-Mar-30	31-Mar-31	31-Mar-32	31-Mar-33	31-Mar-34
Ethanol	Rs./ Litre	72.41	74.22	76.08	72.41	74.22	76.08	72.41	74.22
DDGS	Rs./ Kg	30.55	31.31	32.09	30.55	31.31	32.09	30.55	31.31

Particulars	Unit	31-Mar-27	31-Mar-28	31-Mar-29	31-Mar-30	31-Mar-31	31-Mar-32	31-Mar-33	31-Mar-34
Fusel Oil/ Impure Spirit	Rs./ Litre	54.31	55.67	57.06	54.31	55.67	57.06	54.31	55.67
CO2	Rs./ Litre	2.83	2.90	2.97	2.83	2.90	2.97	2.83	2.90
Annual Increment	%		2.50%	2.50%	2.50%	2.50%	2.50%	2.50%	2.50%

Source: NITCON Research

The selling price as considered for the purpose of financial evaluation has been based as –

- Ethanol – the selling price as considered is based on latest price as declared by the Government of India.
- DDGS – the selling price of DDGS has been considered as of prevailing prices of DDGS in the markets.
- Fusel Oil/ Impure Spirit – the selling price of fusel oil has been considered as the prevailing prices of fusel oil in the domestic markets.
- CO2 – the selling price of CO2 has been considered at prevailing market prices of CO2 in the domestic markets.

Further it should be noted that historically the prices of fuel grade ethanol, as set by Government from time to time have been in sync with the increase in raw material prices. However, the Consultants believe that the cost has been 81rationalized and in future the same is expected to grow at around 2.50% per annum and the same has been considered while undertaking financial evaluation.

Sales Realization

Based on the sales quantity and selling prices as discussed above, the sales realization for the Project as estimated has been presented below –

All Figures in Rs. Crore									
Particulars	**Unit**	**31-Mar-27**	**31-Mar-28**	**31-Mar-29**	**31-Mar-30**	**31-Mar-31**	**31-Mar-32**	**31-Mar-33**	**31-Mar-34**
Ethanol	Rs. Crore	132.54	207.82	225.04	231.99	237.79	243.73	249.83	256.07
DDGS	Rs. Crore	23.30	36.53	39.56	40.78	41.80	42.84	43.91	45.01
Fusel Oil/ Impure Spirit	Rs. Crore	1.05	1.64	1.78	1.83	1.88	1.92	1.97	2.02
CO2	Rs. Crore	2.53	3.97	4.30	4.43	4.54	4.66	4.77	4.89
Total Sales	**Rs. Crore**	**159.42**	**249.96**	**270.67**	**279.03**	**286.01**	**293.16**	**300.49**	**308.00**

Source: NITCON Estimates

Variable Costs

Raw Material Cost

As discussed previously, the main raw material for the project will be Rice/broken rice. The same will be procured from the nearby area and State. The raw material prices as considered as the prevailing prices of raw material (broken rice) in mandi in these regions. The raw material annual requirement and cost for the Project have been presented in the table below –

Particulars	Unit	31-Mar-27	31-Mar-28	31-Mar-29	31-Mar-30	31-Mar-31	31-Mar-32	31-Mar-33	31-Mar-34
Raw Material Cost									
Gains Consumed	Tons	52,153	69,474	73,816	73,816	73,816	73,816	73,816	73,816
Purchase Price	Rs./ Kg	22.75	23.32	23.90	24.50	25.11	25.74	26.38	27.04
Annual Increment	%		2.50%	2.50%	2.50%	2.50%	2.50%	2.50%	2.50%
Raw Material Consumed	**Rs. Crore**	**118.65**	**162.00**	**176.43**	**180.84**	**185.36**	**190.00**	**194.75**	**199.62**
Opening Balance of Raw Material	Rs. Crore	-	25.98	26.63	29.00	29.73	30.47	31.23	32.01
Purchase of Raw Material	**Rs. Crore**	**144.63**	**162.65**	**178.80**	**181.57**	**186.11**	**190.76**	**195.53**	**200.42**
Closing Balance of Raw Material	Rs. Crore	25.98	26.63	29.00	29.73	30.47	31.23	32.01	32.81

Source: NITCON Estimates

Here it should be noted that the Consultants have considered an annual increase in the raw material cost at 2.50% per annum from the second year of operations, in line with the annual increase considered for the selling prices, as both are expected to move in tandem.

Stores and Consumables

The stores and consumables (which includes chemicals, enzymes etc.) as considered have been provided in the exhibit below –

Particulars	Unit	31-Mar-27	31-Mar-28	31-Mar-29	31-Mar-30	31-Mar-31	31-Mar-32	31-Mar-33	31-Mar-34
Urea									
Urea Consumption	Kg/ KL	4.00	4.00	4.00	4.00	4.00	4.00	4.00	4.00
Annual Consumption	Kg	95,127	1,26,720	1,34,640	1,34,640	1,34,640	1,34,640	1,34,640	1,34,640
Purchase Price	Rs./ Kg	16.00	16.40	16.81	17.23	17.66	18.10	18.56	19.02
Cost of Urea	**Rs. Crore**	**0.15**	**0.21**	**0.23**	**0.23**	**0.24**	**0.24**	**0.25**	**0.26**
Liquozyme									
Liquozyme Consumption	Kg/ KL	0.70	0.70	0.70	0.70	0.70	0.70	0.70	0.70
Annnual Consumption	Kg	16,647	22,176	23,562	23,562	23,562	23,562	23,562	23,562
Purchase Price	Rs./ Kg	575.00	589.38	604.11	619.21	634.69	650.56	666.82	683.49
Cost of Liquozyme	**Rs. Crore**	**0.96**	**1.31**	**1.42**	**1.46**	**1.50**	**1.53**	**1.57**	**1.61**

Particulars	Unit	31-Mar-27	31-Mar-28	31-Mar-29	31-Mar-30	31-Mar-31	31-Mar-32	31-Mar-33	31-Mar-34
Saccharifying Enzyme									
S Enzyme Consumption	Kg/ KL	1.30	1.30	1.30	1.30	1.30	1.30	1.30	1.30
Annual Consumption	Kg	30,916	41,184	43,758	43,758	43,758	43,758	43,758	43,758
Purchase Price	Rs./ Kg	565.00	579.13	593.60	608.44	623.65	639.25	655.23	671.61
Cost of S Enzyme	**Rs. Crore**	**1.75**	**2.39**	**2.60**	**2.66**	**2.73**	**2.80**	**2.87**	**2.94**
Antifoam Oil									
Antifoam Oil Consumption	Kg/ KL	2.50	2.50	2.50	2.50	2.50	2.50	2.50	2.50
Annual Consumption	Kg	59,454	79,200	84,150	84,150	84,150	84,150	84,150	84,150
Purchase Price	Rs./ Kg	50.00	51.25	52.53	53.84	55.19	56.57	57.98	59.43
Cost of Antifoam Oil	**Rs. Crore**	**0.30**	**0.41**	**0.44**	**0.45**	**0.46**	**0.48**	**0.49**	**0.50**
NaOH (Caustic Lye)									
NaOH Consumption	Kg/ KL	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00
Annual Consumption	Kg	2,85,380	3,80,160	4,03,920	4,03,920	4,03,920	4,03,920	4,03,920	4,03,920
Purchase Price	Rs./ Kg	28.00	28.70	29.42	30.15	30.91	31.68	32.47	33.28
Cost of NaOH	**Rs. Crore**	**0.80**	**1.09**	**1.19**	**1.22**	**1.25**	**1.28**	**1.31**	**1.34**
Yeast									
Yeast Consumption	Kg/ KL	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30
Annual Consumption	Kg	7,135	9,504	10,098	10,098	10,098	10,098	10,098	10,098
Purchase Price	Rs./ Kg	705.00	722.63	740.69	759.21	778.19	797.64	817.58	838.02
Cost of Yeast	**Rs. Crore**	**0.50**	**0.69**	**0.75**	**0.77**	**0.79**	**0.81**	**0.83**	**0.85**
Evaporation CIP HNO3 (67%)									
HNO3 Consumption	Kg/ KL	4.00	4.00	4.00	4.00	4.00	4.00	4.00	4.00
Annual Consumption	Kg	95,127	1,26,720	1,34,640	1,34,640	1,34,640	1,34,640	1,34,640	1,34,640
Purchase Price	Rs./ Kg	18.00	18.45	18.91	19.38	19.87	20.37	20.87	21.40
Cost of HNO3	**Rs. Crore**	**0.17**	**0.23**	**0.25**	**0.26**	**0.27**	**0.27**	**0.28**	**0.29**

Particulars	Unit	31-Mar-27	31-Mar-28	31-Mar-29	31-Mar-30	31-Mar-31	31-Mar-32	31-Mar-33	31-Mar-34
Evaporation CIP Caustic (48%)									
Caustic Acid Consumption	Kg/ KL	6.00	6.00	6.00	6.00	6.00	6.00	6.00	6.00
Annual Consumption	Kg	1,42,690	1,90,080	2,01,960	2,01,960	2,01,960	2,01,960	2,01,960	2,01,960
Purchase Price	Rs./ Kg	24.00	24.60	25.22	25.85	26.49	27.15	27.83	28.53
Cost of Caustic Acid	**Rs. Crore**	**0.34**	**0.47**	**0.51**	**0.52**	**0.54**	**0.55**	**0.56**	**0.58**
Waste Water Treatment Cost									
Waste Water Treatment Cost	Rs./ KL	0.60	0.62	0.63	0.65	0.66	0.68	0.70	0.71
Waste Water Treatment Cost	**Rs. Crore**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**
Other Stores and Consumables									
Stores and Consumables	Rs./ KL	14.00	14.35	14.71	15.08	15.45	15.84	16.24	16.64
Other Stores and Consumables	**Rs. Crore**	**0.03**	**0.05**	**0.05**	**0.05**	**0.05**	**0.05**	**0.05**	**0.06**
Annual Increment	%		2.50%	2.50%	2.50%	2.50%	2.50%	2.50%	2.50%
Total Stores and Consumables	**Rs. Crore**	**5.00**	**6.83**	**7.44**	**7.63**	**7.82**	**8.01**	**8.21**	**8.42**

Source: NITCON Estimates

Here it should be noted that annual increase in the stores and consumables cost for the Project have been considered at 2.50%, which is in sync with the increase in prices of some of the stores and consumables as proposed to be used.

Power and Fuel Cost

As discussed previously, the Company will be operating its own captive power plant for power and steam requirement. The fuel cost for operating the 5 – 5.4 MW power plant has been provided in the exhibit below –

Particulars	Unit	31-Mar-27	31-Mar-28	31-Mar-29	31-Mar-30	31-Mar-31	31-Mar-32	31-Mar-33	31-Mar-34
Power and Fuel Cost									
Boiler Load	Tons/ Hour	26	26	26	26	26	26	26	26
Working Hours	Hours	24	24	24	24	24	24	24	24
Steam Generated	Tons/ Day	499	499	530	530	530	530	530	530

Particulars	Unit	31-Mar-27	31-Mar-28	31-Mar-29	31-Mar-30	31-Mar-31	31-Mar-32	31-Mar-33	31-Mar-34
Fuel Required per Ton of Steam	Tons/ Ton	0.22	0.22	0.22	0.22	0.22	0.22	0.22	0.22
Fuel Requirement Per Day	Tons	111	111	118	118	118	118	118	118
Cost of Fuel	Rs./ Ton	4,500	4,613	4,728	4,846	4,967	5,091	5,219	5,349
Annual Increment	%		2.50%	2.50%	2.50%	2.50%	2.50%	2.50%	2.50%
Annual Power and Fuel Cost	**Rs. Crore**	**9.89**	**13.51**	**15.63**	**16.02**	**16.42**	**16.83**	**17.25**	**17.68**

Source: NITCON Estimates

Here it should be noted that rice husk will be the primary fuel for the Project. The price of rice husk is expected to increase at about 2.50% per annum in accordance with historical trend and the same has been considered while undertaking financial assessment.

Further use of DG set for overhead use is also expected to be minimal and the cost for same has been accounted for in other manufacturing expenses costs.

Water Cost

Water will be required as part of process and also for overhead requirements. The cost of drawing water from borewell has been considered by the Consultants and has been presented in the exhibit below –

Particulars	Unit	31-Mar-27	31-Mar-28	31-Mar-29	31-Mar-30	31-Mar-31	31-Mar-32	31-Mar-33	31-Mar-34
Water Cost									
Water	KL/ KL of ENA	4.00	4.00	4.00	4.00	4.00	4.00	4.00	4.00
Total Annual Water	KL	95,127	1,26,720	1,34,640	1,34,640	1,34,640	1,34,640	1,34,640	1,34,640
Cost of Water	Rs./ KL	20.00	20.50	21.01	21.54	22.08	22.63	23.19	23.77
Annual Increment	%		2.50%	2.50%	2.50%	2.50%	2.50%	2.50%	2.50%
Annual Water Cost	**Rs. Crore**	**0.19**	**0.26**	**0.28**	**0.29**	**0.30**	**0.30**	**0.31**	**0.32**

Source: NITCON Estimates

An annual increase of 2.50% on the water cost has been considered, however the same is unlikely, as the water cost is generally revised once in 2-3 years' time frame, but the increase as been considered annually to take into account the price increases when it happens.

Repairs and Maintenance Cost

The repairs and maintenance cost of the Project has been considered as a percentage of gross fixed assets. The same as estimated has been tabulated as –

Particulars	Unit	31-Mar-27	31-Mar-28	31-Mar-29	31-Mar-30	31-Mar-31	31-Mar-32	31-Mar-33	31-Mar-34
Repairs and Maintenance Cost	% of GFA	1.00%	1.00%	1.00%	1.00%	1.00%	1.00%	1.00%	1.00%
Annual Increment	%		2.50%	2.50%	2.50%	2.50%	2.50%	2.50%	2.50%
Repairs and Maintenance Cost	**Rs. Crore**	**1.22**	**1.63**	**1.63**	**1.67**	**1.71**	**1.75**	**1.79**	**1.84**

Source: NITCON Estimates

The repairs and maintenance costs have been considered to increase annually by 2.50%, in line with the increase in other costs.

Material Handling and Transport

The Consultants note that the proposed Project will be required to transport raw material from various mandis and warehouses of FCI within a catchment area of 500 Km from the location of the Project. Railway transport cost for the same has been considered for the transport of raw material, utilising 60–65-ton capacity rakes. Considering all these aspects the material handling and transport cost for the Project as estimated has been presented in the table below –

Particulars	Unit	31-Mar-27	31-Mar-28	31-Mar-29	31-Mar-30	31-Mar-31	31-Mar-32	31-Mar-33	31-Mar-34
Raw Material Handling									
RM Handled	Tons	52,153	69,474	73,816	73,816	73,816	73,816	73,816	73,816
Average Freight Cost	Rs./ Ton	650	666	683	700	717	735	754	773
Loading/ Unloading, Last Mile	Rs./ Ton	500	513	525	538	552	566	580	594
Annual Increment	%		2.50%	2.50%	2.50%	2.50%	2.50%	2.50%	2.50%
Raw Material Handling	**Rs. Crore**	**6.00**	**8.19**	**8.92**	**9.14**	**9.37**	**9.60**	**9.84**	**10.09**
Finished Goods Transport									
Finished Goods Handled	KL	20,710	31,680	33,468	33,660	33,660	33,660	33,660	33,660
Transportation Cost	Rs./ KL/ Km	2.00	2.05	2.10	2.15	2.21	2.26	2.32	2.38
Average Kilometrage	Km	500	500	500	500	500	500	500	500
Annual Increment	%		2.50%	2.50%	2.50%	2.50%	2.50%	2.50%	2.50%
Finished Goods Transport	**Rs. Crore**	**2.07**	**3.25**	**3.52**	**3.62**	**3.72**	**3.81**	**3.90**	**4.00**
Material Handling and Transport	**Rs. Crore**	**8.07**	**11.44**	**12.43**	**12.77**	**13.09**	**13.41**	**13.75**	**14.09**

Source: NITCON Estimates

The Consultants note that the transportation cost of finished goods is generally borne by the OMC (single side), however considering the option, that the Project might require transportation of Ethanol from the unit to nearest blending depots or refineries, the Consultants have also considered transportation cost using truck and the quotation for the same has been taken from various transport companies. Based on the same, the finished goods transportation cost as estimates is also included in the table above.

Further it should be noted that the material handling and transport cost for the project has been assumed to annual increase at 2.50% level.

Other Manufacturing Expenses

Other manufacturing expenses for the project have been estimated at 1.16% of the sales and subsequently 2.50% annual increase has been considered. The other manufacturing expenses include miscellaneous expenses like training of manpower, ISO certification etc. The other manufacturing expenses as considered have been presented in the table below –

Particulars	Unit	31-Mar-27	31-Mar-28	31-Mar-29	31-Mar-30	31-Mar-31	31-Mar-32	31-Mar-33	31-Mar-34
Other Manufacturing Expense	% of Sales	1.00%	1.00%	1.03%	1.05%	1.08%	1.10%	1.13%	1.16%
Annual Increment	%		2.50%	2.50%	2.50%	2.50%	2.50%	2.50%	2.50%
Other Manufacturing Expense	**Rs. Crore**	**1.59**	**2.50**	**2.77**	**2.84**	**2.91**	**2.99**	**3.06**	**3.14**

Source: NITCON Estimates

Insurance Cost

The insurance cost for the Project has been considered in line with the prevailing market rates at 0.25% of the gross fixed assets.

Particulars	Unit	31-Mar-27	31-Mar-28	31-Mar-29	31-Mar-30	31-Mar-31	31-Mar-32	31-Mar-33	31-Mar-34
Insurance	% of GFA	0.25%	0.25%	0.25%	0.25%	0.25%	0.25%	0.25%	0.25%
Annual Increment	%		2.50%	2.50%	2.50%	2.50%	2.50%	2.50%	2.50%
Insurance	**Rs. Crore**	**0.31**	**0.42**	**0.42**	**0.43**	**0.44**	**0.45**	**0.46**	**0.47**

Source: NITCON Estimates

Manpower Costs

The total manpower required for the Project is estimated to be 150 personnel. The details of the same has been presented in the table below –

Particulars	Unit	31-Mar-27	31-Mar-28	31-Mar-29	31-Mar-30	31-Mar-31	31-Mar-32	31-Mar-33	31-Mar-34
Managing Director	Nos.	0	0	0	0	0	0	0	0
General Manager	Nos.	1	1	1	1	1	1	1	1
Finance Head	Nos.	1	1	1	1	1	1	1	1
Finance Manager	Nos.	2	2	2	2	2	2	2	2

Particulars	Unit	31-Mar-27	31-Mar-28	31-Mar-29	31-Mar-30	31-Mar-31	31-Mar-32	31-Mar-33	31-Mar-34
Finance Executive	Nos.	2	2	2	2	2	2	2	2
Accounts Manager	Nos.	1	1	1	1	1	1	1	1
Accounts Executive	Nos.	2	2	2	2	2	2	2	2
Marketing Manager	Nos.	2	2	2	2	2	2	2	2
Marketing Executive	Nos.	4	4	4	4	4	4	4	4
HR & Admin	Nos.	2	2	2	2	2	2	2	2
Production and Maintenance									
Plant Manager	Nos.	1	1	1	1	1	1	1	1
Store Manager	Nos.	4	4	4	4	4	4	4	4
Quality Control Staff	Nos.	6	6	6	6	6	6	6	6
Manager (Production)	Nos.	6	6	6	6	6	6	6	6
Supervisors/ Foreman	Nos.	14	14	14	14	14	14	14	14
Security Guards	Nos.	14	14	14	14	14	14	14	14
Skilled Operator/ Workers	Nos.	12	12	12	12	12	12	12	12
Semi-Skilled Operator/ Workers	Nos.	32	32	32	32	32	32	32	32
Helpers / Unskilled Workers	Nos.	44	44	44	44	44	44	44	44
Total Manpower Requirement	Nos.	**150**	**150**	**150**	**150**	**150**	**150**	**150**	**150**
Managing Director	Rs./ Month	-	-	-	-	-	-	-	-
General Manager	Rs./ Month	1,50,000	1,53,750	1,57,594	1,61,534	1,65,572	1,69,711	1,73,954	1,78,303
Finance Head	Rs./ Month	90,000	92,250	94,556	96,920	99,343	1,01,827	1,04,372	1,06,982
Finance Manager	Rs./ Month	35,000	35,875	36,772	37,691	38,633	39,599	40,589	41,604
Finance Executive	Rs./ Month	25,000	25,625	26,266	26,922	27,595	28,285	28,992	29,717
Accounts Manager	Rs./ Month	45,000	46,125	47,278	48,460	49,672	50,913	52,186	53,491
Accounts Executive	Rs./ Month	20,000	20,500	21,013	21,538	22,076	22,628	23,194	23,774
Marketing Manager	Rs./ Month	65,000	66,625	68,291	69,998	71,748	73,542	75,380	77,265
Marketing Executive	Rs./ Month	35,000	35,875	36,772	37,691	38,633	39,599	40,589	41,604
HR & Admin	Rs./ Month	25,000	25,625	26,266	26,922	27,595	28,285	28,992	29,717
Production and Maintenance									
Plant Manager	Rs./ Month	1,00,000	1,02,500	1,05,063	1,07,689	1,10,381	1,13,141	1,15,969	1,18,869
Store Manager	Rs./ Month	35,000	35,875	36,772	37,691	38,633	39,599	40,589	41,604

Particulars	Unit	31-Mar-27	31-Mar-28	31-Mar-29	31-Mar-30	31-Mar-31	31-Mar-32	31-Mar-33	31-Mar-34
Quality Control Staff	Rs./ Month	35,000	35,875	36,772	37,691	38,633	39,599	40,589	41,604
Manager (Production)	Rs./ Month	50,000	51,250	52,531	53,845	55,191	56,570	57,985	59,434
Supervisors/ Foreman	Rs./ Month	30,000	30,750	31,519	32,307	33,114	33,942	34,791	35,661
Security Guards	Rs./ Month	12,000	12,300	12,608	12,923	13,246	13,577	13,916	14,264
Skilled Operator/ Workers	Rs./ Month	25,000	25,625	26,266	26,922	27,595	28,285	28,992	29,717
Semi-Skilled Operator/ Workers	Rs./ Month	18,000	18,450	18,911	19,384	19,869	20,365	20,874	21,396
Helpers / Unskilled Workers	Rs./ Month	12,000	12,300	12,608	12,923	13,246	13,577	13,916	14,264
Annual Increase in Salaries	%		2.50%	2.50%	2.50%	2.50%	2.50%	2.50%	2.50%
Benefits and Perks	%	15%	15%	15%	15%	15%	15%	15%	15%
Manpower Cost	**Rs. Crores**	**3.63**	**4.96**	**5.08**	**5.21**	**5.34**	**5.48**	**5.61**	**5.75**

Source: NITCON Estimates

It should be noted that the Consultants have considered 15% benefits and 2.50% annual increase, up and above the base salary as indicated in table above, while assessing the overall manpower cost as part of financial evaluation.

The salaries as considered for the various personnel, are in line with the prevailing salaries for similar position on other similar distillery Projects coming up in India.

Fixed Costs

The fixed costs as considered for the Project, in line with the industry standards has been presented in the table below –

Particulars	Unit	31-Mar-27	31-Mar-28	31-Mar-29	31-Mar-30	31-Mar-31	31-Mar-32	31-Mar-33	31-Mar-34
Administrative Expenses	% of Sales	0.50%	0.50%	0.50%	0.50%	0.50%	0.50%	0.50%	0.50%
Selling and Distribution Expenses	% of Sales	0.75%	0.75%	0.75%	0.75%	0.75%	0.75%	0.75%	0.75%
Administrative Expenses	**Rs. Crore**	**0.80**	**1.25**	**1.35**	**1.40**	**1.43**	**1.47**	**1.50**	**1.54**
Selling and Distribution Expenses	**Rs. Crore**	**1.20**	**1.87**	**2.03**	**2.09**	**2.15**	**2.20**	**2.25**	**2.31**

Source: NITCON Estimates

Administrative Expenses

The administrative expenses for the project have been considered at 0.50% of net sales and include the following –

1. The salaries of the Directors
2. Rents and Duties
3. Communication Expenses
4. Non-operation Insurance cost
5. Audit and Legal Fee, etc.

Selling and Distribution Expenses

The selling and distribution expenses for the project have been estimated at 0.75% of net sales and includes –

1. Salaries of sales team
2. Discounts and rebates offered on by-products
3. Pilferage, etc.

EBDITA and EBDITA Margin

The EBDITA and EBDITA Margin for the Project as estimated has been tabulated as –

All Figures in Rs. Crores								
Particulars	**31-Mar-27**	**31-Mar-28**	**31-Mar-29**	**31-Mar-30**	**31-Mar-31**	**31-Mar-32**	**31-Mar-33**	**31-Mar-34**
Net Sales	159.42	249.96	270.67	279.03	286.01	293.16	300.49	308.00
Other Income	0.65	0.87	0.92	0.93	0.96	0.89	0.51	-
Total Income	**160.07**	**250.84**	**271.59**	**279.96**	**286.96**	**294.04**	**300.99**	**308.00**
Variable Costs								
Raw Material Cost	144.63	162.65	178.80	181.57	186.11	190.76	195.53	200.42
Stores and Consumables	5.00	6.83	7.44	7.63	7.82	8.01	8.21	8.42
Power and Fuel	9.89	13.51	15.63	16.02	16.42	16.83	17.25	17.68
Water	0.19	0.26	0.28	0.29	0.30	0.30	0.31	0.32
Repairs and Maintenance Cost	1.22	1.63	1.63	1.67	1.71	1.75	1.79	1.84
Material Handling and Transport	8.07	11.44	12.43	12.77	13.09	13.41	13.75	14.09
Other Manufacturing Expenses	1.59	2.50	2.77	2.84	2.91	2.99	3.06	3.14
Insurance	0.31	0.42	0.42	0.43	0.44	0.45	0.46	0.47
Wages	3.63	4.96	5.08	5.21	5.34	5.48	5.61	5.75
Total Variable Costs	**174.53**	**204.19**	**224.49**	**228.42**	**234.13**	**239.99**	**245.99**	**252.14**
Opening Balance - WIP	-	1.89	1.62	1.78	1.81	1.86	1.90	1.95
Sub Total	**174.53**	**206.08**	**226.12**	**230.21**	**235.94**	**241.84**	**247.89**	**254.09**
Closing Balance WIP	1.89	1.62	1.78	1.81	1.86	1.90	1.95	2.00
Opening Balance - Finished Goods	-	21.38	18.85	20.72	21.08	21.61	22.15	22.70
Sub Total	**172.64**	**225.84**	**243.18**	**249.11**	**255.17**	**261.55**	**268.09**	**274.79**

All Figures in Rs. Crores								
Particulars	**31-Mar-27**	**31-Mar-28**	**31-Mar-29**	**31-Mar-30**	**31-Mar-31**	**31-Mar-32**	**31-Mar-33**	**31-Mar-34**
Closing Balance - Finished Goods	21.38	18.85	20.72	21.08	21.61	22.15	22.70	23.27
Total Cost of Production	**151.26**	**206.99**	**222.47**	**228.03**	**233.56**	**239.40**	**245.38**	**251.52**
Fixed Costs								
Administrative Expenses	0.80	1.25	1.35	1.40	1.43	1.47	1.50	1.54
Selling and Distribution Expenses	1.20	1.87	2.03	2.09	2.15	2.20	2.25	2.31
Total Fixed Costs	**1.99**	**3.12**	**3.38**	**3.49**	**3.58**	**3.66**	**3.76**	**3.85**
Total Operating Costs	**153.26**	**210.12**	**225.85**	**231.51**	**237.14**	**243.06**	**249.14**	**255.37**
EBDITA	**6.82**	**40.72**	**45.74**	**48.45**	**49.83**	**50.98**	**51.85**	**52.63**
EBDITA Margin	**4.26%**	**16.23%**	**16.84%**	**17.30%**	**17.36%**	**17.34%**	**17.23%**	**17.09%**

Source: NITCON Estimates

From the table above, the EBDITA and EBDITA Margin of the Company has been estimated to be in range of 4.26% to 17.09%, with average being 15.46%. The same is considered to be in line with stand-alone distilleries norms.

Working Capital Assessment

The working capital holding norms have been considered in line with the experience of the Promoters, operating other distilleries.

Particulars	Unit	Value
Raw Material	Days	60
Stores and Consumables	Days	30
Work in Progress	Days	2
Finished Goods	Days	30
Debtors	Days	30
Trade Creditors	Days	30
Expenses Creditors	Days	30
Margin Money	%	25%

Source: NITCON Estimates

Based on the holding norms as discussed above, the working capital requirement of the Project has been presented in the table below –

All Figures in Rs. Crores								
Particulars	**31-Mar-27**	**31-Mar-28**	**31-Mar-29**	**31-Mar-30**	**31-Mar-31**	**31-Mar-32**	**31-Mar-33**	**31-Mar-34**
Raw Material Cost	11.89	14.79	16.25	16.51	16.92	17.34	17.78	18.22

All Figures in Rs. Crores								
Particulars	31-Mar-27	31-Mar-28	31-Mar-29	31-Mar-30	31-Mar-31	31-Mar-32	31-Mar-33	31-Mar-34
Spares and Consumables	0.41	0.62	0.68	0.69	0.71	0.73	0.75	0.77
Work in Progress	1.89	1.62	1.78	1.81	1.86	1.90	1.95	2.00
Finished Goods	21.38	18.85	20.72	21.08	21.61	22.15	22.70	23.27
Debtors	15.94	22.72	24.61	25.37	26.00	26.65	27.32	28.00
Current Assets	**51.51**	**58.60**	**64.04**	**65.46**	**67.10**	**68.77**	**70.49**	**72.25**
Creditors - Trade	11.89	14.79	16.25	16.51	16.92	17.34	17.78	18.22
Creditors - Expense	2.46	3.78	4.15	4.26	4.37	4.48	4.59	4.70
Current Liabilities	**14.35**	**18.56**	**20.41**	**20.77**	**21.28**	**21.82**	**22.36**	**22.92**
Working Capital Gap	**37.16**	**40.04**	**43.63**	**44.70**	**45.81**	**46.96**	**48.13**	**49.33**
Margin Money	9.29	10.01	10.91	11.17	11.45	11.74	12.03	12.33
Bank Borrowing	**27.87**	**30.03**	**32.72**	**33.52**	**34.36**	**35.22**	**36.10**	**37.00**

Source: NITCON Estimates

It is noted from the exhibit above that the margin money for working capital as required for first year of operations i.e. FY 2026-27 is Rs 9.29 Crores, which as per practice has been considered as part of the overall project cost.

Means of Finance

The overall project cost has been estimated at Rs. 171.85 Crores, which is proposed to be funded in a debt equity ratio of 4: 1. Based on the same, the means of finance for the Project are –

All Figures in Rs. Crore					
Particulars	Original Cost	Already incurred	31-Mar-26	31-Mar-27	Total
Promoters Contribution	28.89	28.89	28.89	5.48	34.37
Term Loan 1	115.57	101.30	109.67	5.90	115.57
Term Loan 2				21.91	21.91
Total Means of Finance	**144.46**	**130.19**	**138.56**	**33.30**	**171.85**

It is noted from the table above that the Promoters will bring in Rs. 34.37 Crores in form of equity/ quasi equity/ unsecured loan. Term Loan of Rs. 137.48 Crores has been sought from Banks/ Lending Institutions. The broad covenants for combined term loans facility have been presented in the exhibit below –

Description	Details
Term Loan Amount	Rs. 137.48 Crores
Interest Rate	12.34% (Rate of Interest from Existing TL of IREDA)
Door to Door Tenure	7.25 Years or 87 Months
Implementation Duration	03 months from Financial Closure (assumed as on 31st March 2026)
Date of Commercial Operations	30th June 2026
Post Construction Moratorium	12 Months
Repayment Duration	6 Years and 24 Quarters
Repayment Mode	24 Accelerated Quarterly Instalments
First Instalment	July, 2027
Last Instalment	April, 2033

Other Assumptions

The other assumptions as considered by the Consultants for evaluation have been tabulated below –

Particulars	Unit	Value
Debt Equity Ratio	Ratio	4.00
Cost of Equity	%	14.00%
Cost of Term Loan	%	12.34%
Cost of Working Capital Loan	%	10.00%
Average Income Tax Rate	%	25.57%

Depreciation Rates

The depreciation rates as considered for evaluation purpose has been tabulated below –

Particulars	Unit	Value
Depreciation - The Companies Act - SLM		
Land and Land Development	%	0.00%
Building and Civil Works	%	3.45%
Plant and Machinery Cost	%	4.17%
Miscellaneous Fixed Assets	%	4.17%
Depreciation - Income Tax Act - WDV		
Land and Land Development	%	0.00%
Building and Civil Works	%	10.00%
Plant and Machinery Cost	%	15.00%
Miscellaneous Fixed Assets	%	15.00%

Source: NITCON Estimates

Income Tax Rates

The income tax rate as considered have been presented in the table below –

All Figures in Rs. Crores									
Description	**31-Mar-26**	**31-Mar-27**	**31-Mar-28**	**31-Mar-29**	**31-Mar-30**	**31-Mar-31**	**31-Mar-32**	**31-Mar-33**	**31-Mar-34**
Income Tax Rate									
Basic Rate	22.00%	22.00%	22.00%	22.00%	22.00%	22.00%	22.00%	22.00%	22.00%
Surcharge	10.00%	10.00%	10.00%	10.00%	10.00%	10.00%	10.00%	10.00%	10.00%
Cess	4.00%	4.00%	4.00%	4.00%	4.00%	4.00%	4.00%	4.00%	4.00%
Effective Corporate Tax Rate	25.17%	25.17%	25.17%	25.17%	25.17%	25.17%	25.17%	25.17%	25.17%
MAT Rate									
Basic Rate	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
Surcharge	10.00%	10.00%	10.00%	10.00%	10.00%	10.00%	10.00%	10.00%	10.00%
Cess	4.00%	4.00%	4.00%	4.00%	4.00%	4.00%	4.00%	4.00%	4.00%
MAT	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%

Source: NITCON Estimates

Key Financial Parameters

The key financial parameters of the Project, have been presented in the exhibit below –

All Figures in Rs. Crore								
Description	31-Mar-27	31-Mar-28	31-Mar-29	31-Mar-30	31-Mar-31	31-Mar-32	31-Mar-33	31-Mar-34
Revenue	160.07	250.84	271.59	279.96	286.96	294.04	300.99	308.00
Total Operating Costs	153.26	210.12	225.85	231.51	237.14	243.06	249.14	255.37
Operating Profit	6.82	40.72	45.74	48.45	49.83	50.98	51.85	52.63
Operating Profit Margin	4.26%	16.23%	16.84%	17.30%	17.36%	17.34%	17.23%	17.09%
Contribution	8.81	43.85	49.12	51.94	53.40	54.64	55.61	56.48
Contribution Margin	5.50%	17.48%	18.09%	18.55%	18.61%	18.58%	18.47%	18.34%
BEP Sales	394.54	166.07	152.85	137.37	122.96	106.72	89.36	77.04
BEP Capacity Utilisation	246.48%	66.21%	56.28%	49.07%	42.85%	36.29%	29.69%	25.01%
Cash Break Even	305.47	128.71	116.74	102.17	87.87	71.58	54.02	41.43
Cash Break Even Margin	190.83%	51.31%	42.99%	36.49%	30.62%	24.34%	17.95%	13.45%
Net Profit Margin	-8.06%	5.91%	7.91%	9.37%	8.46%	9.20%	9.93%	10.39%
Equity Share Capital	34.37	34.37	34.37	34.37	34.37	34.37	34.37	34.37
Unsecured loan from Promoters	11.89	15.89	15.89	15.89	15.89	15.89	15.89	-
Reserves and Surplus	-12.90	1.91	23.39	49.62	73.91	100.97	130.86	162.87
Tangible Net Worth (TNW)	33.36	52.17	73.65	99.88	124.17	151.23	181.12	197.24
Term Loan	115.57	106.57	91.57	73.57	53.57	30.57	7.57	-
Debt Equity Ratio	3.46	2.04	1.24	0.74	0.43	0.20	0.04	-
Total Outside Liability (TOL)	201.59	191.46	179.20	159.76	137.31	110.90	83.53	59.92
TOL/ TNW	6.04	3.67	2.43	1.60	1.11	0.73	0.46	0.30
Closing Cash Balance	2.00	2.13	11.65	23.28	29.57	36.31	50.56	72.95
DSCR	-	1.49	1.44	1.49	1.37	1.36	1.52	5.06
Minimum DSCR	1.36							
Average DSCR	2.77							
NPV	137.74							
IRR	22.44%							
Cost of Capital	10.15%							

Source: NITCON Estimates

Sensitivity Analysis

CIPL is evaluating the project based on the feedstock availability, which can be either broken/ damaged rice or maize/ corn. However, based on the market assessment undertaken by the team from NITCON. Considering the same the Consultants have undertaken sensitivity of the project as below –

Scenario	NPV	IRR	WACC	Min DSCR	Avg. DSCR
	Rs. Crores	%	%	Ratio	Ratio
100% Broken Rice	137.74	22.44%	10.15%	1.36	2.77
5% decrease in capacity utilisation	47.85	15.17%	10.33%	1.00	2.03
5% decrease in selling prices	46.54	15.05%	10.34%	1.00	2.02
5% increase in raw material & stores cost	72.73	17.32%	10.28%	1.13	2.24
1% increase in interest rates	124.93	22.36%	10.77%	1.33	2.68
2% increase in interest rates	113.09	22.31%	11.39%	1.31	2.60
10% increase in hardware cost	124.52	20.80%	10.18%	1.32	2.67
100% Broken Rice	137.74	22.44%	10.15%	1.36	2.77

Source: NITCON Analysis

Conclusions

The Consultants conclude the following from the discussions in the chapter –

1. The average EBDITA Margin of the Project is 15.46%, while the average PAT margin is 6.64%.
2. The Net Present Value of the Project is estimated at Rs. 137.74 crore, while the Internal Rate of Return (IIR) at 22.44% is higher than post-tax cost of capital at 10.15%, indicating the Project is financially viable preposition.
3. The average DSCR for the Project has been estimated at 2.77 and the minimum DSCR is 1.36, indicating fair repayment capacity of the Project.

The project remains viable under adverse case scenarios and found to be Techno-Economically and Financially Viable Project.

Annexure 1 – Financial Statements

Profit and Loss Account

All Figures in Rs. Crores								
Particulars	**31-Mar-27**	**31-Mar-28**	**31-Mar-29**	**31-Mar-30**	**31-Mar-31**	**31-Mar-32**	**31-Mar-33**	**31-Mar-34**
Net Sales	159.42	249.96	270.67	279.03	286.01	293.16	300.49	308.00
Interest on DSRA	0.65	0.87	0.92	0.93	0.96	0.89	0.51	-
Total Income	**160.07**	**250.84**	**271.59**	**279.96**	**286.96**	**294.04**	**300.99**	**308.00**
Variable Costs								
Raw Material Cost	144.63	162.65	178.80	181.57	186.11	190.76	195.53	200.42
Stores and Consumables	5.00	6.83	7.44	7.63	7.82	8.01	8.21	8.42
Power and Fuel	9.89	13.51	15.63	16.02	16.42	16.83	17.25	17.68
Water	0.19	0.26	0.28	0.29	0.30	0.30	0.31	0.32
Repairs and Maintenance Cost	1.22	1.63	1.63	1.67	1.71	1.75	1.79	1.84
Material Handling and Transport	8.07	11.44	12.43	12.77	13.09	13.41	13.75	14.09
Other Manufacturing Expenses	1.59	2.50	2.77	2.84	2.91	2.99	3.06	3.14
Insurance	0.31	0.42	0.42	0.43	0.44	0.45	0.46	0.47
Wages	3.63	4.96	5.08	5.21	5.34	5.48	5.61	5.75
Total Variable Costs	**174.53**	**204.19**	**224.49**	**228.42**	**234.13**	**239.99**	**245.99**	**252.14**
Opening Balance - WIP	-	1.89	1.62	1.78	1.81	1.86	1.90	1.95
Sub Total	**174.53**	**206.08**	**226.12**	**230.21**	**235.94**	**241.84**	**247.89**	**254.09**
Closing Balance WIP	1.89	1.62	1.78	1.81	1.86	1.90	1.95	2.00
Opening Balance - Finished Goods	-	21.38	18.85	20.72	21.08	21.61	22.15	22.70
Sub Total	**172.64**	**225.84**	**243.18**	**249.11**	**255.17**	**261.55**	**268.09**	**274.79**
Closing Balance - Finished Goods	21.38	18.85	20.72	21.08	21.61	22.15	22.70	23.27
Total Cost of Production	**151.26**	**206.99**	**222.47**	**228.03**	**233.56**	**239.40**	**245.38**	**251.52**
Fixed Costs								
Administrative Expenses	0.80	1.25	1.35	1.40	1.43	1.47	1.50	1.54
Selling and Distribution Expenses	1.20	1.87	2.03	2.09	2.15	2.20	2.25	2.31
Total Fixed Costs	**1.99**	**3.12**	**3.38**	**3.49**	**3.58**	**3.66**	**3.76**	**3.85**
Total Operating Costs	**153.26**	**210.12**	**225.85**	**231.51**	**237.14**	**243.06**	**249.14**	**255.37**
EBDITA	**6.82**	**40.72**	**45.74**	**48.45**	**49.83**	**50.98**	**51.85**	**52.63**
EBDITA Margin	**4.26%**	**16.23%**	**16.84%**	**17.30%**	**17.36%**	**17.34%**	**17.23%**	**17.09%**
Depreciation	4.90	6.53	6.53	6.53	6.53	6.53	6.53	6.53

All Figures in Rs. Crores								
Particulars	**31-Mar-27**	**31-Mar-28**	**31-Mar-29**	**31-Mar-30**	**31-Mar-31**	**31-Mar-32**	**31-Mar-33**	**31-Mar-34**
Interest on Term Loan	12.72	16.37	14.46	12.11	9.34	6.12	2.61	0.05
Interest on Working Capital Loan	2.09	3.00	3.27	3.35	3.44	3.52	3.61	3.70
Total Expenditure	**172.98**	**236.02**	**250.11**	**253.51**	**256.44**	**259.23**	**261.89**	**265.65**
Profit Before Tax	**-12.90**	**14.82**	**21.48**	**26.45**	**30.52**	**34.81**	**39.10**	**42.35**
Applicable Tax	-	-	-	0.22	6.23	7.75	9.21	10.35
Profit After Tax	**-12.90**	**14.82**	**21.48**	**26.23**	**24.29**	**27.06**	**29.89**	**32.00**
PAT Margin	-8.06%	5.91%	7.91%	9.37%	8.46%	9.20%	9.93%	10.39%

Balance Sheet

All Figures in Rs. Crore									
Description	**31-Mar-26**	**31-Mar-27**	**31-Mar-28**	**31-Mar-29**	**31-Mar-30**	**31-Mar-31**	**31-Mar-32**	**31-Mar-33**	**31-Mar-34**
Sources of Funds									
Shareholders' Funds									
Equity Capital/ Quasi Equity	28.89	34.37	34.37	34.37	34.37	34.37	34.37	34.37	34.37
Reserves and Surplus	-	-12.90	1.91	23.39	49.62	73.91	100.97	130.86	162.87
Total Shareholders' Funds	**28.89**	**21.47**	**36.28**	**57.76**	**83.99**	**108.28**	**135.34**	**165.23**	**197.24**
Term Liabilities									
Term Loan	109.67	126.98	110.18	89.58	65.78	37.98	9.18	-	-
Unsecured Loan from Promoters	-	11.89	15.89	15.89	15.89	15.89	15.89	15.89	-
Other Term Liabilities									
Total Term Liabilities	**109.67**	**138.87**	**126.07**	**105.47**	**81.67**	**53.87**	**25.07**	**15.89**	**-**
Current Liabilities									
Creditors - Trade	-	21.89	14.79	16.25	16.51	16.92	17.34	17.78	18.22
Creditors - Expense	-	2.46	3.78	4.15	4.26	4.37	4.48	4.59	4.70
Working Capital Loan	-	27.87	30.03	32.72	33.52	34.36	35.22	36.10	37.00
Current Maturities	-	10.50	16.80	20.60	23.80	27.80	28.80	9.18	-
Total Current Liabilities	**-**	**62.72**	**65.39**	**73.73**	**78.09**	**83.44**	**85.83**	**67.64**	**59.92**
Total Sources of Funds	**138.56**	**223.06**	**227.75**	**236.96**	**243.75**	**245.60**	**246.25**	**248.76**	**257.16**
Application of Funds									
Gross Fixed Assets		162.56	162.56	162.56	162.56	162.56	162.56	162.56	162.56
Cumulative Depreciation	-	4.90	11.43	17.96	24.49	31.02	37.55	44.08	50.61
Net Fixed Assets	**-**	**157.66**	**151.13**	**144.60**	**138.07**	**131.54**	**125.01**	**118.48**	**111.95**
CWIP	**138.56**								
MAT Credit	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**
DSRA	**-**	**11.89**	**15.89**	**16.67**	**16.94**	**17.39**	**16.15**	**9.23**	**-**

All Figures in Rs. Crore									
Description	31-Mar-26	31-Mar-27	31-Mar-28	31-Mar-29	31-Mar-30	31-Mar-31	31-Mar-32	31-Mar-33	31-Mar-34
Current Assets									
Inventories	-	35.57	35.88	39.43	40.09	41.10	42.12	43.18	44.25
Debtors	-	15.94	22.72	24.61	25.37	26.00	26.65	27.32	28.00
Cash and Bank Balance	-	2.00	2.13	11.65	23.28	29.57	36.31	50.56	72.95
Total Current Assets	**-**	**53.51**	**60.73**	**75.69**	**88.74**	**96.67**	**105.09**	**121.06**	**145.21**
Expenses Written Off	-	-	-	-	-	-	-	-	-
Total Application of Funds	**138.56**	**223.06**	**227.75**	**236.96**	**243.75**	**245.60**	**246.25**	**248.76**	**257.16**

Cashflow Statement

All Figures in Rs. Crore									
Description	31-Mar-26	31-Mar-27	31-Mar-28	31-Mar-29	31-Mar-30	31-Mar-31	31-Mar-32	31-Mar-33	31-Mar-34
Sources of Funds									
PAT		-12.90	14.82	21.48	26.23	24.29	27.06	29.89	32.00
Depreciation	-	4.90	6.53	6.53	6.53	6.53	6.53	6.53	6.53
Increase in Equity	28.89	5.48	-	-	-	-	-	-	-
Increase in Term Loan	109.67	17.32	-	-	-	-	-	-	-
Increase in Unsecured Loan	-	11.89	4.00	-	-	-	-	-	-
Increase in Long Term Liabilities	-	-	-	-	-	-	-	-	-
Increase in Trade Creditors	-	21.89	-	1.47	0.25	0.41	0.42	0.43	0.44
Increase in Expense Creditors	-	2.46	1.32	0.38	0.11	0.11	0.11	0.11	0.11
Increase in Workings Capital Loan	-	27.87	2.16	2.69	0.80	0.84	0.86	0.88	0.90
Increase in Current Maturities	-	10.50	6.30	3.80	3.20	4.00	1.00	-	-
Decrease in Gross Fixed Assets	-	-	-	-	-	-	-	-	-
Decrease in CWIP	-	138.56	-	-	-	-	-	-	-
Decrease in MAT Credits	-	-	-	-	-	-	-	-	-
Decrease in DSRA	-	-	-	-	-	-	1.24	6.92	9.23
Decrease in Inevntories	-	-	-	-	-	-	-	-	-
Decrease in Debtors	-	-	-	-	-	-	-	-	-
Total Sources	**138.56**	**227.96**	**35.12**	**36.34**	**37.12**	**36.18**	**37.22**	**44.77**	**49.22**
Application of Funds									

All Figures in Rs. Crore									
Description	**31-Mar-26**	**31-Mar-27**	**31-Mar-28**	**31-Mar-29**	**31-Mar-30**	**31-Mar-31**	**31-Mar-32**	**31-Mar-33**	**31-Mar-34**
Decrease in Equity	-	-	-	-	-	-	-	-	-
Decrease in Term Loan	-	-	16.80	20.60	23.80	27.80	28.80	9.18	-
Decrease in Unsecured Loan	-	-	-	-	-	-	-	-	15.89
Decrease in Long Term Liabilities	-	-	-	-	-	-	-	-	-
Decrease in Trade Creditors	-	-	7.10	-	-	-	-	-	-
Decrease in Expense Creditors	-	-	-	-	-	-	-	-	-
Decrease in Working Capital Loan	-	-	-	-	-	-	-	-	-
Decrease in Current Maturities	-	-	-	-	-	-	-	19.62	9.18
Increase in Gross Fixed Assets	-	162.56	-	-	-	-	-	-	-
Increase in CWIP	138.56	-	-	-	-	-	-	-	-
Increase in MAT Credits	-	-	-	-	-	-	-	-	-
Increase in DSRA	-	11.89	4.00	0.79	0.26	0.45	-	-	-
Increase in Inventories	-	35.57	0.31	3.55	0.66	1.00	1.03	1.05	1.08
Increase in Debtors	-	15.94	6.78	1.88	0.76	0.63	0.65	0.67	0.68
Total Uses	**138.56**	**225.96**	**34.99**	**26.82**	**25.49**	**29.89**	**30.48**	**30.52**	**26.83**
Net Cash Flow	-	2.00	0.13	9.52	11.63	6.29	6.74	14.25	22.39
Opening Balance	-	-	2.00	2.13	11.65	23.28	29.57	36.31	50.56
Closing Balance	-	2.00	2.13	11.65	23.28	29.57	36.31	50.56	72.95

Repayment Schedule

Term Loan – 1

Particulars	31-Mar-26	31-Mar-27	31-Mar-28	31-Mar-29	31-Mar-30	31-Mar-31	31-Mar-32	31-Mar-33	31-Mar-34
	All Figures in Rs. Crores								
Interest Rate	12.34%	12.34%	12.34%	12.34%	12.34%	12.34%	12.34%	12.34%	12.34%
Annual Summary									
Opening Balance of Loan	-	109.67	115.57	106.57	91.57	73.57	53.57	30.57	7.57
Addition of Loan	8.36	5.90	-	-	-	-	-	-	-
Repayment of Loan	-	-	9.00	15.00	18.00	20.00	23.00	23.00	7.57
Closing Balance of Loan	109.67	115.57	106.57	91.57	73.57	53.57	30.57	7.57	-
Interest for the Period		10.70	13.75	12.07	10.00	7.64	4.95	2.12	0.04
April									
Opening Balance of Loan		109.67	115.57	106.57	91.57	73.57	53.57	30.57	7.57
Addition of Loan		1.97	-						
Repayment of Loan			-	3.75	4.50	5.00	5.75	5.75	7.57
Closing Balance of Loan	-	111.63	115.57	102.82	87.07	68.57	47.82	24.82	-
Interest for the Period	-	1.14	1.19	1.08	0.92	0.73	0.52	0.28	0.04
May									
Opening Balance of Loan	-	111.63	115.57	102.82	87.07	68.57	47.82	24.82	-
Addition of Loan		1.97	-						
Repayment of Loan									
Closing Balance of Loan	-	113.60	115.57	102.82	87.07	68.57	47.82	24.82	-
Interest for the Period	-	1.16	1.19	1.06	0.90	0.71	0.49	0.26	-
June									
Opening Balance of Loan	-	113.60	115.57	102.82	87.07	68.57	47.82	24.82	-
Addition of Loan	-	1.97	-						
Repayment of Loan									
Closing Balance of Loan	-	115.57	115.57	102.82	87.07	68.57	47.82	24.82	-
Interest for the Period	-	1.18	1.19	1.06	0.90	0.71	0.49	0.26	-

All Figures in Rs. Crores									
Particulars	**31-Mar-26**	**31-Mar-27**	**31-Mar-28**	**31-Mar-29**	**31-Mar-30**	**31-Mar-31**	**31-Mar-32**	**31-Mar-33**	**31-Mar-34**
July									
Opening Balance of Loan	-	115.57	115.57	102.82	87.07	68.57	47.82	24.82	-
Addition of Loan									
Repayment of Loan			3.00	3.75	4.50	5.00	5.75	5.75	
Closing Balance of Loan	-	115.57	112.57	99.07	82.57	63.57	42.07	19.07	-
Interest for the Period	-	1.19	1.17	1.04	0.87	0.68	0.46	0.23	-
August									
Opening Balance of Loan	-	115.57	112.57	99.07	82.57	63.57	42.07	19.07	-
Addition of Loan		-	-						
Repayment of Loan									
Closing Balance of Loan	-	115.57	112.57	99.07	82.57	63.57	42.07	19.07	-
Interest for the Period	-	1.19	1.16	1.02	0.85	0.65	0.43	0.20	-
Sepetember									
Opening Balance of Loan	-	115.57	112.57	99.07	82.57	63.57	42.07	19.07	-
Addition of Loan	-	-	-						
Repayment of Loan									
Closing Balance of Loan	-	115.57	112.57	99.07	82.57	63.57	42.07	19.07	-
Interest for the Period	-	1.19	1.16	1.02	0.85	0.65	0.43	0.20	-
October									
Opening Balance of Loan	-	115.57	112.57	99.07	82.57	63.57	42.07	19.07	-
Addition of Loan		-							
Repayment of Loan			3.00	3.75	4.50	5.00	5.75	5.75	
Closing Balance of Loan	-	115.57	109.57	95.32	78.07	58.57	36.32	13.32	-
Interest for the Period	-	1.19	1.14	1.00	0.83	0.63	0.40	0.17	-
November									
Opening Balance of Loan	-	115.57	109.57	95.32	78.07	58.57	36.32	13.32	-

All Figures in Rs. Crores									
Particulars	**31-Mar-26**	**31-Mar-27**	**31-Mar-28**	**31-Mar-29**	**31-Mar-30**	**31-Mar-31**	**31-Mar-32**	**31-Mar-33**	**31-Mar-34**
Addition of Loan		-							
Repayment of Loan									
Closing Balance of Loan	-	115.57	109.57	95.32	78.07	58.57	36.32	13.32	-
Interest for the Period	-	1.19	1.13	0.98	0.80	0.60	0.37	0.14	-
December									
Opening Balance of Loan	-	115.57	109.57	95.32	78.07	58.57	36.32	13.32	-
Addition of Loan	-	-							
Repayment of Loan									
Closing Balance of Loan	-	115.57	109.57	95.32	78.07	58.57	36.32	13.32	-
Interest for the Period	-	1.19	1.13	0.98	0.80	0.60	0.37	0.14	-
January									
Opening Balance of Loan	101.30	115.57	109.57	95.32	78.07	58.57	36.32	13.32	-
Addition of Loan		-							
Repayment of Loan		-	3.00	3.75	4.50	5.00	5.75	5.75	
Closing Balance of Loan	101.30	115.57	106.57	91.57	73.57	53.57	30.57	7.57	-
Interest for the Period	1.04	1.19	1.11	0.96	0.78	0.58	0.34	0.11	-
February									
Opening Balance of Loan	101.30	115.57	106.57	91.57	73.57	53.57	30.57	7.57	-
Addition of Loan	-	-							
Repayment of Loan		-							
Closing Balance of Loan	101.30	115.57	106.57	91.57	73.57	53.57	30.57	7.57	-
Interest for the Period	1.04	1.19	1.10	0.94	0.76	0.55	0.31	0.08	-
March									
Opening Balance of Loan	101.30	115.57	106.57	91.57	73.57	53.57	30.57	7.57	-
Addition of Loan	8.36	-							
Repayment of Loan		-							
Closing Balance of Loan	109.67	115.57	106.57	91.57	73.57	53.57	30.57	7.57	-

All Figures in Rs. Crores									
Particulars	31-Mar-26	31-Mar-27	31-Mar-28	31-Mar-29	31-Mar-30	31-Mar-31	31-Mar-32	31-Mar-33	31-Mar-34
Interest for the Period	1.08	1.19	1.10	0.94	0.76	0.55	0.31	0.08	-

Term Loan – 2

All Figures in Rs. Crores									
Particulars	31-Mar-26	31-Mar-27	31-Mar-28	31-Mar-29	31-Mar-30	31-Mar-31	31-Mar-32	31-Mar-33	31-Mar-34
Interest Rate	12.34%	12.34%	12.34%	12.34%	12.34%	12.34%	12.34%	12.34%	12.34%
Annual Summary									
Opening Balance of Loan	-	-	21.91	20.41	18.61	16.01	12.21	7.41	1.61
Addition of Loan	-	21.91	-	-	-	-	-	-	-
Repayment of Loan	-	-	1.50	1.80	2.60	3.80	4.80	5.80	1.61
Closing Balance of Loan	-	21.91	20.41	18.61	16.01	12.21	7.41	1.61	-
Interest for the Period		2.03	2.62	2.39	2.11	1.70	1.16	0.50	0.01
April									
Opening Balance of Loan		-	21.91	20.41	18.61	16.01	12.21	7.41	1.61
Addition of Loan		7.30	-						
Repayment of Loan				0.45	0.65	0.95	1.20	1.45	1.61
Closing Balance of Loan	-	7.30	21.91	19.96	17.96	15.06	11.01	5.96	-
Interest for the Period	-	0.08	0.23	0.21	0.19	0.16	0.12	0.07	0.01
May									
Opening Balance of Loan	-	7.30	21.91	19.96	17.96	15.06	11.01	5.96	-
Addition of Loan		7.30	-						
Repayment of Loan									
Closing Balance of Loan	-	14.61	21.91	19.96	17.96	15.06	11.01	5.96	-
Interest for the Period	-	0.11	0.23	0.21	0.18	0.15	0.11	0.06	-
June									
Opening Balance of Loan	-	14.61	21.91	19.96	17.96	15.06	11.01	5.96	-
Addition of Loan	-	7.30	-						

All Figures in Rs. Crores									
Particulars	**31-Mar-26**	**31-Mar-27**	**31-Mar-28**	**31-Mar-29**	**31-Mar-30**	**31-Mar-31**	**31-Mar-32**	**31-Mar-33**	**31-Mar-34**
Repayment of Loan									
Closing Balance of Loan	-	21.91	21.91	19.96	17.96	15.06	11.01	5.96	-
Interest for the Period	-	0.19	0.23	0.21	0.18	0.15	0.11	0.06	-
July									
Opening Balance of Loan	-	21.91	21.91	19.96	17.96	15.06	11.01	5.96	-
Addition of Loan									
Repayment of Loan			0.50	0.45	0.65	0.95	1.20	1.45	
Closing Balance of Loan	-	21.91	21.41	19.51	17.31	14.11	9.81	4.51	-
Interest for the Period	-	0.23	0.22	0.20	0.18	0.15	0.11	0.05	-
August									
Opening Balance of Loan	-	21.91	21.41	19.51	17.31	14.11	9.81	4.51	-
Addition of Loan		-	-						
Repayment of Loan									
Closing Balance of Loan	-	21.91	21.41	19.51	17.31	14.11	9.81	4.51	-
Interest for the Period	-	0.23	0.22	0.20	0.18	0.15	0.10	0.05	-
Sepetember									
Opening Balance of Loan	-	21.91	21.41	19.51	17.31	14.11	9.81	4.51	-
Addition of Loan	-	-	-						
Repayment of Loan									
Closing Balance of Loan	-	21.91	21.41	19.51	17.31	14.11	9.81	4.51	-
Interest for the Period	-	0.23	0.22	0.20	0.18	0.15	0.10	0.05	-
October									
Opening Balance of Loan	-	21.91	21.41	19.51	17.31	14.11	9.81	4.51	-
Addition of Loan		-							
Repayment of Loan			0.50	0.45	0.65	0.95	1.20	1.45	-
Closing Balance of Loan	-	21.91	20.91	19.06	16.66	13.16	8.61	3.06	-

All Figures in Rs. Crores									
Particulars	**31-Mar-26**	**31-Mar-27**	**31-Mar-28**	**31-Mar-29**	**31-Mar-30**	**31-Mar-31**	**31-Mar-32**	**31-Mar-33**	**31-Mar-34**
Interest for the Period	-	0.23	0.22	0.20	0.17	0.14	0.09	0.04	-
November									
Opening Balance of Loan	-	21.91	20.91	19.06	16.66	13.16	8.61	3.06	-
Addition of Loan		-							
Repayment of Loan									
Closing Balance of Loan	-	21.91	20.91	19.06	16.66	13.16	8.61	3.06	-
Interest for the Period	-	0.23	0.22	0.20	0.17	0.14	0.09	0.03	-
December									
Opening Balance of Loan	-	21.91	20.91	19.06	16.66	13.16	8.61	3.06	-
Addition of Loan	-	-							
Repayment of Loan									
Closing Balance of Loan	-	21.91	20.91	19.06	16.66	13.16	8.61	3.06	-
Interest for the Period	-	0.23	0.22	0.20	0.17	0.14	0.09	0.03	-
January									
Opening Balance of Loan	-	21.91	20.91	19.06	16.66	13.16	8.61	3.06	-
Addition of Loan		-							
Repayment of Loan		-	0.50	0.45	0.65	0.95	1.20	1.45	-
Closing Balance of Loan	-	21.91	20.41	18.61	16.01	12.21	7.41	1.61	-
Interest for the Period	-	0.23	0.21	0.19	0.17	0.13	0.08	0.02	-
February									
Opening Balance of Loan	-	21.91	20.41	18.61	16.01	12.21	7.41	1.61	-
Addition of Loan	-	-							
Repayment of Loan		-							
Closing Balance of Loan	-	21.91	20.41	18.61	16.01	12.21	7.41	1.61	-
Interest for the Period	-	0.23	0.21	0.19	0.16	0.13	0.08	0.02	-
March									

All Figures in Rs. Crores									
Particulars	31-Mar-26	31-Mar-27	31-Mar-28	31-Mar-29	31-Mar-30	31-Mar-31	31-Mar-32	31-Mar-33	31-Mar-34
Opening Balance of Loan	-	21.91	20.41	18.61	16.01	12.21	7.41	1.61	-
Addition of Loan	-	-							
Repayment of Loan		-							
Closing Balance of Loan	-	21.91	20.41	18.61	16.01	12.21	7.41	1.61	-
Interest for the Period	-	0.23	0.21	0.19	0.16	0.13	0.08	0.02	-

Term Loan – 1 & 2 Summary

All Figures in Rs. Crores									
Particulars	31-Mar-26	31-Mar-27	31-Mar-28	31-Mar-29	31-Mar-30	31-Mar-31	31-Mar-32	31-Mar-33	31-Mar-34
Interest Rate	12.34%	12.34%	12.34%	12.34%	12.34%	12.34%	12.34%	12.34%	12.34%
Annual Summary									
Opening Balance of Loan	-	109.67	137.48	126.98	110.18	89.58	65.78	37.98	9.18
Addition of Loan	8.36	27.82	-	-	-	-	-	-	-
Repayment of Loan	-	-	10.50	16.80	20.60	23.80	27.80	28.80	9.18
Closing Balance of Loan	109.67	137.48	126.98	110.18	89.58	65.78	37.98	9.18	-
Interest for the Period	-	12.72	16.37	14.46	12.11	9.34	6.12	2.61	0.05
April									
Opening Balance of Loan	-	109.67	137.48	126.98	110.18	89.58	65.78	37.98	9.18
Addition of Loan	-	9.27	-	-	-	-	-	-	-
Repayment of Loan	-	-	-	4.20	5.15	5.95	6.95	7.20	9.18
Closing Balance of Loan	-	118.94	137.48	122.78	105.03	83.63	58.83	30.78	-
Interest for the Period	-	1.21	1.41	1.28	1.11	0.89	0.64	0.35	0.05
May									
Opening Balance of Loan	-	118.94	137.48	122.78	105.03	83.63	58.83	30.78	-
Addition of Loan	-	9.27	-	-	-	-	-	-	-
Repayment of Loan	-	-	-	-	-	-	-	-	-
Closing Balance of Loan	-	128.21	137.48	122.78	105.03	83.63	58.83	30.78	-

All Figures in Rs. Crores									
Particulars	31-Mar-26	31-Mar-27	31-Mar-28	31-Mar-29	31-Mar-30	31-Mar-31	31-Mar-32	31-Mar-33	31-Mar-34
Interest for the Period	-	1.27	1.41	1.26	1.08	0.86	0.60	0.32	-
June									
Opening Balance of Loan	-	128.21	137.48	122.78	105.03	83.63	58.83	30.78	-
Addition of Loan	-	9.27	-	-	-	-	-	-	-
Repayment of Loan	-	-	-	-	-	-	-	-	-
Closing Balance of Loan	-	137.48	137.48	122.78	105.03	83.63	58.83	30.78	-
Interest for the Period	-	1.37	1.41	1.26	1.08	0.86	0.60	0.32	-
July									
Opening Balance of Loan	-	137.48	137.48	122.78	105.03	83.63	58.83	30.78	-
Addition of Loan	-	-	-	-	-	-	-	-	-
Repayment of Loan	-	-	3.50	4.20	5.15	5.95	6.95	7.20	-
Closing Balance of Loan	-	137.48	133.98	118.58	99.88	77.68	51.88	23.58	-
Interest for the Period	-	1.41	1.40	1.24	1.05	0.83	0.57	0.28	-
August									
Opening Balance of Loan	-	137.48	133.98	118.58	99.88	77.68	51.88	23.58	-
Addition of Loan	-	-	-	-	-	-	-	-	-
Repayment of Loan	-	-	-	-	-	-	-	-	-
Closing Balance of Loan	-	137.48	133.98	118.58	99.88	77.68	51.88	23.58	-
Interest for the Period	-	1.41	1.38	1.22	1.03	0.80	0.53	0.24	-
Sepetember									
Opening Balance of Loan	-	137.48	133.98	118.58	99.88	77.68	51.88	23.58	-
Addition of Loan	-	-	-	-	-	-	-	-	-
Repayment of Loan	-	-	-	-	-	-	-	-	-
Closing Balance of Loan	-	137.48	133.98	118.58	99.88	77.68	51.88	23.58	-
Interest for the Period	-	1.41	1.38	1.22	1.03	0.80	0.53	0.24	-
October									

All Figures in Rs. Crores									
Particulars	**31-Mar-26**	**31-Mar-27**	**31-Mar-28**	**31-Mar-29**	**31-Mar-30**	**31-Mar-31**	**31-Mar-32**	**31-Mar-33**	**31-Mar-34**
Opening Balance of Loan	-	137.48	133.98	118.58	99.88	77.68	51.88	23.58	-
Addition of Loan	-	-	-	-	-	-	-	-	-
Repayment of Loan	-	-	3.50	4.20	5.15	5.95	6.95	7.20	-
Closing Balance of Loan	-	137.48	130.48	114.38	94.73	71.73	44.93	16.38	-
Interest for the Period	-	1.41	1.36	1.20	1.00	0.77	0.50	0.21	-
November									
Opening Balance of Loan	-	137.48	130.48	114.38	94.73	71.73	44.93	16.38	-
Addition of Loan	-	-	-	-	-	-	-	-	-
Repayment of Loan	-	-	-	-	-	-	-	-	-
Closing Balance of Loan	-	137.48	130.48	114.38	94.73	71.73	44.93	16.38	-
Interest for the Period	-	1.41	1.34	1.18	0.97	0.74	0.46	0.17	-
December									
Opening Balance of Loan	-	137.48	130.48	114.38	94.73	71.73	44.93	16.38	-
Addition of Loan	-	-	-	-	-	-	-	-	-
Repayment of Loan	-	-	-	-	-	-	-	-	-
Closing Balance of Loan	-	137.48	130.48	114.38	94.73	71.73	44.93	16.38	-
Interest for the Period	-	1.41	1.34	1.18	0.97	0.74	0.46	0.17	-
January									
Opening Balance of Loan	101.30	137.48	130.48	114.38	94.73	71.73	44.93	16.38	-
Addition of Loan	-	-	-	-	-	-	-	-	-
Repayment of Loan	-	-	3.50	4.20	5.15	5.95	6.95	7.20	-
Closing Balance of Loan	101.30	137.48	126.98	110.18	89.58	65.78	37.98	9.18	-
Interest for the Period	1.04	1.41	1.32	1.15	0.95	0.71	0.43	0.13	-
February									
Opening Balance of Loan	101.30	137.48	126.98	110.18	89.58	65.78	37.98	9.18	-
Addition of Loan	-	-	-	-	-	-	-	-	-
Repayment of Loan	-	-	-	-	-	-	-	-	-

All Figures in Rs. Crores									
Particulars	31-Mar-26	31-Mar-27	31-Mar-28	31-Mar-29	31-Mar-30	31-Mar-31	31-Mar-32	31-Mar-33	31-Mar-34
Closing Balance of Loan	101.30	137.48	126.98	110.18	89.58	65.78	37.98	9.18	-
Interest for the Period	1.04	1.41	1.31	1.13	0.92	0.68	0.39	0.09	-
March									
Opening Balance of Loan	101.30	137.48	126.98	110.18	89.58	65.78	37.98	9.18	-
Addition of Loan	8.36	-	-	-	-	-	-	-	-
Repayment of Loan	-	-	-	-	-	-	-	-	-
Closing Balance of Loan	109.67	137.48	126.98	110.18	89.58	65.78	37.98	9.18	-
Interest for the Period	1.08	1.41	1.31	1.13	0.92	0.68	0.39	0.09	-

Depreciation

Depreciation – Companies Act – SLM

All Figures in Rs. Crores								
Particulars	**31-Mar-27**	**31-Mar-28**	**31-Mar-29**	**31-Mar-30**	**31-Mar-31**	**31-Mar-32**	**31-Mar-33**	**31-Mar-34**
Opening Block	**138.56**	**162.56**	**162.56**	**162.56**	**162.56**	**162.56**	**162.56**	**162.56**
Land and Land Development	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Building and Civil Works	24.86	28.12	28.12	28.12	28.12	28.12	28.12	28.12
Plant and Machinery Cost	112.20	132.85	132.85	132.85	132.85	132.85	132.85	132.85
Miscellaneous Fixed Assets	0.49	0.59	0.59	0.59	0.59	0.59	0.59	0.59
Gross Fixed Assets	**24.00**	**-**	**-**	**-**	**-**	**-**	**-**	**-**
Land and Land Development	-	-	-	-	-	-	-	-
Building and Civil Works	3.26	-	-	-	-	-	-	-
Plant and Machinery Cost	20.65	-	-	-	-	-	-	-
Miscellaneous Fixed Assets	0.09	-	-	-	-	-	-	-
Gross Fixed Assets	**162.56**	**162.56**	**162.56**	**162.56**	**162.56**	**162.56**	**162.56**	**162.56**
Land and Land Development	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Building and Civil Works	28.12	28.12	28.12	28.12	28.12	28.12	28.12	28.12
Plant and Machinery Cost	132.85	132.85	132.85	132.85	132.85	132.85	132.85	132.85
Miscellaneous Fixed Assets	0.59	0.59	0.59	0.59	0.59	0.59	0.59	0.59
Depreciation	**4.90**	**6.53**	**6.53**	**6.53**	**6.53**	**6.53**	**6.53**	**6.53**
Land and Land Development	-	-	-	-	-	-	-	-
Building and Civil Works	0.73	0.97	0.97	0.97	0.97	0.97	0.97	0.97
Plant and Machinery Cost	4.16	5.54	5.54	5.54	5.54	5.54	5.54	5.54
Miscellaneous Fixed Assets	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02
Cumulative Depreciation	**4.90**	**11.43**	**17.96**	**24.49**	**31.02**	**37.55**	**44.08**	**50.61**
Land and Land Development	-	-	-	-	-	-	-	-
Building and Civil Works	0.73	1.70	2.67	3.64	4.61	5.58	6.55	7.52
Plant and Machinery Cost	4.16	9.69	15.23	20.76	26.30	31.83	37.37	42.90
Miscellaneous Fixed Assets	0.02	0.04	0.07	0.09	0.12	0.14	0.17	0.19
Net Fixed Assets	**157.66**	**151.13**	**144.60**	**138.07**	**131.54**	**125.01**	**118.48**	**111.95**

All Figures in Rs. Crores								
Particulars	31-Mar-27	31-Mar-28	31-Mar-29	31-Mar-30	31-Mar-31	31-Mar-32	31-Mar-33	31-Mar-34
Land and Land Development	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Building and Civil Works	27.39	26.42	25.45	24.48	23.52	22.55	21.58	20.61
Plant and Machinery Cost	128.69	123.16	117.62	112.09	106.55	101.02	95.48	89.95
Miscellaneous Fixed Assets	0.57	0.55	0.52	0.50	0.47	0.45	0.42	0.40

Depreciation - Income Tax Act - WDV

All Figures in INR Crores								
Particulars	31-Mar-27	31-Mar-28	31-Mar-29	31-Mar-30	31-Mar-31	31-Mar-32	31-Mar-33	31-Mar-34
Opening Block	**138.56**	**162.56**	**162.56**	**162.56**	**162.56**	**162.56**	**162.56**	**162.56**
Land and Land Development	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Building and Civil Works	24.86	28.12	28.12	28.12	28.12	28.12	28.12	28.12
Plant and Machinery Cost	112.20	132.85	132.85	132.85	132.85	132.85	132.85	132.85
Miscellaneous Fixed Assets	0.49	0.59	0.59	0.59	0.59	0.59	0.59	0.59
Gross Fixed Assets	**24.00**	**-**	**-**	**-**	**-**	**-**	**-**	**-**
Land and Land Development	-	-	-	-	-	-	-	-
Building and Civil Works	3.26	-	-	-	-	-	-	-
Plant and Machinery Cost	20.65	-	-	-	-	-	-	-
Miscellaneous Fixed Assets	0.09	-	-	-	-	-	-	-
Gross Fixed Assets	**162.56**	**162.56**	**162.56**	**162.56**	**162.56**	**162.56**	**162.56**	**162.56**
Land and Land Development	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Building and Civil Works	28.12	28.12	28.12	28.12	28.12	28.12	28.12	28.12
Plant and Machinery Cost	132.85	132.85	132.85	132.85	132.85	132.85	132.85	132.85
Miscellaneous Fixed Assets	0.59	0.59	0.59	0.59	0.59	0.59	0.59	0.59
Depreciation	**22.83**	**19.54**	**16.74**	**14.34**	**12.29**	**10.54**	**9.04**	**7.76**
Land and Land Development	-	-	-	-	-	-	-	-
Building and Civil Works	2.81	2.53	2.28	2.05	1.85	1.66	1.49	1.35
Plant and Machinery Cost	19.93	16.94	14.40	12.24	10.40	8.84	7.52	6.39
Miscellaneous Fixed Assets	0.09	0.08	0.06	0.05	0.05	0.04	0.03	0.03

All Figures in INR Crores								
Particulars	31-Mar-27	31-Mar-28	31-Mar-29	31-Mar-30	31-Mar-31	31-Mar-32	31-Mar-33	31-Mar-34
Cumulative Depreciation	**22.83**	**42.37**	**59.11**	**73.45**	**85.75**	**96.29**	**105.33**	**113.09**
Land and Land Development	-	-	-	-	-	-	-	-
Building and Civil Works	2.81	5.34	7.62	9.67	11.52	13.18	14.67	16.02
Plant and Machinery Cost	19.93	36.87	51.26	63.50	73.90	82.75	90.26	96.65
Miscellaneous Fixed Assets	0.09	0.16	0.23	0.28	0.33	0.37	0.40	0.43
Net Fixed Assets	**139.73**	**120.19**	**103.45**	**89.11**	**76.81**	**66.27**	**57.23**	**49.47**
Land and Land Development	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Building and Civil Works	25.31	22.78	20.50	18.45	16.61	14.95	13.45	12.11
Plant and Machinery Cost	112.92	95.98	81.59	69.35	58.95	50.10	42.59	36.20
Miscellaneous Fixed Assets	0.50	0.43	0.36	0.31	0.26	0.22	0.19	0.16

Tax Assessment

All Figures in Rs. Crores								
Description	31-Mar-27	31-Mar-28	31-Mar-29	31-Mar-30	31-Mar-31	31-Mar-32	31-Mar-33	31-Mar-34
Profit Before Tax	-12.90	14.82	21.48	26.45	30.52	34.81	39.10	42.35
Add: Dep. As per Comp. Act	4.90	6.53	6.53	6.53	6.53	6.53	6.53	6.53
Less: Dep. As per IT Act	22.83	19.54	16.74	14.34	12.29	10.54	9.04	7.76
Revised PBT	-30.83	1.80	11.27	18.64	24.76	30.80	36.59	41.12
Set- Off of Losses	-	1.80	11.27	17.76	-	-	-	-
Taxable Income	-	-	-	0.88	24.76	30.80	36.59	41.12
Income Tax Rate								
Basic Rate	22.00%	22.00%	22.00%	22.00%	22.00%	22.00%	22.00%	22.00%
Surcharge	10.00%	10.00%	10.00%	10.00%	10.00%	10.00%	10.00%	10.00%
Cess	4.00%	4.00%	4.00%	4.00%	4.00%	4.00%	4.00%	4.00%
Effective Corporate Tax Rate	25.17%	25.17%	25.17%	25.17%	25.17%	25.17%	25.17%	25.17%
MAT Rate								
Basic Rate	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
Surcharge	10.00%	10.00%	10.00%	10.00%	10.00%	10.00%	10.00%	10.00%
Cess	4.00%	4.00%	4.00%	4.00%	4.00%	4.00%	4.00%	4.00%
MAT	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%

All Figures in Rs. Crores								
Description	**31-Mar-27**	**31-Mar-28**	**31-Mar-29**	**31-Mar-30**	**31-Mar-31**	**31-Mar-32**	**31-Mar-33**	**31-Mar-34**
Income Tax	-	-	-	0.22	6.23	7.75	9.21	10.35
MAT	-	-	-	-	-	-	-	-
Higher Tax of IT and MAT	-	-	-	0.22	6.23	7.75	9.21	10.35
Less: MAT Credit	-	-	-	-	-	-	-	-
Applicable Tax	-	-	-	0.22	6.23	7.75	9.21	10.35

Financial Ratios Calculations

NPV and IRR

All Figures in Rs. Crore											
Description	**31-Mar-26**	**31-Mar-27**	**31-Mar-28**	**31-Mar-29**	**31-Mar-30**	**31-Mar-31**	**31-Mar-32**	**31-Mar-33**	**31-Mar-34**	**31-Mar-35**	**31-Mar-36**
Initial Capital Investment											
Capital Investment	-138.56	-24.00	-	-	-	-	-	-	-	-	-
Increase/ Decrease in Working Capital		-9.29	-2.88	-3.59	-1.07	-1.12	-1.14	-1.17	-1.20	-4.23	-1.31
Total Initial Capital Investment	**-138.56**	**-33.30**	**-2.88**	**-3.59**	**-1.07**	**-1.12**	**-1.14**	**-1.17**	**-1.20**	**-4.23**	**-1.31**
Operating Cash Flow											
PAT	-	-12.90	14.82	21.48	26.23	24.29	27.06	29.89	32.00	33.31	35.46
Depreciation	-	4.90	6.53	6.53	6.53	6.53	6.53	6.53	6.53	6.53	6.53
Interest Coverage	-	14.82	19.37	17.73	15.34	10.17	7.49	4.76	2.83	2.77	2.75
Total Operating Cashflow	**-**	**6.82**	**40.72**	**45.74**	**48.10**	**40.99**	**41.08**	**41.18**	**41.36**	**42.61**	**44.75**
Terminal Cashflow											
Salvage Value											98.89
Working Capital Release											167.12
Total Terminal Cashflow	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**266.01**
Total Free Cashflow	**-138.56**	**-26.48**	**37.85**	**42.15**	**47.03**	**39.87**	**39.94**	**40.01**	**40.16**	**38.37**	**309.45**

All Figures in Rs. Crore											
Description	31-Mar-26	31-Mar-27	31-Mar-28	31-Mar-29	31-Mar-30	31-Mar-31	31-Mar-32	31-Mar-33	31-Mar-34	31-Mar-35	31-Mar-36
NPV	137.74										
IRR	22.44%										
Post tax Cost of Capital	10.15%										

DSCR

All Figures in Rs. Crore								
Description	31-Mar-27	31-Mar-28	31-Mar-29	31-Mar-30	31-Mar-31	31-Mar-32	31-Mar-33	31-Mar-34
Debt Service Coverage Ratio								
PAT	-12.90	14.82	21.48	26.23	24.29	27.06	29.89	32.00
Depreciation	4.90	6.53	6.53	6.53	6.53	6.53	6.53	6.53
Interest on Term Loan/ Secured Loan	12.72	16.37	14.46	12.11	9.34	6.12	2.61	0.05
Total	4.72	37.72	42.47	44.87	40.16	39.71	39.03	38.58
Interest on Term Loan/ Secured Loan	12.72	16.37	14.46	12.11	9.34	6.12	2.61	0.05
Repayment of Loan	-	9.00	15.00	18.00	20.00	23.00	23.00	7.57
Total	12.72	25.37	29.46	30.11	29.34	29.12	25.61	7.62
DSCR	-	1.49	1.44	1.49	1.37	1.36	1.52	5.06
Min DSCR	1.36							
Average DSCR	2.77							
Interest Coverage Ratio								
EBIT	1.91	34.19	39.21	41.92	43.30	44.45	45.32	46.10
Interest for the Period	14.82	19.37	17.73	15.47	12.78	9.64	6.22	3.75
Interest Coverage Ratio	0.13	1.76	2.21	2.71	3.39	4.61	7.28	12.30

Weighted Average Cost of Capital

Loans	Amount	ROI	Tax Rate	Post tax cost of capital (%)	Proportion	WACC
Equity/ Quasi Equity/ Unsecured	34.37	14.00%	0.00%	14.00%	20.00%	2.80%
Term Loan	137.48	12.34%	25.57%	9.18%	80.00%	7.35%
Total	**171.85**					**10.15%**

Payback Period

All Figures in Rs. Crore					
Description	31-Mar-27	31-Mar-28	31-Mar-29	31-Mar-30	31-Mar-31
Project Duration (Years)	0	1	2	3	4
Cumulative Cashflow	-165.04	-127.19	-85.04	-38.01	1.86
Payback Period (Months)	Not Yet	Not Yet	Not Yet	Not Yet	37.90

Break Even Point

All Figures in Rs. Crore								
Description	31-Mar-27	31-Mar-28	31-Mar-29	31-Mar-30	31-Mar-31	31-Mar-32	31-Mar-33	31-Mar-34
Total Expenses	153.26	210.12	225.85	231.51	237.14	243.06	249.14	255.37
Total Fixed Costs (incl Dep and Int)	21.71	29.03	27.65	25.48	22.88	19.83	16.51	14.13
Total Variable Cost	151.26	206.99	222.47	228.03	233.56	239.40	245.38	251.52
Contribution	8.81	43.85	49.12	51.94	53.40	54.64	55.61	56.48
Contribution Margin	5.50%	17.48%	18.09%	18.55%	18.61%	18.58%	18.47%	18.34%
Break Even Sales	394.54	166.07	152.85	137.37	122.96	106.72	89.36	77.04
Break Even Capacity Utilisation	246.48 %	66.21%	56.28%	49.07%	42.85%	36.29%	29.69%	25.01%
Cash Break Even	305.47	128.71	116.74	102.17	87.87	71.58	54.02	41.43
Cash Break Even Margin	190.83 %	51.31%	42.99%	36.49%	30.62%	24.34%	17.95%	13.45%

Intentionally Left Blank for Notes